PART I: Technical Foundation — Manufacturing, Supply Chain, and Regulatory Framework

Chapter 1

This chapter reproduces the full content of the main technical report on high-temperature permanent magnets for defense and security applications, including magnet types, manufacturing, machining waste, supply chain structure, competitive landscape, and qualification/testing requirements.

Executive Summary

Defense and security systems rely on rare-earth permanent magnets as enabling components across major platforms, including the F-35 Lightning II, Virginia and Columbia class submarines, and unmanned aerial vehicles (UAVs). Reported platform-level demand is large, with the F-35 requiring more than 900 lb of rare earth elements (REEs), an Arleigh Burke DDG-51 destroyer requiring about 5,200 lb, and a Virginia class submarine needing about 9,200 lb, underscoring the strategic importance of secure supply chains for these materials and components.

For high-temperature operation, samarium-cobalt (SmCo) magnets (SmCo5 and Sm2Co17 families) are emphasized in the provided sources as preferred for high-temperature applications (e.g., 200–550 ℃) because of their thermal stability and corrosion resistance. High-temperature NdFeB is positioned as a separate engineering path where coercivity is maintained at elevated temperatures (e.g., 150–200 ℃) via heavy rare earth (HREE) additions (Dy/Tb) or via grain boundary diffusion processes that enhance coercivity without relying on bulk HREE loading.

Across both magnet types, the core industrial route described in the sources is powder metallurgy: alloy preparation, powder production, particle alignment and pressing, sintering and heat treatment, machining, and then magnetization. The machining and finishing step is consistently highlighted as the dominant source of material loss and scrap risk, with one public report stating machining recovery rates of 60–90% depending on final magnet complexity (i.e., 10–40% material becomes waste). The waste stream (magnet swarf) is described as often recycled back into process streams, and it can contain machine cutting fluids and magnetic powder because cutting is performed before magnetization.

From a defense supply-chain perspective, the sources emphasize that REE and critical material supply chains are global and multi-tiered, and that China dominates mining and processing stages for rare earths and other critical materials. U.S. policy and acquisition frameworks are tightening disclosure and sourcing requirements, including a DFARS-related expanded prohibition that GAO reports will go into effect on January 1, 2027, and a requirement for contractors to disclose the source of certain rare earth permanent magnets used in DoD systems.

Magnet Types and Grades

SmCo and high-temperature NdFeB in the provided sources are presented less as commodity items and more as tightly specified engineering materials chosen to match elevated-temperature reliability requirements and environmental constraints. In defense and security contexts, these magnets sit in actuators, motors, and electromechanical subsystems within major platforms, contributing to the large REE usage cited for systems such as the F-35 and naval vessels/submarines.

SmCo grades. The sources describe SmCo magnets (SmCo5 and Sm2Co17) as offering "excellent hard magnetic properties, thermal stability, and corrosion resistance," and as a preferred/ultimate choice for high-temperature applications (e.g., 200–550 ℃). Reported performance ranges include a broad SmCo (BH)max range of 16–33 MGOe, reflecting variability across compositions and processing.

A specific grade example in the dataset is an Sm2O3-doped Sm2(Co,Fe,Cu,Zr)17 magnet reported as "EEC 2:17-27," with (BH)max of 26–28 MGOe, intrinsic coercivity Hci of 34.1–36.1 kOe, and Br of 10.5–10.8 kGs. Commercial supplier data in the provided sources also lists VACOMAX SmCo properties including Br minimum values in the range 0.85–1.17 T (as presented), HcB minimum values around 600 kA/m (as presented), and typical (BH)max values in the range 160–265 kJ/m³ (20–33 MGOe), with maximum operating temperature stated as up to 350 ℃ for some alloys.

Thermal stability is further reflected in reversible temperature coefficient examples: a "standard Sm2Co17 magnet" is reported with a reversible temperature coefficient of about −350 ppm/K between room temperature and 200 ℃, and SmCo5 is reported with RTC(Br) about −450 ppm/K.

High-temperature NdFeB and coercivity engineering. The sources describe a conventional high-temperature NdFeB approach of adding Dy/Tb to improve coercivity and avoid demagnetization at high operating temperatures (150–200 ℃). One cited example (in a non-defense end-use context) reports that sintered NdFeB magnets used for wind turbines and electric/hybrid vehicle motors "usually contain about 10 wt% Dy/Tb," illustrating the scale of HREE loading sometimes used to preserve coercivity at elevated temperatures.

A second approach emphasized in the dataset is grain boundary diffusion (GBD), described as a "well-established method" that "considerably enhances the coercivity" of sintered NdFeB and optimizes microstructures, including approaches using DyH3 nanoparticles (to fabricate "Dy-less" high-coercivity magnets) and diffusion sources such as DyCo alloys or Dy films deposited by magnetron sputtering. Quantitatively, one study reports that increasing diffusion temperature from 700 to 950 ℃ increased Hcj from 1339.3 to 1767.1 kA/m while Jr increased from 1.380 T to 1.439 T. Another report states that Dy diffusion with a primary diffusion heat treatment at 900 ℃ for 6 h increased coercivity from 1073 to 1418 kA/m versus the original magnet.

Manufacturing Process Chain

The manufacturing chain described for sintered rare-earth magnets is dominated by powder metallurgy steps and high-temperature thermal processing, with the sources explicitly describing the general flow as alloy preparation, powder production, particle alignment and pressing, sintering and heat treatment, machining, and finally magnetizing. For SmCo specifically, one cited process statement describes manufacturing "by powder metallurgical technique involving isostatic pressing and sintering in inert atmosphere at temperatures range of 1140–1170 ℃."

Raw materials and powder production. Powder production routes in the sources include hydrogen decrepitation (HD), described as a process where a material absorbs hydrogen gas in its interstitial sites, as well as crushing/milling and jet milling routes for Sm2(Co,Fe,Cu,Zr)17 powders. One example reports Sm2(Co,Fe,Cu,Zr)17 powder prepared from book mold ingots by crushing and milling, then ground to an average particle size of about 7 ± 2 μm using an AFG jet mill.

Alignment and pressing. The sources show strong-field alignment as a critical step prior to compaction, including an example where blended powder mixtures were aligned using a 4 T pulsed magnetic field. Pressing approaches include isostatic pressing into green compacts, with one report citing approximately 240 MPa and another citing about 250 MPa after alignment in a magnetic field of 1300 kA/m.

Sintering, densification, and heat treatment. For SmCo sintering, the dataset includes a statement of inert-atmosphere sintering in the 1140–1170 ℃ range for powder-metallurgical routes. Another Sm2(Co,Fe,Cu,Zr)17 example reports sintering at about 1200 ℃ and holding slightly below sintering temperature for homogenization. Density is treated as a key acceptance/quality target, with one source stating that "the density [is required] to be > 95% of theoretical density."

Post-sinter heat treatment is documented as multi-step thermal processing, including quenching and annealing at 850 ℃ followed by slow cooling to 400 ℃ at 0.7 K/min, a multi-hour hold, and a final quench in one reported Sm2(Co,Fe,Cu,Zr)17 route. The sources also emphasize that sintering and heat treatment cause substantial shrinkage that must be designed into tooling and machining allowances, reporting about 7–25% linear shrinkage overall and anisotropic shrinkage ranges of 17–25% parallel to the magnetic alignment direction and 7–13% perpendicular to it (depending on composition and sintering temperatures).

Diffusion processing for high-coercivity NdFeB. For high-temperature NdFeB, grain boundary diffusion is described in the sources as a route to produce high-performance magnets and to enhance coercivity and thermal stability, using diffusion sources including DyH3 nanoparticles, DyCo alloys for infiltration, and Dy films deposited by magnetron sputtering. The diffusion temperature dependence and coercivity improvements (e.g., 700–950 ℃ leading to Hcj 1339.3⟶1767.1 kA/m; or 900 ℃/6 h yielding 1073⟶1418 kA/m) provide quantitative anchors for the thermal budgets involved in these processes.

Coatings and corrosion protection. The sources differentiate corrosion behavior by material family, noting that SmCo magnets are naturally corrosion resistant and that coating is often optional for SmCo in some applications. For NdFeB, a coating specification reference lists Ni-Cu-Ni (standard triple layer) with a temperature note of ~200 ℃, indicating the kinds of coatings referenced for neodymium magnets in higher-temperature environments.

Machining, Finishing, and Waste Generation

The sources consistently identify machining as both a major technical risk (brittleness-driven fracture and chipping) and the dominant stage for material loss when converting sintered blocks into end-use geometries. A general process description states that after sintering and heat treatment, magnets are machined and then magnetized, which also aligns with the description that cutting is performed before magnetization in the context of swarf generation.

How parts are finished and delivered. For sintered rare-earth magnets in general, one source describes that "finished magnets are then machined" using wire saws, wire electrical discharge machining (EDM), or abrasive wheel slitting machines, implying that suppliers can deliver machined/finished geometries rather than only sintered blocks. The same overarching process description places magnetization after machining, which is consistent with the machining-before-magnetization statement in the NdFeB supply chain report's swarf discussion.

Methods and tolerances. For high-precision shaping, wire EDM is highlighted in the provided sources as enabling tight tolerances, with one tolerance reference stating that wire EDM can achieve tolerances in the range of ±0.005 mm to ±0.02 mm. Surface grinding is also described as a common finishing step for sintered magnets to achieve precise dimensions, with tolerances typically in the range of ±0.02 mm to ±0.05 mm.

Kerf and gap mechanics in wire EDM are quantified in the provided sources, including a reported wire-workpiece gap usually ranging from 0.025 to 0.075 mm. Kerf stability is described in one experimental study as having standard deviation in the range of 1.1–3.1 μm across a set of 27 cuts, and the same source provides an empirical mean kerf relationship $K_m(\mu m) = 0.203 \cdot V + 104$ (as reported).

Waste and scrap mechanisms. Material loss is described as being greatest during machining of the sintered block into end-use geometries; a U.S. public report states that machining has a recovery rate of 60–90% depending on size and complexity of the final magnet, implying 10–40% waste during machining for many geometries. That same report notes that although considerable material is lost, the resulting waste (magnet swarf) is often recycled and returns to the process flow stream.

SmCo introduces an additional scrap driver due to brittleness. One source attributes "magnet production loss of up to 20–30% in volume" to brittleness effects, and a separate processing reference describes Sm-Co magnets as the

most brittle with machining failure rate up to 20%. In a wire EDM example, 1 of 7 reference magnets (about 14.3%) had fracture problems while preparing rectangular bars, illustrating the brittleness-driven yield risk in finishing operations.

Waste stream forms and composition. The NdFeB magnet supply chain report provides a global-scale estimate that the volume of the swarf recycling stream in 2020 was about 9% of NdFeB magnets produced globally. That report further notes that swarf streams "can contain some machine cutting fluids and magnetic powder," and explicitly links this to the fact that cutting is performed before magnetization.

Waste generation summary. The table below consolidates the waste-related quantitative points that are explicitly stated in the provided sources.

Waste and scrap metric	Reported value	Context and notes
Machining recovery rate	60–90% recovery (10–40% waste)	Described as the greatest material loss step when machining sintered blocks to end-use specifications, varying with size and complexity.
SmCo production loss linked to brittleness	Up to 20–30% volume loss	Attributed to brittleness in SmCo-based magnets in one report.
Sm-Co machining failure rate	Up to 20%	Described as a machining failure rate for brittle Sm-Co magnets in a manufacturing process reference.
WEDM fracture example	~14.3% (1 of 7) fracture problems	Example observed during wire EDM preparation of SmCo bars for bending tests.
NdFeB swarf stream (global, 2020)	~9% of production	Framed as the swarf recycling stream volume estimate in 2020 for NdFeB magnets globally.
Swarf composition indicator	Cutting fluids + magnetic powder possible	Explained as a consequence of cutting being performed before magnetization.

Supply Chain Structure and Logistics

The supply chain for rare earths and other critical materials relevant to permanent magnets is described by GAO as spread across the globe, with China dominating the mining and processing stages, and as a complex multi-tier process from raw material in the ground to a DoD weapon system. In response, DoD is reported to be actively working to establish a "Mine-to-Magnet" supply chain for rare earth materials, and GAO reports approximately $439 million in DoD awards since 2020 to vendors to reestablish domestic rare earth supply chains.

Supply chain flow. A defense-relevant "raw material to installed component" flow, using phrasing and step ordering described in the sources, can be expressed as:

Mining and extraction ⟶ separation to high-purity oxides/metals ⟶ alloy preparation ⟶ powder production (including routes such as hydrogen decrepitation) ⟶ particle alignment and pressing ⟶ sintering and heat treatment ⟶ machining ⟶ magnetization ⟶ lot qualification and inspection ⟶ delivery/installation into defense subsystems.

Within the manufacturing portion of this chain, specific operations are documented with quantitative anchors such as isostatic pressing around 240–250 MPa after magnetic alignment, and sintering regimes such as 1140–1170 ℃ (SmCo, inert atmosphere) or about 1200 ℃ (reported Sm2(Co,Fe,Cu,Zr)17 example), followed by multi-step heat treatments (e.g., 850 ℃ anneal and controlled cool to 400 ℃).

Defense traceability and sourcing pressures. GAO reports that DoD completed DFARS rulemaking for an expanded prohibition that will go into effect on January 1, 2027, and that DoD will have to require contractors to disclose the source of certain rare earth permanent magnets used in DoD systems. In addition, the Federal Register text cited in the provided sources notes that DFARS authorizes DoD to issue individual and class nonavailability determinations if compliant materials are unavailable (DFARS 225.7018-4). The same Federal Register excerpt explicitly references "recycled NdFeB magnets milled and sintered in the United States," indicating that U.S.-processed recycled feedstock is contemplated within DFARS discussions of compliant sourcing pathways.

Logistics implications of machining waste. The sources indicate that machining creates substantial swarf streams that are often recycled, and that these streams can contain cutting fluids and magnetic powder, implying that collection, storage, and handling logistics must manage both metallic/magnetic solids and process fluids. Because machining is described as the "greatest material loss" step and because recovery rate varies (60–90%), the magnitude of swarf handling and potential recycling flows is sensitive to end-use geometry complexity and tolerance requirements.

Competitive Landscape and Certifications

The competitive landscape in the provided sources highlights a mix of specialized magnet manufacturers and vertically integrated magnetic-materials companies, with supplier differentiation frequently expressed in terms of in-house alloy/powder capability, machining/assembly capability, and ability to meet defense compliance expectations such as ITAR registration and AS9100 quality management systems. In parallel, broader policy and market context emphasizes that the United States imports large percentages of some extracted and processed critical mineral resources and related products, framing the business case for domestic capacity expansions and "mine-to-magnet" initiatives.

The table below summarizes defense-relevant attributes that are explicitly stated in the provided sources.

Company	What they do (as stated in sources)	Defense-relevant compliance and quality indicators	Notes on integration and capability
Electron Energy Corporation (EEC)	Specializes in RECo5 and RE2TM17 type sintered magnets. Purchases pure raw materials (Sm, Co, Fe, Cu, Zr) for in-house production of SmCo 1:5 and 2:17 alloys.	ITAR compliant and DDTC registered. Certified to AS9100D and ISO 9001:2015.	Joined Magnetic Holdings, LLC (parent of Dexter and MCE). A press release describes this as creating a group of leading permanent magnet firms with a "vertically integrated manufacturing process" to create a robust and resilient supply chain.

Company	What they do (as stated in sources)	Defense-relevant compliance and quality indicators	Notes on integration and capability
Arnold Magnetic Technologies	Described as a U.S.-based industrial magnet manufacturer specializing in high-performance permanent magnets and assemblies for demanding markets such as aerospace and defense. Based in Rochester, NY with an operating history of over 130 years and serving aerospace and defense among other end-markets.	States that all locations in its RECOMA supply chain are registered with ITAR and can adhere to DFAR clauses. Lists ISO 9001:2015 and AS 9100 Rev D.	States it has established domestic supply chains for rare earth materials to ensure compliance with DoD requirements. A defense-facing example also cites its RECOMA 35E SmCo grade for 150–250 ℃ and high-speed applications (100,000 rpm+).
VACUUMSCHMELZE (VAC)	Described as one of the leading global manufacturers of advanced magnetic materials and related products. Provides VACOMAX SmCo material made from rare-earths and cobalt.	(No explicit ITAR/AS9100 statements in the provided quotes.)	States it has state-of-the-art facilities for further processing methods including wire-EDM. Announced an MoU with Ucore and e-VAC Magnetics to evaluate a collaborative supply agreement for high-purity rare earth oxides.
MP Materials	Described (in the provided source) as presenting a "mine-to-magnet story," with NdPr metal online, trial sintered magnets running, and guidance to ~1,000 t/yr of finished magnets with a ramp beginning late 2025.	(No defense certifications in the provided quote.)	Positioned in the provided source as a domestic capacity expansion relevant to supply chain resilience for NdFeB magnets.

Qualification and Testing Requirements

The sources describe a qualification and acceptance ecosystem that combines magnetic property verification, dimensional and flux checks, materials and defect inspection, and environmental/corrosion testing tied to coatings and corrosion susceptibility. In defense procurement contexts, these technical requirements interact with compliance frameworks such as ITAR/DDTC registration and DFARS sourcing requirements and disclosures described by GAO and supplier statements.

Magnetic property qualification. A manufacturing process reference states that full second-quadrant demagneti-

zation curves are required from a representative sample from each batch of magnets, and that a batch is typically defined as the load from the final heat treatment furnace. The same reference notes that it may also be necessary to measure side-to-side flux using a Hall probe and Gauss meter.

Magnetic measurement instrumentation and field levels in the provided sources include vibrating sample magnetometer (VSM) measurements with a maximum field of 9 T in one study, and demagnetization curve measurements using a permagraph with maximum applied field reported as 1500 kA/m after magnetization by a 6 T field in another report. A further source describes evaluation using a pulse-excitation-type BH tracer or a superconducting magnet-based VSM with magnetic field range of approximately 4775 $kA \cdot m^{-1}$ (as stated).

For elevated-temperature magnetic measurements, an Arnold measurement white paper notes that some hysteresisgraphs have heated pole shoes or a heated sample holder enabling measurements up to +200 ℃, and that a pulse field magnetometer (PFM) can measure magnetic properties in a temperature range of −40 to 220 ℃ (depending on the manufacturer).

Mechanical and defect inspection. Mechanical testing examples in the provided sources include flexural strength determination via three-point bending on an Instron 5944 universal test machine in one SmCo-focused study. Materials quality checks in a separate study include oxygen content measurement using an Eltra ON 900 oxygen/nitrogen analyzer and density measurement by Archimedes' principle.

Non-destructive inspection is also described. One source states that applying ultrasonic testing (UT) allows users to analyze internal defects such as inclusions, porosity, and microcracks, and can identify locations of internal anomalies within Sm-Co sintered magnets. The same source argues UT can improve manufacturing efficiency and machinability, reduce scrap, and increase the use of what would be post-production waste while enhancing reuse and recycling at end-of-life disposition.

Corrosion, coatings, and environmental testing. The sources indicate corrosion susceptibility varies by magnet system, with one source stating "these magnets have a poor corrosion resistance and must be protected," and reporting a salt spray test example using 5% NaCl solution continuously sprayed at 35 ± 2 ℃ for 72 hours on magnetized and demagnetized samples. Coating standards and methods are also cited: MIL-C-26074E requires that coated test specimens be exposed to neutral salt-spray testing in accordance with ASTM B117 for 100 hours for electroless nickel coatings (as stated in the cited document). Parylene is cited as a military specification conformal coating under MIL-I-46058.

Corrosion-performance impacts from diffusion processing are also referenced in the dataset, with one report stating that Tafel tests reveal Dy-diffused magnets have higher corrosion resistance with more positive equilibrium potential and lower corrosion current density (as described).

Compliance and sourcing requirements. Supplier certifications and registrations are explicitly cited in the dataset for U.S. defense-facing suppliers, including Arnold's statements of ISO 9001:2015 and AS 9100 Rev D, and that all locations in its Samarium Cobalt RECOMA supply chain are registered with ITAR and can adhere to DFAR clauses. EEC likewise states ITAR compliance and DDTC registration and lists certification to AS9100D and ISO 9001:2015.

At the policy level, GAO reports DFARS rulemaking completion for an expanded prohibition effective January 1, 2027, and that DoD will have to require contractor disclosure of the source of certain rare earth permanent magnets used in DoD systems. The Federal Register excerpt cited in the dataset also notes DFARS authority for nonavailability determinations when compliant materials are unavailable (DFARS 225.7018-4).

Gaps and Recommendations

The provided sources offer multiple explicit entry points for process-improvement and waste-management interventions, particularly at the machining stage where the largest reported material losses occur and where swarf streams can contain both cutting fluids and magnetic powders. They also explicitly describe inspection and testing practices (e.g., batch-defined demagnetization curves and UT for internal defects) that can be used to reduce scrap and support qualification confidence, which is directly relevant to both cost and sustainability in defense-grade supply chains.

A wastewater-focused provider targeting defense-grade magnet manufacturing can align offerings to the waste stream and compliance characteristics described in the sources. The following recommendations are grounded in the waste and testing descriptions explicitly cited.

- Prioritize treatment and separation solutions for swarf streams that may include cutting fluids and fine magnetic powders, because the NdFeB supply chain report explicitly states swarf "can contain" both cutting fluids and magnetic powder, and that cutting occurs before magnetization.
- Quantify and design for high variability in swarf volume driven by machining yield, because a public report states machining recovery ranges from 60–90% depending on magnet size and complexity (10–40% waste), and because SmCo brittleness can drive additional losses and failures (e.g., up to 20–30% volume loss attributed to brittleness in one source and up to 20% machining failure rates in another).
- Integrate inspection-enabled quality control offerings where appropriate, because UT is described as a method to locate internal anomalies such as inclusions, porosity, and microcracks in Sm-Co sintered magnets and as enabling reduced scrap and increased use of post-production waste and recycling opportunities.

DFARS-related disclosure and sourcing requirements described by GAO indicate that procurement teams should treat traceability and source documentation as schedule-critical deliverables, not merely contractual boilerplate. Supplier-facing recommendations supported by the provided sources include:

- Plan qualification around batch-based sampling defined by final heat treatment furnace loads and include full second-quadrant demagnetization curves and flux checks where needed, because these are explicitly described as batch-level requirements and practices in a manufacturing process reference.
- Where corrosion risk is material (e.g., in magnets described as having poor corrosion resistance), align coating selection and validation with salt-spray practices and military coating specifications cited in the sources (e.g., ASTM B117 salt-spray tests and MIL-C-26074E for electroless nickel; MIL-I-46058 for parylene).

Policy actions in the sources reinforce that defense magnet supply chains are a national priority, including DoD's stated effort to establish a "Mine-to-Magnet" supply chain and the reported $439 million in awards since 2020 to reestablish domestic rare earth supply chains. Congressional proposals cited in the dataset further include bills directing DOE consortium research on REE extraction and establishing DoD grant programs authorizing $50 million per year for four years to encourage domestic development of critical minerals, reflecting sustained legislative interest in reducing vulnerability associated with globalized supply chains. Additional proposals would create federally chartered entities to mitigate high costs and radioactive byproducts potentially associated with REE extraction from minerals found with radioactive elements.

Chapter 2

This chapter reproduces the full content of the waste-stream characterization report focused on machining swarf and sludge from NdFeB and SmCo magnet finishing operations, including waste fractions, swarf chemistry, particle sizes, fluid contamination, recycling routes, and wastewater implications.

1. Overview of Machining Waste Generation

The manufacturing of high-performance rare earth permanent magnets (REPMs) is a subtractive process that generates a substantial amount of waste, particularly during the final machining and finishing stages. Sintered magnet blocks cannot be produced to precise final shapes, necessitating extensive grinding, cutting, and shaping to meet the tight dimensional tolerances required for defense and high-technology applications. This machining waste, known as swarf or sludge, represents a significant material and economic loss, but also a valuable secondary resource for critical rare earth elements (REEs).

The fraction of raw material converted to waste varies widely depending on the final magnet geometry, with estimates ranging from 6% to as high as 73%. More commonly, manufacturing scrap, including swarf and cutting sludge, is reported to account for 20% to 30% of the starting alloy mass. This manufacturing waste is dominated by fine-particle sludge; in some analyses, sludge constitutes over 70% of the total waste generated. The global volume of this waste stream is significant, with the swarf recycled in 2020 estimated to be equivalent to 9% of the total NdFeB magnets produced globally that year. While defense-specific factors like tighter tolerances are understood to increase waste generation, quantitative data directly linking tolerance levels to a specific increase in waste fraction is not detailed in the available literature.

2. Swarf Chemical Composition

The chemical and phase composition of machining swarf is critical to its recycling potential. The swarf largely mirrors the composition of the parent magnet alloy but is heavily oxidized and contaminated with machining media.

NdFeB Swarf. Neodymium-Iron-Boron (NdFeB) swarf is a complex mixture containing valuable REEs and a high concentration of iron.

- **Elemental Composition**: NdFeB magnets contain approximately 30-32 wt.% REEs. Analysis of swarf and scrap material shows a composition of roughly 63-72 wt.% iron, 20-27 wt.% neodymium, 3-6 wt.% praseodymium, and 0.5-7 wt.% dysprosium, along with about 1 wt.% boron. Minor additions of cobalt (up to 5 wt.%), copper, aluminum, and niobium may also be present.
- **Phase Composition**: The primary magnetic phase in the swarf is Nd2Fe14B, which can account for 96-98% of the original magnet's structure. Swarf is also characterized by the presence of REE-rich grain boundary phases.
- **Oxide Content**: Swarf is heavily oxidized due to the high surface area of fine particles and exposure to air and cutting fluids. Oxygen content in recycled material can be as high as 2000–50,000 ppm, compared to 300-400 ppm in primary alloys. Oxidation leads to the decomposition of the primary Nd2Fe14B phase into various oxides, including Nd2O3, Fe2O3, and complex oxides like NdFeO3.

SmCo Swarf. Samarium-Cobalt (SmCo) swarf is characterized by high concentrations of samarium and cobalt, with a simpler REE profile than NdFeB.

- **Elemental Composition**: The composition varies based on the alloy stoichiometry (SmCo5 or Sm2Co17). SmCo5 alloys contain approximately 36 wt.% REEs (primarily Sm) and 63 wt.% Co. Sm2Co17 alloys contain around 25 wt.% Sm, 60-70 wt.% Co, and significant additions of iron (10-15 wt.%), copper (5-10 wt.%), and zirconium. One detailed analysis of Sm2Co17 scrap powder found 42.3 wt.% Co, 22.8 wt.% Sm, and 22.4 wt.% Fe.
- **Phase Composition**: The dominant phases in SmCo swarf are intermetallic compounds such as Sm2Co17 and FeCo.

3. Particle Size Distributions

Machining processes like grinding and cutting generate swarf with a wide range of particle sizes, which significantly impacts handling, oxidation, and recycling efficiency.

- **Milled Powders**: Studies involving planetary ball milling of NdFeB scrap show a progressive reduction in particle size with time. The median particle diameter (d50) can be reduced from over 20 µm to below 5 µm after 15 hours of grinding. The d90 value (representing the coarsest 10% of particles) can decrease from over 65 µm to around 10 µm in the same timeframe.
- **Hydrogen Processed Powders**: Hydrogen decrepitation (HD) produces a friable powder with particle sizes typically ranging from 6 µm to 600 µm. The more advanced Hydrogenation-Disproportionation-Desorption-Recombination (HDDR) process results in much finer, nanograined powders with grain sizes around 300 nm.
- **Sieved Fractions**: In recycling workflows, specific mesh sizes are used for separation. Powders are often sieved to <90 µm or <45 µm for processing. For example, sieving hydrogen-processed powder at 90 µm can effectively separate the brittle magnet material from ductile nickel coatings.

4. Cutting Fluid Contamination

Machining sludge is inherently contaminated with the fluids used as coolants and lubricants during grinding and cutting operations. This contamination complicates recycling by introducing organic and inorganic impurities.

- **Fluid Composition**: A common type of coolant is a water-miscible, mineral oil-based fluid, such as Mobilcut 230. This type of fluid can contain mineral oil, boric acid (10-20 wt.%), ethanol (1-5 wt.%), and various oleates and oils. Ethanol is also used as a grinding medium to reduce friction and prevent surface oxidation.
- **Contamination Levels**: The amount of residual coolant in grinding slurries can be substantial, varying from less than 1% to over 20% by weight. This contamination introduces unwanted elements, primarily carbon and additional oxygen, which can interfere with metallurgical recovery processes.
- **Removal**: Pre-treatment steps are often required to remove these organic impurities. This can include washing with solvents like acetone or thermal pre-conditioning via pyrolysis to volatilize the organic compounds before melting. Pyrolysis at 600℃ for 15 minutes has been shown to reduce volatiles to below detectable limits.

5. Rare Earth Recovery Processes and Yields

A variety of metallurgical techniques have been developed to recover REEs and other valuable metals like cobalt from magnet swarf. These can be broadly categorized into hydrometallurgical, pyrometallurgical, and hydrogen-based routes.

Hydrometallurgical Routes. Hydrometallurgy, which uses aqueous solutions to leach and separate metals, is a flexible and widely studied approach for REE recovery, capable of producing high-purity products.

- **Leaching**: The first step involves dissolving the swarf in an acid. Strong mineral acids like sulfuric acid (H2SO4), hydrochloric acid (HCl), and nitric acid (HNO3) are commonly used. This process can achieve high leaching efficiencies, often exceeding 95-99% for REEs. However, it often co-dissolves large amounts of iron, complicating downstream separation. Alternative "greener" reagents like organic acids (citric acid) and acid-free copper salt solutions are also being explored.
- **Separation and Purification**: After leaching, the REEs are separated from iron and other impurities. This is typically done through selective precipitation (using agents like oxalic acid or sodium sulfate) or solvent extraction. Solvent extraction uses organic chemicals like D2EHPA mixed with kerosene to selectively pull REEs from the aqueous solution.

- **Recovery Yields**: Laboratory-scale hydrometallurgical processes report high recovery rates. For example, various studies have demonstrated REE recovery efficiencies of 97-99%. Element-specific recoveries can be very high, with some processes achieving >98% for Nd, >98% for Pr, and >97% for Dy.

Pyrometallurgical Routes. Pyrometallurgy uses high temperatures to smelt, refine, and separate metals. These methods are often faster and can handle larger capacities than hydrometallurgical routes.

- **Methods**: Common techniques include liquid metal extraction (using molten magnesium or copper to selectively dissolve REEs), molten salt extraction (using chloride or fluoride salts), and oxidative slagging (where REEs are oxidized and collected in a slag phase). Temperatures for these processes range from 700℃ to over 1500℃.
- **Recovery Yields**: Pyrometallurgical methods can also achieve high recovery. Liquid magnesium extraction has shown nearly 100% efficiency for Nd, though lower for Dy (~60%). Molten MgCl2 extraction has recovered ~80% of Nd and Dy. Oxidative smelting followed by leaching has demonstrated >99% overall REE recovery in some studies.

Hydrogen Processing. Hydrogen-based processes use hydrogen gas to break down the brittle magnet material into a powder, facilitating separation and direct recycling.

- **Methods**: The two main techniques are Hydrogen Decrepitation (HD) and Hydrogenation-Disproportionation-Desorption-Recombination (HDDR). HD uses hydrogen at or near room temperature to create a friable, demagnetized powder. HDDR is a more complex process involving higher temperatures (750-950℃) that produces a fine, anisotropic powder suitable for high-performance bonded or re-sintered magnets.
- **Recovery Yields**: These "short-loop" recycling methods are highly efficient in terms of material recovery, with processes like HPMS (Hydrogen Processing of Magnet Scrap) reporting recovery rates of 90-98% for the magnetic alloy powder.

6. SmCo Waste Streams

While much of the recycling research focuses on NdFeB magnets, SmCo magnets are critical for high-temperature defense applications, and their waste streams are also a valuable source of cobalt and samarium.

- **Composition**: SmCo swarf composition depends on the alloy type. SmCo5 alloys contain roughly 36 wt.% REEs (mostly Sm) and 63 wt.% Co. The more common high-temperature Sm2Co17 grades contain ~25 wt.% Sm, 60-70 wt.% Co, and significant amounts of Fe (10-15 wt.%), Cu (5-10 wt.%), and Zr. One detailed analysis of Sm2Co17 scrap powder reported 42.3 wt.% Co, 22.8 wt.% Sm, and 22.4 wt.% Fe.
- **Recovery Processes**: Recycling of SmCo is considered simpler than NdFeB due to the less complex mixture of REEs. Hydrometallurgical routes are common, using acids like HNO3 or H2SO4 to leach the metals. Selective leaching can be effective; for example, oxidative leaching with 3M HNO3 at 60℃ achieved 99% dissolution of Sm and Co while leaving iron as a solid residue.
- **Recovery Yields**: Studies have demonstrated high recovery efficiencies for both samarium and cobalt. One process reported >98% recovery of Sm with >99% purity. Another achieved precipitation efficiencies of 98.7% for Sm and 96.2% for Co. Microwave-assisted acid digestion has also shown quantitative recovery (>95%) of all elements from SmCo waste.

7. Commercial Recycling Landscape

While much of the research on REE magnet recycling remains at the laboratory or pilot scale, a growing number of companies are commercializing these technologies.

- **Urban Mining Company (Noveon Magnetics)**: Based in Texas, this company uses a hydrogen-based process (Magnet-to-Magnet) to recycle scrap magnets directly into new, high-performance sintered NdFeB magnets.
- **Cyclic Materials**: A Canadian company that has developed a hydrometallurgical process to recover REEs, as well as copper, steel, and aluminum, from magnetic waste streams.
- **HyProMag**: A UK-based company (with operations in Germany and the US) commercializing the HPMS process developed at the University of Birmingham. They are focused on producing recycled NdFeB alloy powders for new magnet production, with a pilot plant operational and larger facilities planned.
- **Other Players**: Other notable entities include **Momentum Technologies** (licensing a membrane-assisted solvent extraction process from Oak Ridge National Laboratory), **TdVib LLC** (licensing an acid-free dissolution process from Ames Laboratory), and established chemical companies like **Solvay** that are involved in REE recovery from various waste streams. In China, companies like **Ganzhou YouLi Technology** and **Ji'an XinTai Technology** use conventional hydrometallurgical routes for REO recovery.

8. Implications for Wastewater Treatment

The hydrometallurgical recycling of magnet swarf generates significant wastewater streams that require specialized treatment. These streams are typically acidic and contain high concentrations of dissolved metals and process chemicals.

- **Wastewater Composition**: The primary liquid waste is the "pregnant leach solution" (PLS) after the initial acid leaching step. Analysis of a PLS from NdFeB waste shows high concentrations of dissolved metals, including iron (up to 62.9 g/L), neodymium (20.5 g/L), and dysprosium (6.4 g/L), along with other elements like cobalt, boron, and praseodymium in the g/L range.
- **pH and Chemical Load**: The wastewater is highly acidic, with a final pH often between 1.0 and 3.0 after leaching. The process consumes large volumes of strong acids (H2SO4, HCl, HNO3), bases (caustics), and precipitating agents like oxalic acid. The final effluent can contain residual salts such as ammonium chloride and ammonium nitrate from pH adjustment steps.
- **Treatment Challenges and Opportunities**: A key treatment step is the removal of iron, which is the most abundant metal. This is often done by raising the pH to precipitate iron as a hydroxide or jarosite. This process itself requires careful pH control and the addition of chemicals like calcium hydroxide. The resulting wastewater still contains dissolved REEs and other metals that must be recovered or removed. The high chemical load and complex metal mixture present a challenge for conventional wastewater treatment but also an opportunity for technologies that can selectively recover valuable elements from these dilute, complex streams. The economic viability of hydrometallurgical recycling is heavily dependent on the cost of wastewater treatment and the ability to recycle process chemicals like solvents and acids.

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Machining Waste Stream Characterization for Defense-Grade Rare Earth Permanent Magnets

1. Overview of Machining Waste Generation

The manufacturing of high-performance rare earth permanent magnets (REPMs) is a subtractive process that generates a substantial amount of waste, particularly during the final machining and finishing stages. Sintered magnet blocks cannot be produced to precise final shapes, necessitating extensive grinding, cutting, and shaping to meet the tight dimensional tolerances required for defense and high-technology applications. This machining waste, known

as swarf or sludge, represents a significant material and economic loss, but also a valuable secondary resource for critical rare earth elements (REEs).

The fraction of raw material converted to waste varies widely depending on the final magnet geometry, with estimates ranging from 6% to as high as 73%. More commonly, manufacturing scrap, including swarf and cutting sludge, is reported to account for 20% to 30% of the starting alloy mass. This manufacturing waste is dominated by fine-particle sludge; in some analyses, sludge constitutes over 70% of the total waste generated. The global volume of this waste stream is significant, with the swarf recycled in 2020 estimated to be equivalent to 9% of the total NdFeB magnets produced globally that year. While defense-specific factors like tighter tolerances are understood to increase waste generation, quantitative data directly linking tolerance levels to a specific increase in waste fraction is not detailed in the available literature.

2. Swarf Chemical Composition

The chemical and phase composition of machining swarf is critical to its recycling potential. The swarf largely mirrors the composition of the parent magnet alloy but is heavily oxidized and contaminated with machining media.

NdFeB Swarf

Neodymium-Iron-Boron (NdFeB) swarf is a complex mixture containing valuable REEs and a high concentration of iron.

- **Elemental Composition**: NdFeB magnets contain approximately 30-32 wt.% REEs. Analysis of swarf and scrap material shows a composition of roughly 63-72 wt.% iron, 20-27 wt.% neodymium, 3-6 wt.% praseodymium, and 0.5-7 wt.% dysprosium, along with about 1 wt.% boron. Minor additions of cobalt (up to 5 wt.%), copper, aluminum, and niobium may also be present.
- **Phase Composition**: The primary magnetic phase in the swarf is Nd2Fe14B, which can account for 96-98% of the original magnet's structure. Swarf is also characterized by the presence of REE-rich grain boundary phases.
- **Oxide Content**: Swarf is heavily oxidized due to the high surface area of fine particles and exposure to air and cutting fluids. Oxygen content in recycled material can be as high as 2000–50,000 ppm, compared to 300-400 ppm in primary alloys. Oxidation leads to the decomposition of the primary Nd2Fe14B phase into various oxides, including Nd2O3, Fe2O3, and complex oxides like NdFeO3.

SmCo Swarf

Samarium-Cobalt (SmCo) swarf is characterized by high concentrations of samarium and cobalt, with a simpler REE profile than NdFeB.

- **Elemental Composition**: The composition varies based on the alloy stoichiometry (SmCo5 or Sm2Co17). SmCo5 alloys contain approximately 36 wt.% REEs (primarily Sm) and 63 wt.% Co. The more common high-temperature Sm2Co17 grades contain ~25 wt.% Sm, 60-70 wt.% Co, and significant additions of iron (10-15 wt.%), copper (5-10 wt.%), and zirconium. One detailed analysis of Sm2Co17 scrap powder reported 42.3 wt.% Co, 22.8 wt.% Sm, and 22.4 wt.% Fe.
- **Phase Composition**: The dominant phases in SmCo swarf are intermetallic compounds such as Sm2Co17 and FeCo.

3. Particle Size Distributions

Machining processes like grinding and cutting generate swarf with a wide range of particle sizes, which significantly impacts handling, oxidation, and recycling efficiency.

- **Milled Powders**: Studies involving planetary ball milling of NdFeB scrap show a progressive reduction in particle size with time. The median particle diameter (d50) can be reduced from over 20 µm to below 5 µm after 15 hours of grinding. The d90 value (representing the coarsest 10% of particles) can decrease from over 65 µm to around 10 µm in the same timeframe.
- **Hydrogen Processed Powders**: Hydrogen decrepitation (HD) produces a friable powder with particle sizes typically ranging from 6 µm to 600 µm. The more advanced Hydrogenation-Disproportionation-Desorption-Recombination (HDDR) process results in much finer, nanograined powders with grain sizes around 300 nm.
- **Sieved Fractions**: In recycling workflows, specific mesh sizes are used for separation. Powders are often sieved to <90 µm or <45 µm for processing. For example, sieving hydrogen-processed powder at 90 µm can effectively separate the brittle magnet material from ductile nickel coatings.

4. Cutting Fluid Contamination

Machining sludge is inherently contaminated with the fluids used as coolants and lubricants during grinding and cutting operations. This contamination complicates recycling by introducing organic and inorganic impurities.

- **Fluid Composition**: A common type of coolant is a water-miscible, mineral oil-based fluid, such as Mobilcut 230. This type of fluid can contain mineral oil, boric acid (10-20 wt.%), ethanol (1-5 wt.%), and various oleates and oils. Ethanol is also used as a grinding medium to reduce friction and prevent surface oxidation.
- **Contamination Levels**: The amount of residual coolant in grinding slurries can be substantial, varying from less than 1% to over 20% by weight. This contamination introduces unwanted elements, primarily carbon and additional oxygen, which can interfere with metallurgical recovery processes.
- **Removal**: Pre-treatment steps are often required to remove these organic impurities. This can include washing with solvents like acetone or thermal pre-conditioning via pyrolysis to volatilize the organic compounds before melting. Pyrolysis at 600℃ for 15 minutes has been shown to reduce volatiles to below detectable limits.

5. Rare Earth Recovery Processes and Yields

A variety of metallurgical techniques have been developed to recover REEs and other valuable metals like cobalt from magnet swarf. These can be broadly categorized into hydrometallurgical, pyrometallurgical, and hydrogen-based routes.

Hydrometallurgical Routes

Hydrometallurgy, which uses aqueous solutions to leach and separate metals, is a flexible and widely studied approach for REE recovery, capable of producing high-purity products.

- **Leaching**: The first step involves dissolving the swarf in an acid. Strong mineral acids like sulfuric acid (H2SO4), hydrochloric acid (HCl), and nitric acid (HNO3) are commonly used. This process can achieve high leaching efficiencies, often exceeding 95-99% for REEs. However, it often co-dissolves large amounts of iron, complicating downstream separation. Alternative "greener" reagents like organic acids (citric acid) and acid-free copper salt solutions are also being explored.
- **Separation and Purification**: After leaching, the REEs are separated from iron and other impurities. This is typically done through selective precipitation (using agents like oxalic acid or sodium sulfate) or solvent

extraction. Solvent extraction uses organic chemicals like D2EHPA mixed with kerosene to selectively pull REEs from the aqueous solution.

- **Recovery Yields**: Laboratory-scale hydrometallurgical processes report high recovery rates. For example, various studies have demonstrated REE recovery efficiencies of 97-99%. Element-specific recoveries can be very high, with some processes achieving >98% for Nd, >98% for Pr, and >97% for Dy.

Pyrometallurgical Routes

Pyrometallurgy uses high temperatures to smelt, refine, and separate metals. These methods are often faster and can handle larger capacities than hydrometallurgical routes.

- **Methods**: Common techniques include liquid metal extraction (using molten magnesium or copper to selectively dissolve REEs), molten salt extraction (using chloride or fluoride salts), and oxidative slagging (where REEs are oxidized and collected in a slag phase). Temperatures for these processes range from 700℃ to over 1500℃.
- **Recovery Yields**: Pyrometallurgical methods can also achieve high recovery. Liquid magnesium extraction has shown nearly 100% efficiency for Nd, though lower for Dy (~60%). Molten MgCl2 extraction has recovered ~80% of Nd and Dy. Oxidative smelting followed by leaching has demonstrated >99% overall REE recovery in some studies.

Hydrogen Processing

Hydrogen-based processes use hydrogen gas to break down the brittle magnet material into a powder, facilitating separation and direct recycling.

- **Methods**: The two main techniques are Hydrogen Decrepitation (HD) and Hydrogenation-Disproportionation-Desorption-Recombination (HDDR). HD uses hydrogen at or near room temperature to create a friable, demagnetized powder. HDDR is a more complex process involving higher temperatures (750-950℃) that produces a fine, anisotropic powder suitable for high-performance bonded or re-sintered magnets.
- **Recovery Yields**: These "short-loop" recycling methods are highly efficient in terms of material recovery, with processes like HPMS (Hydrogen Processing of Magnet Scrap) reporting recovery rates of 90-98% for the magnetic alloy powder.

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Chapter 3

This chapter reproduces the full content of the DFARS compliance pathway report, including definitions of covered materials and countries, the phased implementation through 2026 and the expanded prohibition effective January

1, 2027, nonavailability determination procedures, recycled NdFeB exceptions, domestic capacity initiatives, and GAO-identified compliance gaps.

Regulatory Overview

The U.S. Department of Defense (DoD) regulates the procurement of rare earth permanent magnets through a framework designed to strengthen the domestic industrial base and reduce reliance on foreign supply chains. This regulatory structure is primarily implemented through the Defense Federal Acquisition Regulation Supplement (DFARS), with its authority stemming from legislative mandates in various National Defense Authorization Acts (NDAA). Key regulations include DFARS 225.7018, "Restriction on acquisition of certain magnets, tantalum, and tungsten," and the corresponding contract clause, DFARS 252.225-7052.

These regulations prohibit the DoD from acquiring "covered materials" that originate from specific "covered countries". The list of covered countries includes the People's Republic of China, the Russian Federation, the Democratic People's Republic of North Korea, and the Islamic Republic of Iran.

The scope of "covered materials" is explicitly defined to include:

- Samarium-cobalt (SmCo) magnets
- Neodymium-iron-boron (NdFeB) magnets
- Tantalum metals and alloys
- Tungsten metal powder
- Tungsten heavy alloy, including any finished or semi-finished components containing it

This framework is a direct response to national security concerns regarding supply chain vulnerabilities, particularly the concentration of rare earth mining and processing in China.

Current Requirements and the 2027 Expanded Prohibition

The DFARS restrictions are being implemented in a phased approach, with a significant expansion of scope scheduled for January 1, 2027.

Current Rules (Effective through December 31, 2026): Through the end of 2026, the prohibition focuses on the final stages of metal and magnet production. Contractors are barred from delivering covered materials that were "melted or produced" in a covered country. For magnets, this specifically includes the melting of constituent elements to form the alloy (e.g., samarium with cobalt) and all subsequent manufacturing steps such as powder formation, pressing, sintering, and magnetization.

Expanded Prohibition (Effective January 1, 2027): A major shift occurs on January 1, 2027, when the prohibition expands to cover the entire upstream supply chain. From this date, contractors may not deliver any covered material that was "mined, refined, separated, melted, or produced" in a covered country. This change means that even if all magnet manufacturing steps occur in a compliant country, the use of rare earth oxides or metals that were mined or refined in a covered country will render the final product non-compliant. For SmCo and NdFeB magnets, this restriction explicitly includes the entire supply chain from the mining of ore or feedstock through the production of the finished magnet.

New legislative mandates also require enhanced transparency. The FY2023 NDAA requires contractors to disclose the provenance of permanent magnets containing rare earths, and if unable to do so, they must establish a supply chain tracking system. DOD is developing a corresponding DFARS rule, with a proposed version expected in late 2024.

Nonavailability Determination Procedures

The DFARS provides a waiver mechanism for instances where compliant materials are not available. This process, detailed in DFARS 225.7018-4, allows for "nonavailability determinations," which can be issued on an individual or class basis.

- **Individual Determinations:** These apply to a single contract and are authorized by the head of the contracting activity. The request must include supporting documentation and a written certification from the requiring activity that explains with specificity why alternatives are unacceptable. Templates and procedural guidance are available in the PGI.
- **Class Determinations:** These apply to more than one contract and require the approval of the Under Secretary of Defense for Acquisition and Sustainment (USD(A&S)). Before issuing a class determination, the USD(A&S) must publish a notice of intent on SAM.gov at least 30 days prior, soliciting information from interested parties on the availability of compliant materials.

In all cases, the nonavailability determination can be made if compliant materials of satisfactory quality and quantity, in the required form, cannot be procured as needed at a reasonable price.

Recycled NdFeB Feedstock as a Compliant Pathway

The DFARS framework provides a specific and significant exception for recycled materials, creating a viable compliance pathway for NdFeB magnets. A neodymium-iron-boron magnet manufactured from recycled material is exempt from the sourcing restrictions, provided that both the **milling of the recycled material and the sintering of the final magnet take place in the United States**.

This statutory exception is a key enabler for the emerging domestic rare earth recycling industry. Companies are leveraging this provision to create DFARS-compliant feedstock for the defense industrial base.

Several companies are actively pursuing this pathway:

- **Noveon Magnetics (formerly Urban Mining Company):** Explicitly markets its NdFeB magnets, produced from recycled feedstock at its Texas facility, as DFARS-compliant. The company's patented Magnet-to-Magnet (M2M) process reuses end-of-life magnets, bypassing the mining and refining steps that are often tied to non-compliant sources.
- **Cyclic Materials:** This company is scaling up its capacity to produce recycled mixed rare earth oxides (rMREO) and has established a partnership with magnet manufacturer Vacuumschmelze (VAC) to recycle 100% of the manufacturing scrap from VAC's new U.S. facility. This creates a closed-loop, domestic source of feedstock.
- **HyProMag USA:** This venture is developing a facility to produce DFARS-compliant recycled NdFeB magnets using a hydrogen-based recycling process, with plans to begin full operations by 2027.
- **ReElement Technologies:** This company processes recycled magnets and provides traceability tokens to support contractor compliance, positioning its refined oxides as a domestic feedstock solution.

The broader Buy American Act framework further supports this, as one of its implementing regulations considers scrap that is "generated, collected, and prepared for processing in the United States" to be domestic material.

Domestic Production Capacity Development

To address the systemic reliance on foreign sources, the DoD has launched a major initiative to reestablish a domestic "mine-to-magnet" supply chain. Since 2020, DoD has invested over $439 million through programs like the Manufacturing Capability Expansion and Investment Prioritization (MCEIP) directorate to fund every critical stage

of the supply chain, from sourcing and separation to metallization, alloying, and magnet manufacturing. The stated goal is to create a sustainable supply chain capable of supporting all U.S. defense requirements by 2027.

Key vendors and investments include:

- **MP Materials**: As the operator of the only active rare earth mine in the U.S. at Mountain Pass, California, MP Materials has received significant DoD support, including a $400 million equity investment and a $150 million loan. This funding supports the expansion of its Texas magnetics facility and the construction of a new 10,000-metric-ton-per-year plant, with DoD committing to a 10-year offtake agreement and a price floor for NdPr products.
- **Lynas USA, LLC**: Awarded a combined $288 million to establish a second domestic, commercial-scale rare earth oxide production facility by 2026.
- **Noveon Magnetics**: Received over $35 million in DoD awards to support the commercialization of its recycled magnet production facility in Texas, which has a stated capacity of 2,000 metric tons per year.
- **E-VAC Magnetics (a subsidiary of VAC)**: Awarded $94.1 million to establish a commercial-scale rare earth magnet manufacturing plant, expected to be completed by 2025.

These investments are intended to create a fully domestic ecosystem, ensuring that every critical node—from mining ore to producing finished magnets—can occur within the United States.

GAO Audit Findings and Compliance Gaps

The Government Accountability Office (GAO) has repeatedly highlighted significant risks in the DoD's rare earth supply chain and identified gaps in its compliance and mitigation efforts. A central finding is the high potential for harm to national security stemming from overreliance on foreign sources, particularly China, for mining and processing.

Key compliance gaps identified by GAO include:

- **Lack of Implementation**: As of 2024, the DoD had not yet established clear policies or guidance to implement the statutory restriction on selling materials from the National Defense Stockpile to adversary nations.
- **Limited Visibility**: The DoD has limited insight into buyers' affiliations with adversary nations, often relying on “buyer self-attestation” for location and end-use information, a significant compliance vulnerability.
- **Fragmented Approach**: Past GAO reports found that the DoD's approach to identifying and mitigating risks was fragmented across different offices and that the department lacked the data to model its own requirements for over 40% of its designated strategic materials.

In response, the GAO has recommended that the DoD establish policies to prevent prohibited sales from the stockpile and enhance its methods for verifying buyer information. The DoD is also in the process of developing a new DFARS rule that will require contractors to disclose the provenance of permanent magnets, a direct response to legislative mandates aimed at improving supply chain transparency.

Supplier Compliance Strategies

In response to the shifting regulatory landscape, key magnet suppliers to the defense sector are adopting various compliance strategies.

- **Securing Compliant International Feedstock**: Some suppliers are establishing supply chains that source raw materials and semi-finished magnet blocks from allied or neutral countries. Arnold Magnetic Technologies, for instance, established a supply chain with a German NdFeB supplier to provide blocks for final machining

and finishing in the United States. Similarly, a consortium including Electron Energy Corporation (EEC) and Dexter Magnetic Technologies announced they are actively working with compliant global suppliers outside of China.

- **Domestic Vertical Integration**: The most robust strategy involves creating a fully domestic supply chain. The partnership between MP Materials and the DoD is the prime example, aiming for a complete "mine-to-magnet" ecosystem within the U.S..
- **Leveraging Recycled Feedstock**: As a direct pathway to DFARS compliance, companies are heavily investing in recycling. Noveon Magnetics bases its entire business model on producing DFARS-compliant magnets from recycled feedstock in its Texas facility. The partnership between Vacuumschmelze (VAC) and Cyclic Materials to recycle all of VAC's manufacturing scrap from its new South Carolina plant is another key example, creating a closed-loop system for a major new domestic producer.
- **Proactive Compliance and Certification**: Companies are proactively working towards the 2027 deadline. Arnold Magnetic Technologies states it is already fully compliant for its SmCo magnet applications. The EEC-led group is targeting mid-2026 for full DFARS 252.225-7052 compliance across its portfolio.

Implications and Recommendations

The transition to a fully DFARS-compliant rare earth magnet supply chain by 2027 presents significant risks and opportunities for the defense industrial base. Defense prime contractors face the risk of supply disruptions and non-compliance if their subcontractors fail to secure compliant magnet sources. A proactive approach to supply chain management is essential, and contractors should already be engaging with their magnet suppliers to verify sourcing and map their transition plans for the 2027 deadline.

The role of recycled magnets is paramount in closing the compliance gap. The explicit DFARS exception for NdFeB magnets milled and sintered in the U.S. from recycled feedstock provides the most direct and near-term pathway for establishing compliant domestic production without reliance on foreign mining operations. This creates a clear market opportunity for companies specializing in rare earth recovery from scrap and end-of-life products.

For a wastewater treatment technology provider, this industrial shift presents several opportunities. The re-shoring of rare earth refining and recycling, particularly through hydrometallurgical processes, will generate new and complex industrial wastewater streams. These streams will contain dissolved rare earth elements, various acids, and other chemical reagents used in the leaching and separation processes. Technology providers who can offer efficient and cost-effective solutions for treating this wastewater—potentially including the recovery of valuable rare earth elements from the waste stream itself—will be well-positioned to serve this growing domestic industry.

Finally, the nonavailability determination process will likely be a critical tool during the transition period. Contractors and suppliers facing unavoidable gaps in compliant sourcing should begin aggregating the necessary data on market availability, technical requirements, and cost to support potential waiver requests through the procedures outlined in DFARS 225.7018-4.

PART II: ECONOMICS, COMPLIANCE TIMELINE, AND BUSINESS CASE

Chapter 4

Recoverable Rare Earth Value from Defense Grade Magnet Machining Swarf

1. Input assumptions

This economic model uses two reference machining-swarf compositions as representative feedstocks for high-performance permanent magnets (NdFeB and SmCo) and treats them as "typical" rather than universal, because actual chemistry varies by grade and program requirements.

The reference compositions applied throughout the calculations are:

- **NdFeB swarf**: ~25 wt% Nd, ~5 wt% Pr, ~3 wt% Dy, ~65 wt% Fe, ~1 wt% B.
- **SmCo (Sm2Co17) swarf**: ~23 wt% Sm, ~50 wt% Co, ~15 wt% Fe, ~7 wt% Cu, ~2 wt% Zr.

Because the value model is intended to be relevant to "high-temperature"/high-coercivity magnet grades, the dysprosium content is treated as a key sensitivity variable. One source notes that NdFeB magnets for high-temperature applications can include **up to 11 wt% Dy** (as an alloying addition) to maintain coercivity at elevated temperature, implying that Dy content can materially exceed the 3 wt% reference value in some high-temperature magnets.

2. Commodity price inputs

The table below summarizes price points used as inputs, emphasizing 2024–2025 where available and retaining some well-cited early-2026 spot assessments where they provide missing coverage for specific oxides (e.g., terbium) or regional benchmarks. NdPr mixed-oxide prices are included because NdPr oxide is a common traded/intermediate form and can be a useful proxy when individual oxide splitting is not modeled.

Material	Price point used (USD/kg)	Date and basis	Source
Neodymium oxide (Nd oxide, 99.5–99.9% FOB)	55.27–56.27	May 2024 average, FOB	Argus sample report
Neodymium oxide (Nd oxide, 99.5–99.9% FOB)	55.77	Argus index table value, FOB China	Argus sample report
Praseodymium oxide (Pr oxide, 99.5–99.9% FOB)	55.36–56.36	May 2024 average, FOB	Argus sample report
Praseodymium oxide (Pr oxide, 99.5–99.9% FOB)	54.58	Argus index table value, FOB China	Argus sample report
NdPr oxide (99% NdPr oxide, ex-works)	50.94–51.35	June 4, 2024, ex-works (yuan/kg converted in-report)	Argus sample report
Dysprosium oxide (Dy oxide, 99.5% FOB export)	265–270	June 2024 export assessment, FOB China	Argus sample report
Terbium oxide (Tb oxide, 99.99% FOB export)	815–835	June 2024 export assessment, FOB	Argus sample report
Samarium oxide (Sm oxide, 99.5% FOB)	2.01	Argus index table value, FOB China	Argus sample report
Samarium metal (Sm metal, min 99% FOB)	12.25	Argus index table value, FOB China	Argus sample report

Material	Price point used (USD/kg)	Date and basis	Source
Cobalt metal (U.S. spot)	33.83	January 2025 U.S. spot price	USGS monthly commodity summary
Cobalt metal (LME average)	23.63	January 2025 LME average price	USGS monthly commodity summary

3. NdFeB swarf value model

3.1 Approach Two complementary bottom-up valuation approaches are presented:

1) **Element-to-oxide price approach** that values recovered Nd oxide, Pr oxide, and Dy oxide using 2024 oxide price points and the reference swarf metal composition (Nd/Pr/Dy wt%).

2) **Process-yield approach** that uses a published techno-economic model of swarf-to-mixed-REO production capacity (100 t swarf/year ⟶ 32 t REO/year), coupled with reported REO composition ranges and mixed-REO selling-price assumptions. This pathway avoids the need to model explicit metal-to-oxide stoichiometric conversions in the prose model and stays closer to documented industrial mass-balance assumptions.

Both approaches are aligned with literature indicating that high REE recovery is achievable. For example, an acid-free dissolution route reported ~97% REE recovery and >99.5% purity products, with scenario leached efficiencies of 96–99%.

3.2 Element-to-oxide price approach Using the **reference NdFeB swarf composition** (per 1 tonne of swarf: 250 kg Nd, 50 kg Pr, 30 kg Dy) and treating the recoverable value as dominated by Nd oxide, Pr oxide, and Dy oxide, the gross value is computed using a representative recovery yield of ~97% (consistent with published REE recovery levels).

- **Nd value (oxide-priced)**: 250 kg Nd × 97% recovery × $55.27–56.27/kg Nd oxide ≈ **$13,400–$13,650 per tonne swarf** (oxide price range applied).
- **Pr value (oxide-priced)**: 50 kg Pr × 97% recovery × $55.36–56.36/kg Pr oxide ≈ **$2,685–$2,733 per tonne swarf**.
- **Dy value (oxide-priced)**: 30 kg Dy × 97% recovery × $265–270/kg Dy oxide ≈ **$7,711–$7,857 per tonne swarf**.

Summing only these oxide-priced REE components yields an indicative gross recoverable value of roughly **$23,800–$24,200 per tonne of NdFeB swarf** under the reference 3 wt% Dy composition and 2024 oxide price points.

This result is directionally consistent with published observations that dysprosium can be a major revenue driver in high-Dy feedstocks, because Dy oxide is priced at a materially higher level than Nd oxide in the oxide market data shown above.

3.3 Process-yield approach using mixed REO output A published TEA-scale mass balance modeled an industrial operation processing **100 metric tons/year of NdFeB magnet swarf** to produce **32 metric tons/year of REOs**, implying a **0.32 tonne REO output per tonne of swarf input** (i.e., 320 kg mixed REO per tonne swarf).

That same analysis reported that the mixed REO output composition contains **73–89% Nd2O3**, **9–11% Pr2O3**, and **0–18% Dy2O3** (mixed together).

Applying those fractions to 320 kg mixed REO per tonne swarf gives oxide-mass ranges per tonne swarf:

- Nd oxide in mixed REO: 320 kg × 73–89% = **234–285 kg** Nd oxide per tonne swarf.

- Pr oxide in mixed REO: 320 kg × 9–11% = **29–35 kg** Pr oxide per tonne swarf.
- Dy oxide in mixed REO: 320 kg × 0–18% = **0–58 kg** Dy oxide per tonne swarf.

Valuing these oxides using 2024 Argus price points yields a gross-oxide-value envelope (still excluding processing costs):

- Nd oxide value: (234–285 kg) × $55.27–56.27/kg ≈ **$12,900–$16,000 per tonne swarf**.
- Pr oxide value: (29–35 kg) × $55.36–56.36/kg ≈ **$1,606–$1,973 per tonne swarf**.
- Dy oxide value: (0–58 kg) × $265–270/kg ≈ **$0–$15,660 per tonne swarf** (Dy-rich case).

Under this process-yield framing, the **Dy fraction** dominates the upside, which matches the TEA statement that in a best-case scenario (6% Dy in swarf), **dysprosium oxide can represent the largest share of total economic value (52%)**.

4. SmCo swarf value model

4.1 Approach For SmCo swarf, the dataset contains (i) a reference swarf composition and (ii) at least one Sm recovery-rate datapoint from SmCo5 scrap. The model therefore focuses on **cobalt metal value** (dominant by mass and price) and a conservative valuation of samarium using a directly cited **samarium metal** price point to avoid requiring explicit Sm-to-Sm2O3 stoichiometric conversion in the prose model.

4.2 Gross recoverable value per tonne Per 1 tonne SmCo swarf (reference): 230 kg Sm metal and 500 kg Co metal are present by mass fraction.

For **samarium**, one SmCo5 scrap study reported obtaining samarium sulfate hydrate (96.5% purity) with a **samarium recovery rate of 87.1%**; this is used as an indicative recovery yield for Sm-bearing scrap, recognizing that Sm2Co17 may behave differently in practice.

For **cobalt**, the dataset does not provide a SmCo-specific cobalt recovery yield, so a conservative "high-yield" assumption consistent with published magnet-scrap processing yield ranges is used. One industry/market report describes hydrogen processing of magnet scrap (HPMS) as enabling alloy recovery with **90–98% yield**; the lower bound (90%) is applied as a conservative proxy recovery for cobalt-bearing fractions in a high-yield recycling route.

Applying these yields and 2025 cobalt prices gives:

- **Sm value (metal-priced)**: 230 kg Sm × 87.1% × $12.25/kg Sm metal ≈ **$2,456 per tonne swarf**.
- **Co value (metal-priced)**: 500 kg Co × 90% × $23.63–33.83/kg Co ≈ **$10,634–$15,224 per tonne swarf** (LME avg vs U.S. spot, Jan 2025).

Under these inputs, the gross recoverable value for SmCo swarf (Sm+Co only, excluding Fe/Cu/Zr credits) is approximately **$13,100–$17,700 per tonne** depending primarily on cobalt price basis and achieved recovery yields.

5. Processing cost estimates

Processing cost estimates in the dataset span a wide range because they reflect different system boundaries (swarf-to-oxide, whole-magnet recycling, and different process chemistries). A TEA of an acid-free dissolution route reported total cost of **~$11 per kg of swarf processed** (and ~$29–$36 per kg of mixed REO produced) at a modeled scale, which implies **~$11,000 per tonne swarf** processed as a central cost benchmark for swarf-focused hydrometallurgical processing.

A separate industry article gives a much higher, less specific estimate that **recycling 1 ton of NdFeB magnets costs ~$50–$70 per kilogram**, which would imply **$50,000–$70,000 per tonne**; the same article also reports hydrometallurgy can achieve recovery rates up to ~95% and that pyrometallurgy can yield ~70–85% recovery, illustrating that cost and yield vary materially by route and process design.

From a wastewater-generation standpoint (relevant for downstream handling cost and treatment requirements), the same article states that hydrometallurgy can produce **10–15 tons of wastewater per ton of rare earth recovered**.

6. Net recoverable value

Net value is presented as **gross recoverable value minus processing cost**, using the TEA-derived **$11,000/tonne swarf** processing-cost benchmark as the primary reference case.

6.1 NdFeB swarf net value Using the element-to-oxide price approach gross value of **~$23,800–$24,200 per tonne** (reference 3 wt% Dy case), subtracting ~$11,000/tonne processing cost yields a net value of approximately **$12,800–$13,200 per tonne swarf** in the TEA-like cost regime.

Under the mixed-REO process-yield envelope, gross oxide value spans widely depending on Dy content (because Dy oxide fraction can range from 0% to 18% of the mixed REO output). This implies net value can range from strongly positive to marginal depending on whether the feedstock is Dy-lean or Dy-rich, consistent with TEA statements that Dy content materially shifts the economic value distribution and can dominate value in best-case scenarios.

6.2 SmCo swarf net value For SmCo swarf, using the gross Sm+Co value range of **~$13,100–$17,700 per tonne** and applying the same $11,000/tonne swarf processing-cost benchmark as a proxy yields an indicative net of approximately **$2,100–$6,700 per tonne** (highly sensitive to cobalt price basis and achieved recovery).

6.3 Comparison to primary ore concentration Recycling economics are frequently framed around the much higher "grade" of secondary feedstocks compared with primary ores. One article states that mining one tonne of rare earth ore typically yields only **3–5 kilograms of usable material**, highlighting why high-grade scrap can be attractive even when processing costs are substantial.

In contrast, rare-earth magnet scrap streams can contain on the order of tens of percent rare earth content, such as the reference NdFeB swarf composition above with substantial Nd/Pr/Dy fractions by mass, and multiple sources report NdFeB magnets can contain "up to ~30%" critical rare earth elements in the magnet alloy.

7. Volume context

The potential annual swarf feedstock volume depends on magnet production volumes and the fraction lost as machining and process scrap. One industry-facing source states that **Nd-Fe-B magnet production generates 20–30% scrap**, while **Sm-Co production generates 15–30% waste** as sludge and scrap, implying a structurally significant secondary stream relative to finished magnet output.

A TEA-scale analysis estimated **130,000–150,000 metric tons of NdFeB magnets** are produced per year globally. Applying the 20–30% production scrap fraction as an indicative range suggests that the global NdFeB manufacturing system could generate on the order of **tens of thousands of tonnes** of scrap/swarf annually (with exact mass depending on whether the scrap fraction is referenced to input alloy or output magnets).

8. Strategic value premium

Beyond commodity value, defense-relevant magnet scrap can carry an additional "strategic" premium when it supports domestic, traceable supply chains and reduces dependence on concentrated upstream refining. One market source states that **China controls ~70% of global rare earth oxide refining capacity**, which increases geopolitical and supply-chain risk premiums for non-Chinese supply pathways, including recycling.

Downstream demand-side premiums for recycled-content magnets can also elevate realized value. A market report notes that OEMs may pay **10–20% premiums for magnets verified as recycled content**, creating a potential margin uplift for compliant recyclers relative to commodity-only valuation approaches.

Policy-linked price support mechanisms can further raise the effective value of domestically produced rare-earth material streams. One recycling-focused article claims the U.S. Department of Defense has **guaranteed minimum prices of $110/kg** to incentivize domestic production, indicating that certain strategic programs may value domestic supply above spot-market economics for some periods and materials.

Chapter 5

DFARS Rare Earth Magnet Compliance Timeline Roadmap for Defense Primes 2024 2028

Executive summary

January 1, 2027 is the inflection point when DFARS restrictions for samarium-cobalt (SmCo) and neodymium-iron-boron (NdFeB) magnets expand upstream to cover the full supply chain (mined, refined, separated, melted, or produced) in covered countries, not just melting/production stages. Because qualification, re-sourcing, and documentation updates typically take multiple contracting cycles, multiple industry sources urge primes to secure DFARS-compliant magnet supply chains in advance of the statutory deadline, with some suppliers publicly targeting "mid-2026" readiness as a practical cutover point.

The May 30, 2024 final rule is already effective and establishes the staged requirements and the governing contract clause (DFARS 252.225-7052). Between now and the end of 2026, the restriction is narrower (focused on material melted/produced in covered countries), but it still requires primes to ensure traceability through alloying/melting and subsequent magnet manufacturing steps (powder formation, pressing, sintering/bonding, magnetization) for SmCo and NdFeB magnets.

Domestic and allied capacity is expanding, but the "mine-to-magnet" buildout is still in progress and includes schedule and permitting risks; DoD itself has stated that establishing new domestic rare earth sources can take "a decade or longer." The roadmap below therefore emphasizes (1) near-term supply chain auditing and provenance tracking, (2) accelerated second-source qualification, (3) use of the recycled NdFeB exception where applicable (U.S. milling + U.S. sintering), and (4) early preparation of nonavailability-determination packages when compliant supply cannot be ensured by late 2026.

Timeline

The timeline below consolidates binding regulatory milestones, expected provenance disclosure implementation windows, and major domestic capacity events relevant to magnet sourcing decisions for defense primes.

2024 On May 30, 2024, DoD issued and made effective a final rule that amended DFARS 225.7018 and the associated clause DFARS 252.225-7052 (Restriction on the Acquisition of Certain Magnets, Tantalum, and Tungsten). The final rule implements a staged restriction: through December 31, 2026, contractors may not deliver covered material

"melted or produced" in covered countries; beginning January 1, 2027 the restriction expands to materials "mined, refined, separated, melted, or produced" in covered countries (the expansion date was extended to January 1, 2027 by NDAA FY2024 Section 854).

GAO reports that DoD completed rulemaking for the revised prohibition in May 2024, but that separate provenance/provenance-disclosure rulemaking for permanent magnets was still developing, with a DoD expectation (as of the GAO report) to publish a proposed DFARS rule for public comment in October 2024.

2025 Regulatory implementation expectations begin to tighten in 2025. GAO describes procurement regulation implementation (including magnet source disclosure requirements) as expected in the 2025–2027 window, and notes that the statutory effective date for the new disclosure requirement is "not earlier than June 2025." GAO further notes that if a contractor cannot make the required disclosure, it will be required to establish a supply chain tracking system to make the disclosure "to the fullest extent possible."

In parallel, domestic magnet capacity milestones reported in public sources indicate that new U.S. production is starting to come online. MP Materials reported (Jan 22, 2025) that commercial NdPr metal production was already online, trial production of sintered NdFeB magnets was underway, first deliveries were on track for year-end, and the facility targeted ~1,000 metric tons/year of finished magnets with a gradual ramp beginning in late 2025.

For additional U.S. capacity, DoD's ManTech reporting indicates that (via MCEIP awards) E-VAC Magnetics is expected to establish commercial-scale rare-earth magnet manufacturing capability by 2025 (award value $94.1 million), and that Noveon Magnetics has established a rare earth magnet manufacturing facility in San Marcos, Texas (award value $28.8 million).

2026 2026 is the "execution year" for supplier transitions under the DFARS staged approach, with multiple sources recommending completion of alternative sourcing and qualification actions before the end of 2026 to avoid a delivery cliff in 2027.

Several supplier-side milestones cluster around 2026. Arnold Magnetic Technologies states that DFARS updates take effect January 1, 2027 and that it aims to achieve compliance by mid-2026. Dexter Magnetic Technologies, Magnetic Component Engineering, and Electron Energy Corporation likewise describe a joint effort to achieve DFARS 252.225-7052 compliance for NdFeB and SmCo magnets by mid-2026, ahead of the January 1, 2027 deadline, including work with compliant global suppliers to source rare earth materials.

On the production side, eVAC Magnetics reports that its Sumter, South Carolina facility opened in fall 2025 and will "quickly scale" to 2,000 metric tons/year by the first quarter of 2026, with stated ability to scale up further in later years; the same statement also notes that the Sumter facility sources NdPr feedstock from MP Materials' Mountain Pass mine through a strategic partnership. Separately, MP Materials' reporting indicates commissioning of a heavy rare earth separation capability targeted for mid-2026, with a Dy/Tb circuit nameplate capacity of 200 MT/year (within a facility designed to process ~3,000 MT/year of feedstock) intended to support planned production of 10,000 MT/year of high-performance NdFeB magnets.

2027 January 1, 2027 is the binding compliance deadline for the expanded prohibition: contractors "shall not deliver" covered material mined, refined, separated, melted, or produced in any covered country, or end items manufactured in any covered country that contain a covered material. For SmCo magnets, DFARS explicitly states that, effective January 1, 2027, the restriction includes the entire supply chain from mining/production of cobalt and samarium ore or feedstock (including recycled material) through production of finished magnets. For NdFeB magnets, DFARS explicitly states that, effective January 1, 2027, the restriction includes the entire supply chain from

mining of neodymium, iron, and boron through production of finished magnets.

The expanded rule also changes how the COTS exception works. Legal commentary notes that, effective January 1, 2027, COTS items will be exempt only if they are less than 50% covered material by weight (not just tungsten), tightening practical compliance requirements for magnet-containing COTS subsystems and assemblies.

Because compliant supply may not exist in sufficient quality/quantity for specific applications, DFARS provides an exception mechanism via nonavailability determinations. DFARS and its implementing materials describe both individual and class nonavailability determinations under DFARS 225.7018-4 when compliant materials are unavailable. For class determinations, USD(A&S) must issue a public notice of intent at least 30 days prior to issuing the determination.

Some domestic production expansions are also described as initiating volumes in 2027. MP Materials' November 2025 presentation describes an expansion of Independence with "initial volumes starting in 2027."

2028 A key late-decade capacity event is MP Materials' planned second domestic magnet manufacturing facility ("10X Facility"), expected to begin commissioning in 2028; MP states that once the new facility is completed and commissioning begins, its total U.S. rare earth magnet manufacturing capacity will reach an estimated 10,000 metric tons. MP further states that, for 10 years following the construction of the 10X Facility, DoD has agreed to ensure that 100% of the magnets produced at the 10X Facility will be purchased by defense and commercial customers, and that DoD has entered into a 10-year price floor commitment of $110/kg for MP's NdPr products stockpiled or sold.

However, some non-U.S. but non-China supply chain options may have timelines extending beyond 2027. One analysis of Lynas' Texas refinery project outlines commissioning beginning in Q4 2027 and commercial production in Q2 2028, with a target to produce 5,000 metric tons/year of magnet-grade NdPr, underscoring that some diversified upstream capacity may arrive after the DFARS 2027 compliance deadline.

Regulatory milestones

The table below summarizes the "what changes when" elements that matter most for magnet procurement and supplier qualification.

Date or window	What changes	Practical implication for primes
May 30, 2024	Final rule issued and effective; DFARS clause 252.225-7052 updated.	Treat as "go-live" for contract language updates and internal compliance program mobilization (policies, supplier questionnaires, traceability expectations).
Through Dec 31, 2026	Prohibition focuses on covered material "melted or produced" in covered countries for deliveries; for SmCo/NdFeB, includes melting and all subsequent phases (powder formation, pressing, sintering/bonding, magnetization).	Immediate focus on traceability through alloying/melting and the magnet-manufacturing chain, even before 2027 upstream expansion.

Date or window	What changes	Practical implication for primes
Not earlier than June 2025 to 2027	Magnet source disclosure requirements expected to start taking effect between 2025 and 2027; if contractor cannot disclose, it must establish a supply chain tracking system to disclose to fullest extent possible.	Build data systems and supplier flowdown now; don't assume a manual, late-stage affidavit will be sufficient if disclosures cannot be made.
Jan 1, 2027	Expanded prohibition covers "mined, refined, separated, melted, or produced" in covered countries; SmCo and NdFeB explicitly defined as covered materials and covered supply chains (SmCo includes recycled feedstock).	Must have compliant sources (or approved nonavailability determination) for upstream feedstock/provenance, not just for final magnet manufacturing location.
Jan 1, 2027	COTS exemption narrowed; COTS items exempt only if <50% covered material by weight.	Magnet-heavy COTS subsystems may become non-exempt and require the same provenance diligence as non-COTS assemblies.
Ongoing	Class nonavailability determinations require USD(A&S) approval and a public notice at least 30 days prior; individual determinations can be made by the head of the contracting activity for one contract.	If you anticipate a waiver need for 2027 deliveries, plan backward from the 30-day notice requirement and the time required to assemble market research and supporting analysis.

Domestic capacity coming online

The capacity timeline below is organized by facility and limited to milestones explicitly reported in the provided sources.

MP Materials MP Materials' magnetics plans provide a staged domestic capability picture. MP's Jan 22, 2025 release indicates trial production of automotive-grade sintered NdFeB magnets, first deliveries targeted for year-end, and a gradual production ramp beginning in late 2025 toward ~1,000 metric tons/year of finished magnets. MP's filings also state the company anticipates manufacturing NdFeB permanent magnets by the end of 2025 and that Independence is designed for ~1,000 MT/year of finished magnets at initial design capacity.

For heavy rare earths (relevant to high-coercivity/high-temperature NdFeB grades that rely on Dy/Tb), MP reports commissioning a heavy rare earth separation facility at Mountain Pass in mid-2026 and describes a Dy/Tb circuit nameplate capacity of 200 MT/year within a facility designed to process ~3,000 MT/year of feedstock, intended to support planned production of 10,000 MT/year of high-performance NdFeB magnets.

For larger-scale domestic magnet capacity, MP states it will construct a second U.S. magnet facility ("10X Facility") expected to begin commissioning in 2028, bringing total U.S. magnet manufacturing capacity to an estimated 10,000 metric tons once completed. MP further states DoD has a 10-year price floor commitment of $110/kg for NdPr

products and a 10-year commitment ensuring 100% of magnets produced at the 10X Facility are purchased by defense and commercial customers.

VAC and eVAC Magnetics eVAC reports that its Sumter, South Carolina magnet manufacturing facility opened in fall 2025 and that it will quickly scale to 2,000 metric tons/year by the first quarter of 2026, with an ability to scale to higher levels in subsequent years. The same announcement states that eVAC's South Carolina facility sources key raw materials via a strategic partnership with MP Materials, which supplies NdPr feedstock from Mountain Pass, California.

Noveon Magnetics DoD ManTech reporting indicates Noveon Magnetics has established a rare earth magnet manufacturing facility in San Marcos, Texas, supported by a $28.8 million MCEIP award, but does not provide a commissioning date or capacity in the cited excerpt.

Lynas USA DoD ManTech reporting indicates Lynas USA received a combined $288 million in MCEIP funding to establish a second domestic, commercial-scale oxide production capability by 2026. Other analyses describe Lynas' Texas project timelines extending into 2027–2028 for commissioning and commercial production, reflecting that some upstream processing capacity may not align with the 2027 compliance deadline for magnet deliveries.

Compliance pathway decision matrix

The options below are not mutually exclusive; primes typically pursue multiple pathways in parallel to reduce schedule and certification risk.

Pathway	When it works best	What must be true by late 2026	Key actions and evidence packets
Transition to compliant (non-covered-country) primary supply	Programs with sufficient lead time to qualify new magnet sources before Jan 1, 2027.	Supply chain provenance for SmCo and NdFeB magnets must avoid covered-country mining/refining/separation/melting/production for 2027 deliveries.	Review supply chains and identify alternative sources; suppliers and advisors explicitly [illegible] supply chain review and alternative sourcing in 2024–2026.
Use the recycled NdFeB exception	NdFeB magnets and end items containing NdFeB magnets where recycled feedstock is available and processing can be done domestically.	Milling of recycled material and sintering of the final NdFeB magnet must take place in the United States to qualify for the exception.	Designate recycled feedstock pathways; work with procuring activity on demonstrating compliance to exceptions.

Pathway	When it works best	What must be true by late 2026	Key actions and evidence packets
Seek nonavailability determination	Critical applications where compliant sources of satisfactory quality/quantity "in the required form" cannot be procured when needed at a reasonable price.	Government decisionmakers must have sufficient market research and analysis to support individual or class determinations; class determinations require public notice at least 30 days prior.	Compile Part A data on alternative sources and detailed market research; template-based packages are explicitly contemplated in PGI 225.7018-4.

Action items

The action plan below is organized as a time-phased set of tasks aligned to the DFARS dates and reported implementation windows.

2024 actions Defense primes should treat the May 30, 2024 final rule as the trigger to mobilize a program-level compliance roadmap, since it establishes the staged restriction and amends DFARS 252.225-7052 in contract language. Legal and policy commentary recommends immediate awareness and preparation steps in 2024, followed by supply chain review in 2025 and supplier transition in 2026.

2025 actions Primes should begin structured supply chain reviews (down to covered material provenance) and implement tracking systems capable of supporting future source-disclosure requirements, since GAO indicates disclosure implementation is expected between 2025 and 2027 and states that contractors unable to disclose must establish a supply chain tracking system to disclose "to the fullest extent possible." In the same period, primes should proactively qualify emerging domestic sources as they begin trial production and early shipments, such as MP's Independence facility ramp beginning in late 2025 and first deliveries targeted by year-end 2025.

2026 actions By 2026, primes should be executing supplier transitions and qualification campaigns, because the restriction expands upstream on January 1, 2027 and will apply to the entire supply chain for SmCo and NdFeB magnets. Industry advisors explicitly recommend identifying and securing alternative sources in 2026 due to the expanded scope and reduced applicability of the COTS exemption.

Primes should also prepare "waiver-ready" nonavailability determination packages if compliant sources cannot be secured for 2027 deliveries. Multiple sources recommend aggregating data to support nonavailability determinations if noncompliance is anticipated by January 1, 2027, and DFARS/PGI materials specify that class determinations require at least 30 days of public notice and that determination packages require detailed market research and alternative-source analysis.

Q4 2026 checkpoint By the end of 2026, primes should have either completed a transition to compliant sources or be in an advanced stage of a government-approved nonavailability determination path, because the contract-delivery prohibition expands on January 1, 2027 and applies directly to contractor deliveries under covered contracts.

2027 actions Starting January 1, 2027, primes must ensure magnet-containing deliverables do not contain covered material mined/refined/separated/melted/produced in covered countries (unless an exception applies). For SmCo, the explicit inclusion of "recycled material" within the covered supply chain means primes should not assume that recycled SmCo feedstock is exempt in the way recycled NdFeB can be exempt (under the narrow U.S. milling + sintering condition).

2028 actions In 2028, primes should reassess their long-term sourcing position in light of new domestic capacity expected to begin commissioning (e.g., MP's 10X Facility), and consider whether long-term offtake/price-stability arrangements can reduce program risk as demand grows and DFARS compliance becomes a baseline requirement for many magnet-containing systems.

Risk assessment

The largest compliance risk is a schedule mismatch between (a) the hard DFARS January 1, 2027 expanded prohibition for magnet supply chains and (b) the time required to build and qualify compliant upstream and downstream capacity. DFARS rulemaking itself can take "a year or longer," and DoD assessed that it can take "a decade or longer" to establish new domestic sources for rare earths, which makes reliance on late-breaking domestic capacity a high-risk strategy for 2027 program deliveries.

A second risk is that supply chain disclosure/provenance requirements may tighten during 2025–2027, and GAO signals that contractors unable to disclose sources will need supply chain tracking systems to comply "to the fullest extent possible." This creates an operational risk where programs may be blocked not only by material provenance, but also by incomplete documentation systems and weak subcontract flowdown practices.

A third risk is the reduced applicability of COTS-based compliance shortcuts. Effective January 1, 2027, legal commentary notes the COTS exemption is narrowed to items containing less than 50% covered material by weight, which can pull previously "commercial" subassemblies into the scope of origin/provenance diligence if magnets are a substantial fraction of mass (e.g., dense motor components).

Finally, primes should plan for the contingency that nonavailability determinations may be necessary for specific applications. The DFARS framework allows such determinations when compliant covered materials of satisfactory quality/quantity in the required form cannot be procured at a reasonable price, but class determinations require public notice at least 30 days prior and require extensive supporting documentation and market research.

Chapter 6

Wastewater Treatment Chemistry for Rare Earth Contaminated Machining Coolant Streams

Summary

Rare-earth-element-bearing wastewaters are commonly treated with a combination of physicochemical separation, pH-driven precipitation, and (where needed) membrane and sorption polishing to reduce toxicity and enable reuse or compliant discharge. Treatment performance is highly sensitive to solution chemistry, especially pH, oxidation state (e.g., Fe(II) vs Fe(III)), co-ions, ionic strength, initial target-element concentration, and the presence of organic substances that can interfere with precipitation and separations. For machining coolant and metalworking-fluid streams specifically, the presence of oils/organics and surfactants creates fouling and process-integration challenges that typically require dedicated pre-treatment before membranes or adsorption/ion-exchange recovery steps are viable at scale.

Stream characterization

Machining coolants (especially water-mix cutting fluids) are commonly prepared as oil-in-water emulsions, with concentrates diluted to form emulsions on the order of 2–10% (v/v) oil in water, which strongly shapes separation chemistry and downstream membrane fouling behavior. The literature explicitly treats machining coolant as a high-volume wastewater category where precipitation is attractive because it can be cost-effective for large volumes, including high-salinity streams such as machining coolant, even when more selective methods would be too costly for bulk treatment. Work focused on rare-earth recovery and removal also notes that real industrial waste streams (including machining slurries/coolants) are more complex than synthetic solutions because cutting fluids and organic contaminants introduce matrix effects that complicate REE recovery and can require pre-treatment or pre-concentration.

Where the objective is coolant reuse rather than discharge, process performance is often evaluated on suspended solids, organics, surfactants, and phosphorus metrics that correlate with coolant stability and downstream equipment reliability (e.g., total suspended solids removal, COD/TOC reduction). One study of grinding-fluid recycling used 1000 L baths filled with aqueous grinding fluids (e.g., 2–5% KFL plus 1–2% FC 430 for Zn–Al parts, or 2–5% ZF 322 for steel parts) and noted that such fluids are periodically cleaned using coagulant addition and centrifugation in current practice, illustrating the combination of oil/surfactant and fine-solid contamination typical of industrial coolant loops.

Chemical precipitation

Hydroxide precipitation windows Across multiple sources, REE hydroxide precipitation is consistently described as a high-pH process in which REEs precipitate as hydroxides when pH is raised into the near-neutral to alkaline range (often cited roughly pH 8–10, depending on matrix and concentration). Experimental pH-onset ranges for hydroxide precipitation differ by element; one reported set of tested REEs begins to precipitate as hydroxides at high pH values including La (pH 7.82–8.10), Ce (pH 7.60–7.80), Pr (pH 7.35–7.89), Eu (pH 6.82–7.30), Tb (~7.20), and Y (pH 6.75–6.83). Consistent with this, Pourbaix-based reasoning in precipitation-focused work also notes that REEs tend to form rare-earth hydroxides in slightly alkaline solutions (e.g., pH > 7.5).

Staged pH control for Fe and Al removal Because iron and aluminum often precipitate at lower pH than REEs, staged neutralization is a common selectivity strategy in REE-bearing aqueous systems, where Fe is removed first at mildly acidic pH before raising pH for REE hydroxides. Practical guidance and review-level summaries describe Fe precipitation occurring first (often around pH ~3–4) prior to REE precipitation at higher pH, enabling separation by pH control when oxidation state is managed appropriately. In a staged precipitation study, more than 95% of iron and aluminum were removed at pH values of ~3.0 and ~4.5, respectively, while lanthanum removal increased gradually as pH was elevated, underscoring narrower pH windows for Fe/Al versus broader pH response for REEs.

Where Fe is present as Fe(II), oxidation to Fe(III) can materially change precipitation and co-precipitation behavior; for example, one study reported that REE removal was noticeably elevated at pH 4.0 after adding H2O2 to convert ferrous to ferric ions, which caused nearly complete precipitation of iron and enhanced REE removal via interaction with ferric precipitates. However, competitive species such as aluminum can suppress REE adsorption onto ferric precipitates, which can reduce apparent REE removal under otherwise similar oxidation and pH conditions.

Oxalate precipitation for recovery at low pH When the objective is REE recovery as a solid product (rather than simple immobilization), oxalic-acid precipitation is widely used because rare-earth oxalates have low solubility, but achieving high recovery often requires substantial reagent consumption and careful pH control. Multiple sources

report that oxalic-acid-based REE precipitation can be optimized at low acidic pH ranges, with pH ~1.25–2.0 repeatedly identified as a window that maximizes recovery and precipitate purity while limiting co-precipitation of contaminants. One dataset indicates that, as pH rises above ~3, precipitation of contaminants (Al, Ca, Fe, Mn, Si) increases and reduces final REE product purity, while a pH around 1.5–2 is recommended to keep contaminants low and recovery high.

Oxalic acid speciation also explains the pH sensitivity of the precipitation kinetics and selectivity, with H2C2O4 dominating at pH < 1, HC2O4– between pH 1–4, and C2O4^{2–} at pH > 4, which changes complexation and precipitation equilibria in multi-ion solutions. In one analysis, a typical oxalic acid efficiency (OAE) of ~60% was observed, with ~40% of added oxalic acid consumed by precipitating/complexing non-REE elements, directly tying wastewater matrix composition to reagent costs and secondary waste generation. Where iron is present, Fe(III) can compete strongly for oxalate (forming Fe–oxalate complexes) and reduce REE precipitation efficiency, reinforcing the importance of upstream Fe removal or strict stoichiometry control in oxalate routes.

Phosphate and carbonate precipitation Beyond hydroxides and oxalates, phosphate-based precipitation is used both for REE recovery and for broader metal immobilization, and phosphate excess can yield high REE recovery ratios under appropriate conditions. One phosphate precipitation study reports that selective separation of REEs from transition elements was achieved at pH values below 2.5, while also emphasizing that Fe is removed first by oxidizing Fe(II) to Fe(III) and precipitating Fe(III) hydroxides using strong alkalis prior to REE recovery steps.

Membrane separation

Membrane unit operations are commonly deployed as pre-treatment and pre-concentration steps for REE-bearing wastewaters, including ultrafiltration and nanofiltration, to remove particulates and/or concentrate dissolved/complexed REEs prior to chemical separation or sorption-based capture. The membrane-technology literature classifies nanofiltration (NF) as having pore sizes below ~2 nm and ultrafiltration (UF) as having pore sizes in the ~2–50 nm range, and uses these in tandem with upstream conditioning (e.g., pH adjustment and complexation) to improve rejection of REE species.

A representative performance datapoint for REE rejection comes from micellar-enhanced ultrafiltration (MEUF), where an UP020 membrane achieved high REE rejection (~97%) at pH = 3.5, 25 ℃, 3 bar operating pressure, 60 min contact time, and SDS concentration of 8.0 mM. A separate membrane-performance summary reports NF270 with MWCO ~180 ± 20 Da operating at pH 1 with rejection >98%, illustrating that tighter NF can achieve high rejection in acidic matrices (subject to fouling and scaling management). Polymer-complexation approaches are also reported, where addition of PAA (e.g., 30 mg/L) enabled >90% REE rejection at pH values between 8 and 9, consistent with the use of complexation to shift REE speciation toward membrane-rejectable forms.

For industrial coolant recycling, a two-stage filtration train has been demonstrated for grinding fluids: an initial bag filtration at 5 μm followed by low-pressure ultrafiltration at ~4 bar using polyacrylonitrile membranes with a 30 kDa cut-off in cross-flow batch mode at 25 ± 1 ℃. In those experiments, ultrafiltration completely removed TSS, and reductions across turbidity, COD, TOC, surfactants, and total phosphorus ranged from ~80% to almost 100%, supporting UF as a robust coolant-loop clarification step even when the objective is not REE capture per se. However, broader reviews emphasize that membrane fouling and stability (including fouling from oils/organics) are key constraints for industrial coolant streams, reinforcing the need for pre-treatment specifically targeting emulsified oil and organic load before NF/UF deployment for REE recovery.

Ion exchange and adsorption

Ion exchange and adsorption are repeatedly identified as promising or mature routes for concentrating and selectively capturing REEs from dilute aqueous streams, particularly when paired with upstream pre-concentration or particulate removal steps. A practical example of ion exchange for REE separation reports use of Lewatit MDS 200 H, a strongly acidic gel-type cation-exchange resin, with an operating pH range of 2–14 and maximum operating temperature of 120 ℃, and provides quantitative operating data for column flow, loading, and elution strategies using NH4EDTA eluent chemistry. Under reported optimal conditions (initial pH 6.0 and 0.02 mol·L−1 NH4EDTA), the system achieved 98.4% heavy-REE purity in the initial stage (0–10 bed volumes), illustrating the potential for high-selectivity polishing when feed chemistry is controlled and compatible with chelant-based elution.

For adsorption, zeolites are a documented low-cost sorbent option for certain REE ions; one study reports over 90% removal of lanthanum and cerium using zeolite as a sorbent and provides relevant material properties for Zeolite 13X (e.g., 4–8 mesh particle size, BET surface area 576 m^2/g, average pore size ~20 Å). The adsorption literature also highlights high-capacity engineered sorbents (e.g., EDTA-chitosan modified materials) with reported maximum adsorption capacities for REEs reaching hundreds of mg/g in laboratory systems, indicating the technical feasibility of high-capacity capture media for polishing or pre-concentration (subject to real-matrix limitations).

Solvent extraction

Solvent extraction remains a standard, widely used approach for REE separation from aqueous solutions in hydrometallurgical flowsheets, but it is typically most suitable after leaching or concentration steps rather than direct treatment of very dilute waters, and it can generate organic waste streams that complicate wastewater management. Solvent-extraction systems commonly use organophosphorus extractants; one example reports use of D2EHPA and TBP in an organic diluent (Elixore 205) at an organic-to-aqueous (O:A) ratio of 1:1, with stripping of loaded organic using high-concentration nitric acid (6 mol/L). For membrane-solvent-extraction hybrids, supported liquid membranes (SLMs) are explicitly described as a modification of solvent extraction that combines extraction and stripping in one unit operation, with one parameterized example using 10% v/v DEHPA in kerosene and a 5 mol·L−1 HNO3 stripping solution.

Because machining-coolant wastewaters can be high-volume and may carry substantial organic load (surfactants/oil), extraction-precipitation strategies have been proposed as potentially suitable for large-scale applications in treating rare-earth-contaminated coolants while reducing chemical oxygen demand relative to conventional liquid–liquid extraction approaches in some reported comparisons.

Cobalt-specific treatment

Cobalt can co-occur with REEs in relevant aqueous systems and can partition into treatment solids; one study notes that treatment solids can concentrate critical metals such as cobalt into oxide/hydroxide waste products, but does not provide a single universal pH window for cobalt removal analogous to the REE hydroxide window. In a staged-neutralization approach, one report states that neutralizing to pH 8 to 8.5 precipitated all REEs and co-precipitated most cobalt and manganese as a pre-concentrate material, indicating that cobalt can be captured in the same high-pH precipitation step used for REEs in some matrices. Conversely, in oxalate precipitation systems designed to recover REE oxalates, cobalt behavior can differ; one experimental report found cobalt concentrations did not change significantly during oxalic acid precipitation and concluded cobalt did not co-precipitate with REE oxalates under the tested conditions, enabling separation by filtration of REE oxalate solids from cobalt-bearing liquor.

These contrasting behaviors imply that cobalt management strategy should be selected based on the chosen REE recovery route and desired product form, rather than assumed to be inherently coupled to REE precipitation in all

systems.

Metalworking fluid pre-treatment

Effective treatment of REE-contaminated machining coolants generally begins with physical and physicochemical steps to remove free oil, tramp oil, and suspended solids, because these constituents foul membranes and interfere with adsorption/ion exchange performance. Industrial guidance on metalworking-fluid recycling describes practical unit operations including sump vacuum and filtration to remove chips/solids, skimmers and coalescers to remove tramp oil before it emulsifies, and centrifuges to separate pure coolant from emulsified tramp oil and both large and small contaminants. Consistent with this, another summary notes that many centrifuges can remove particulate matter as small as ~2 microns, which can be relevant for capturing fine metallic fines and abrasive particles typical of grinding operations.

Electrocoagulation is also described as a metalworking-fluid wastewater treatment method for removal of oils and contaminants, and one industry-facing source reports that electrocoagulation equipment followed by a filter press can reduce COD in cutting-fluid wastewaters by ~80%. For sites where disposal/haulage costs dominate, evaporative treatment is described as a practical route for spent coolant management, with one vendor claiming evaporators can provide a payback of less than one year versus hauling and help meet stringent spent-coolant disposal regulations (noting that performance and economics are site-specific).

Integrated treatment train design

An integrated treatment train for rare-earth-contaminated machining coolants typically combines (1) oil and solids removal, (2) pH and redox conditioning, (3) bulk metal removal by precipitation/coagulation, and (4) polishing and/or recovery operations (membrane concentration, ion exchange/adsorption, or selective precipitation to a product). The table below summarizes a pragmatic chemistry-first decision structure, emphasizing pH setpoints and the role of pre-treatment.

Objective	Primary chemistry lever	Typical pH regime	Enabling steps and constraints
Bulk immobilization and discharge compliance	Hydroxide precipitation via pH raise	REE hydroxides typically precipitate at pH ~8–10 (matrix dependent)	Stage Fe removal first at ~pH 3–4 before REE precipitation to reduce co-precipitation losses
REE recovery as oxalate product	Oxalic acid precipitation	Optimal recovery/purity often at pH ~1.25–2.0 with sufficient residence time and dosing (e.g., 30–40 g/L, >2 h)	Oxalic acid efficiency can be ~60% with substantial consumption by non-REEs, so impurity control and pre-removal of Fe are critical
Coolant-loop reuse with reduced disposal	Filtration and UF clarification	UF with 30 kDa at ~4 bar in cross-flow can fully remove TSS and reduce COD/TOC/surfactants substantially	Oils/surfactants remain key; upstream coagulation/centrifugation is commonly used in practice

Objective	Primary chemistry lever	Typical pH regime	Enabling steps and constraints
REE pre-concentration from dilute streams	Membranes (NF/UF) and/or ion exchange	MEUF example: pH 3.5, 3 bar, SDS 8 mM, 97% rejection on UP020	Fouling from oils/organics requires dedicated pre-treatment in coolant matrices
High-selectivity separation or polishing	Ion exchange or engineered sorbents	Example IX system operated with initial pH 6.0 and NH4EDTA eluent to reach high HREE purity	Resin selection and regeneration chemistry matter; strong-acid resin examples report broad operating pH ranges and acid regeneration options

In practice, pH-based precipitation and coagulation steps are typically paired with physical solids separation (clarification, filtration, centrifugation) because pH adjustment is used both to drive metal hydroxide formation and to facilitate suspended solids removal via aggregation and settling. Where downstream membranes are included, the literature emphasizes that pH changes can induce scaling (e.g., Ca/silicate precipitation) and that pretreatment is needed to prevent membrane shutdown, which is directly relevant to coolant streams containing hardness ions and surfactants.

Finally, effluent targets and required treatment intensity are explicitly described as site-specific, with regulatory discharge limits and effluent quality objectives determining the necessary level of REE and co-metal removal rather than any universal REE discharge standard. Where numeric standards are available for analogous regulated parameters, examples include rare-earth-industry effluent standards such as GB 26451-2011 limits (phosphorus 5 mg/L, COD 100 mg/L) used in one study to judge compliance after oxidation and precipitation treatment, illustrating how non-metal metrics (COD) can become the binding constraint even when metal/phosphorus targets are met.

Implications for Algamol case development

This chemistry review provides a parameter set that can be used to scope an Algamol facility business case around REE-contaminated coolant treatment by bounding required pH windows, identifying pre-treatment needs, and identifying where higher selectivity steps (ion exchange/adsorption or membrane pre-concentration) become feasible after organics control. In particular, the need for organics and emulsified oil removal before membrane or adsorption-based recovery is explicitly emphasized in coolant-focused reviews, which should be treated as a front-end requirement when estimating system footprint, consumables, and operational complexity at a machining site. Benchmarking against alternative onsite strategies (e.g., evaporators marketed for spent coolant disposal) can also inform the economic baseline, given claims of payback driven by reduced hauling and compliance support, even though detailed cost data remain site-specific and would need onsite measurement and validation.

Chapter 7

Algamol Wastewater Treatment Business Case Defense Grade Rare Earth Magnet Machining Facility

Summary

Defense-grade permanent magnet machining and finishing (e.g., grinding, EDM, slicing) generates coolant and sludge streams that are frequently expensive to manage offsite and are often constrained by hazardous-waste classi-

fication and local sewer discharge requirements. Across multiple industry sources, offsite hazardous waste disposal is commonly summarized as \$0.50–\$3.00/gal (about \$130–\$800/m^3), with additional variability by jurisdiction and waste characteristics. Reported per-gallon disposal pricing can be substantially higher for small-volume drum pick-ups (e.g., ~\$8–\$9/gal), reinforcing the cost penalty for small and mid-sized operations without in-house treatment or recycling.

A pragmatic Algamol deployment concept for this segment is a modular on-site treatment train that (1) pre-separates tramp oil and solids (e.g., ultrafiltration) and (2) applies an algae-based capture step intended to reduce dissolved metal loads and concentrate metals into a manageable biosludge stream for offsite recycling or downstream recovery. A representative small-system cost envelope cited in the compiled business-case inputs is total CAPEX ~\$110k–\$310k with annual OPEX ~\$28k–\$73k/year (inclusive of chemicals, energy, labor, membrane replacement, and sludge handling). If such a system replaces (or meaningfully reduces) offsite hauling of coolant/wastewater at \$130–\$800/m^3, simple paybacks can be short (often <2 years) even before assigning any value to recovered cobalt/rare earths, because disposal costs can dominate lifecycle economics at small industrial sites.

Target customer profile

The target customer is a defense-sector magnet machining and finishing facility or subcontractor that handles high-temperature rare-earth magnets (SmCo and high-coercivity NdFeB) as finished-to-print parts or precision-machined blanks, and therefore runs multiple wet machining steps requiring recirculating coolant and periodic sump change-out and sludge handling. In practice, these sites can include integrated magnet manufacturers and machining houses prioritized for pilots such as EEC/Permag (Landisville, PA), Arnold (multiple sites), and VAC (Sumter, SC).

Operationally, coolant streams are typically high-water-content emulsions; one source notes coolant can be up to 10% concentrate mixed with 90% water, and another notes the remaining liquid after removing sludge and oils is often ~90–98% water. For grinding operations on RE-magnet materials, published process data indicate coolant can be applied at ~5% concentration with a throughput of ~30–50 L/min per wheel, and that grinding slurries can retain from <1% to >20% cooling agent after on-site solid–liquid separation and recirculation.

For business-case sizing, an indicative wastewater generation range used in the business-case inputs is ~5–50 m^3/day depending on facility size and operational intensity. For comparison, metalworking facilities with smaller footprints can report far lower volumes (e.g., ~2 m^3/month in one automotive metalworking wastewater case study), highlighting that volumes are extremely site-specific and often depend on coolant management practices and recirculation discipline.

Wastewater characterization

A defensible near-term characterization is to treat magnet-machining coolant wastewater as an oily, metal-bearing wastewater with substantial solids (swarf/abrasives) and variable oil content, rather than as a single uniform stream with fixed dissolved-metal concentrations. The most operationally relevant descriptors for treatment selection and permitting are (a) the emulsion/oil fraction, (b) suspended solids and fine metallic particles, and (c) pH and discharge acceptability for sewer or surface-water pathways.

From the available sources, key measurable characteristics and constraints include:

- **pH envelope for sewer acceptability**: one guidance document notes spent cutting fluids are generally acceptable for discharge to municipal sanitary sewers when pH is between 6.0 and 9.0 (subject to local approval).
- **Oil limits**: the same source gives an example acceptance criterion that tramp oil concentrations do not exceed 100 mg/L for discharge to municipal sanitary sewers (again subject to local sewer authority).

- **Analytical comparators**: an industrial coolant disposal resource provides an example table of measured results and "legal limits" for several parameters, including fats/oil/grease and total petroleum hydrocarbons, and TCLP metals comparators (e.g., total chromium result 0.008 mg/L with a 5.0 mg/L comparator).

For rare-earth magnet grinding specifically, the "slurry" is not only wastewater; it is a multi-phase mixture of metallic swarf, abrasives from the grinding media, and coolant, with remaining coolant content varying widely depending on separation effectiveness. In one study of NdFeB grinding slurry handling, up to 5% of the magnetic alloy can end up in the slurry sink, which directly links tighter machining and finishing requirements to higher metal-bearing waste generation.

The provided sources do not supply reliable, defense-magnet-specific dissolved concentration ranges for Nd/Pr/Dy/Sm/Co/Fe in coolant wastewater, and one review explicitly notes it does not address wastewater generation rates from machining facilities or grinding coolant usage, reinforcing the need for site sampling rather than generic concentration assumptions.

Current cost baseline

Cost drivers The "cost of doing nothing" is typically a combination of coolant purchase/replacement, offsite hauling and disposal (especially if classified hazardous), minimum pickup charges, and recurring compliance/testing burdens associated with hazardous classification and discharge approvals. Coolant itself can be expensive (reported up to $50/gal in some contexts), which increases the economic value of extending coolant life through recycling or separation-based maintenance programs.

Disposal costs vary widely by location, waste type, and packaging (bulk vs drum), and sources report a broad range of disposal prices from $0.25/gal to $6/gal and beyond (including higher "expensive" categories). One industry resource summarizes offsite hazardous disposal as $0.50–$3.00/gal ($130–$800/m³), and user discussions show that small-volume drum pickups can cost on the order of ~$490 per 55-gal drum (~$8.9/gal).

Baseline cost range Using the commonly cited $130–$800/m³ offsite hazardous disposal range, annual offsite disposal cost scales linearly with wastewater volume and can dominate total operating cost at mid-sized machining sites. The table below shows the implied annual hauling/disposal spend for two representative facility scales (excluding internal labor, manifests, sampling, and minimum pickup charges).

Facility scale	Wastewater volume assumption	Annual volume	Offsite disposal cost range	Implied annual disposal spend
Scenario A	5 m³/day	1,825 m³/yr	$130–$800/m³	~$237k–$1.46M/yr
Scenario B	20 m³/day	7,300 m³/yr	$130–$800/m³	~$949k–$5.84M/yr

These ranges are consistent with the general observation that disposal costs depend strongly on jurisdiction and materials processed, and that high-cost regions can see materially higher per-gallon charges.

Proposed system

A defensible modular treatment concept for a defense magnet machining facility pairs well-established physical separation with an Algamol algae-based step that is positioned to support metals capture/recovery from wastewater,

with a final pH/discharge polishing step for sewer or reuse pathways where permitted. This modularization is important because multiple sources highlight that basic requirements often begin with pH neutralization, and that local sewer utilities must be consulted before discharge of metalworking liquids.

Process flow A practical process flow consistent with the source evidence is:

1. **Equalization and oil/solids management** (tank + skimming/filtration), to dampen variability and remove free/tramp oils and large solids before membranes or biosorption.
2. **Ultrafiltration pre-treatment** to break emulsions and remove fine suspended solids, producing a treated aqueous phase plus a concentrated oily phase that can be sent for oil recovery or incineration.
3. **Algamol bioreactor module** as the primary metal-capture step; Algamol states its technology supports the recovery of rare earth metals from wastewater, while broader literature indicates algae-based REE biosorption is promising but still early-stage for real wastes and generally benefits from mildly acidic operating conditions (often pH 3–6, with a commonly reported optimum around pH 5).
4. **Polishing and discharge conditioning** (pH adjustment and final filtration) to meet site-specific discharge requirements (e.g., sewer acceptance pH 6–9, tramp oil targets).
5. **Sludge handling** to dewater and manage captured solids; depending on composition and TCLP results, sludge may be disposed as hazardous or non-hazardous, with very different unit costs.

This approach aligns with the observation that precipitation/neutralization plus settling is a standard metals-removal paradigm for aqueous metal-bearing streams, even if the ultimate objective is recovery rather than disposal.

Capital cost estimate

On the low end, small coolant ultrafiltration units are reported at ~$5,000–$13,000 for equipment sized to ~100–300 gallons/day, which can support pilots, point-of-use recycling, or small shops but is typically insufficient as the sole treatment method for larger defense machining facilities generating multi-m^3/day streams.

For a more comprehensive modular on-site system (UF + bioreactor + pH dosing + controls/installation), the business-case inputs cite an estimated total CAPEX of ~$110k–$310k, with an illustrative component breakdown of UF membrane unit $30k–$80k, bioreactor module $50k–$150k, pH adjustment/chemical dosing $10k–$30k, and controls/piping/installation $20k–$50k.

These figures are directionally consistent with broader industrial wastewater guidance that places small industrial system CAPEX in the ~$50k–$500k range depending on complexity and customization, and with vendor commentary that system upgrades can require investments such as ~$65k–$200k and that complex multi-stage systems can reach ~$400k to >$1.5M depending on volume and requirements.

Operating cost estimate

The business-case inputs provide an estimated annual OPEX range of ~$28k–$73k/year, with the cost drivers itemized as chemicals (NaOH/acid/nutrients) $5k–$15k/year, membrane replacement $5k–$10k/year, energy $3k–$8k/year, labor (part-time operator) $10k–$25k/year, and sludge disposal/processing $5k–$15k/year.

Independent sources reinforce why OPEX is often the dominant cost driver over lifecycle horizons; for example, one wastewater treatment provider notes that operating costs (including disposal/recycling, electricity, staff/analytics/maintenance) can materially exceed the initial investment over time, and that disposal and recycling alone can represent a large fraction of overall lifecycle cost.

Where discharge to a POTW is feasible after treatment, unit "discharge" costs can be far lower than hazardous hauling; an EPA unit-cost compendium lists discharge to POTW (sewer) at $1.50 per 1,000 gallons (exclusive of the facility's pretreatment costs and any local surcharges), which is roughly $0.40/m³. However, the same compendium also lists a cost for trucking offsite to POTW management at $0.07/gal (excluding transportation), illustrating that logistics and packaging can still impose meaningful costs even when the endpoint is a municipal facility rather than hazardous disposal.

Revenue streams

The most defensible "revenue" mechanism in the near-term business case is **avoided disposal**, because offsite hazardous disposal costs are directly cited and can be very large at multi-m³/day volumes. Additional potential value can come from **recovered metals**, but the evidence base specific to "magnet machining coolant" recovery is currently thinner than for solid swarf and grinding slurries, so metal-recovery revenues should be treated as upside dependent on sampling and downstream refining partnerships.

Still, multiple sources support the plausibility of metal value capture from magnet-related wastes:

- **Cobalt value from SmCo streams**: the business-case inputs cite an indicative cobalt value of $24–$34/kg for recovered cobalt from SmCo streams, which frames the potential upside when cobalt-rich solids are captured into a concentrated sludge rather than diluted into offsite liquid disposal.
- **Rare earth concentrate value benchmark**: the business-case inputs cite a benchmark value of ~$24k–$30k per "tonne of equivalent swarf" for recovered REE concentrate, which can be used as a planning proxy for high-REE solids streams if Algamol's biosludge can be routed to a recycler/refiner.
- **Grinding slurry as an REE-bearing waste**: a NdFeB grinding slurry study notes that grinding slurries are mixtures of swarf/abrasives/coolant and that up to 5% of alloy can end up in this sink, implying a direct feedstock for recovery if captured and segregated.
- **Commercial feasibility of high recovery/purity in related routes**: literature summaries report recovery methods such as membrane-assisted solvent extraction at ~95% REE recovery and molten salt electrolysis producing neodymium at 99.78% purity, indicating that high recovery and high purity are feasible somewhere in the downstream chain even if not performed on-site at the machine shop.

ROI and payback analysis

This ROI model intentionally separates (A) "avoided disposal" economics (typically bankable) from (B) "recovered metals" economics (typically upside), because many sources emphasize classification and local regulatory context as the decisive cost drivers rather than the theoretical value of dilute dissolved metals.

Scenario A For a facility generating 5 m³/day (1,825 m³/year), the implied annual offsite disposal spend at $130–$800/m³ is ~$237k–$1.46M/year. If Algamol's modular system is implemented with annual OPEX ~$28k–$73k/year, the annual "avoided disposal minus treatment OPEX" benefit is approximately ~$164k–$1.43M/year (ignoring residual offsite costs for any unrecovered concentrate and ignoring potential metal revenues). With CAPEX ~$110k–$310k, simple payback is therefore approximately ~0.1–1.9 years across the disposal-cost range, which is consistent with third-party wastewater industry statements that some clients see ROIs as fast as ~1½ years and that typical ranges can be ~3–4 years depending on context and system scope.

Scenario B For a facility generating 20 m³/day (7,300 m³/year), the implied annual offsite disposal spend at $130–$800/m³ is ~$949k–$5.84M/year. Using the same OPEX estimate ($28k–$73k/year) and CAPEX ($110k–$310k), simple

paybacks compress to weeks-to-months in the high hauling-cost case and remain well below 1 year in most cases, even before any recovery crediting, because the baseline is dominated by high unit disposal costs applied to large annual volumes.

Sensitivities The dominant sensitivities that determine real-world payback are (1) whether the waste can be reclassified away from hazardous through proper characterization and segregation, (2) whether on-site recycling can materially reduce disposal volumes (e.g., via ultrafiltration and sump management), and (3) the local jurisdiction's hauling/disposal and sewer acceptance constraints. A qualitative case example illustrates the power of classification and recycling: one client reduced haul-away costs from ~$275–$300 per 55-gallon barrel to ~$90 per barrel by relabeling for recycling, and other sources report >80% disposal-cost reduction after implementing cutting fluid recycling systems, demonstrating that the "disposal avoidance" lever can be very large in practice.

Strategic value proposition

Beyond direct cost savings, Algamol can position a defense-magnet machining wastewater solution around three strategic dimensions that are evidence-supported in the source set.

First, **supply chain and critical-minerals circularity** creates a strategic rationale to keep REE- and cobalt-bearing residues on-site or in domestic recycling loops; defense supply chain policy is increasingly focused on magnet and rare earth supply vulnerabilities, and DoD has invested >$439M since 2020 to establish domestic rare earth element supply chains. A wastewater-to-concentrate pathway that supports downstream recovery can be framed as an enabling step in "mine-to-magnet" resiliency—especially for high-value residues like SmCo-related cobalt-bearing solids.

Second, **technology differentiation via biosorption** is plausible but must be communicated honestly as an emerging pathway: multiple reviews note algae-based or algal-sorbent REE recovery from real wastes is still in its infancy, even though promising lab and pilot performance exists (e.g., reported removal up to 86% at pH 5 in one study, and REE sorption efficiencies in the high-90% range in other cited work). A credible commercialization narrative therefore depends on a staged pilot program with real coolant and sludge samples, consistent with calls for laboratory results to be tested with actual manufacturing wastewater.

Third, **regulatory and permitting risk management** is intrinsically valuable because discharge permits are case-by-case and local sewer authorities require consultation, and hazardous waste determinations create testing and reporting burdens. Even without claiming universal discharge limits, Algamol can offer a compliance-oriented value proposition centered on pH management and oil/solids control to align with commonly cited sewer acceptance criteria (pH 6–9; tramp oil ≤100 mg/L) and to reduce hazardous-waste designation risk through segregation and pre-treatment.

Go-to-market recommendations

A defensible go-to-market plan in the defense magnet machining niche should be structured as a pilot-to-scale program that converts "unknown" site-specific chemistry into bankable economic and compliance outcomes. Near-term recommended steps include:

- **Prioritize pilots at named defense-relevant targets** (EEC/Permag, Arnold, VAC Sumter) to access real streams and establish security/compliance operating procedures early.
- **Instrument the pilot for cost and compliance outcomes**, including routine sampling and metals panels; published unit cost compendia provide representative costs for sampling and metals analyses (e.g., ~$36–$45 per sample and ~$89 for RCRA metals 6010; ~$171 for a "priority 17 metals" panel).

- **Design for upstream separation first**, because ultrafiltration is a well-established method in coolant management and produces a concentrated oil phase that is typically managed via oil recovery or incineration, reducing the load to downstream biological or adsorption steps.
- **Commercialize with a TCO-first offer**, aligning with industry messaging that hidden and ongoing costs can exceed equipment price in a few years and that the most economical solution is not always the cheapest upfront, which supports lease/service models with performance guarantees tied to disposal avoidance.

Implementation gaps

Several gaps in the available evidence base matter directly for Algamol's technical de-risking and the credibility of a defense-sector business case.

The most material gap is a lack of **published, defense-magnet-machining-specific coolant wastewater composition data**, particularly dissolved Nd/Pr/Dy/Sm/Co/Fe concentrations, which prevents a rigorous bottom-up recovered-metal revenue forecast from wastewater alone. A second gap is the limited availability of **industrial-scale case studies** for algae-based REE recovery on real manufacturing effluents; the literature contains promising lab and pilot results, but also emphasizes the early maturity of algal sorbents for real wastes and the need for industrial validation.

Finally, the waste form often most relevant to rare earth value capture may be **grinding slurries and solid residues**, not the aqueous phase alone; one European-focused study notes that there is no established industry for recycling NdFeB grinding slurries in Europe and that slurries are often stored and exported to China or disposed as hazardous due to oily components, emphasizing both a market gap and a potential domestic opportunity for integrated capture and recovery solutions. A pilot program should therefore explicitly treat slurry management (segregation, dewatering, and routing to recyclers) as a core part of the value proposition, rather than positioning the solution solely as a water discharge compliance tool.

PART III Regulatory landscape facility models and market strategy

Chapter 8 Consolidated executive briefing

Strategic Briefing High Temperature Rare Earth Permanent Magnets in the Defense Sector Key Findings and Algamol Market Opportunity

1. The Defense Magnet Market Rare-earth permanent magnets are enabling components across major defense platforms and subsystems, with DoD explicitly citing their importance across aircraft, submarines, and UAVs. The scale of rare-earth usage at the platform level is substantial; DoD cites that an F-35 requires more than 900 lb of rare earth elements (REEs), an Arleigh Burke DDG-51 destroyer about 5,200 lb, and a Virginia-class submarine about 9,200 lb, which underscores why magnet availability and provenance have become procurement-critical.

For high-temperature, high-reliability defense environments (roughly 150–550℃), the source base points to two primary material families. Samarium-cobalt (SmCo5 and Sm2Co17) magnets are repeatedly described as offering strong hard-magnetic properties, thermal stability, and corrosion resistance, and as preferred for high-temperature applications (e.g., 200–550℃). High-temperature NdFeB is positioned as viable mainly by coercivity-engineering: heavy rare earth (HREE) additions (Dy/Tb) to prevent demagnetization around 150–200℃, plus microstructure engineering such as grain-boundary diffusion (GBD) to raise coercivity without bulk HREE loading.

Quantitatively, SmCo magnets are reported to span roughly 16–33 MGOe in (BH)max across grades and processing conditions. A specific Sm2Co17-grade example (EEC 2:17-27) is reported at (BH)max 26–28 MGOe with intrinsic coercivity 34.1–36.1 kOe and Br 10.5–10.8 kGs, which is directionally consistent with the "high-temperature" use case emphasized for SmCo in defense subsystems. Commercial SmCo grade data (e.g., VACOMAX) also explicitly cites (BH)max in the ~20–33 MGOe range and maximum operating temperature up to ~350℃ for some alloys, providing a practical "catalog anchor" for high-temperature procurement and qualification decisions.

On the supply side, the public record emphasizes structural concentration risk in upstream REE supply chains. GAO summarizes rare-earth supply chains as global and multi-tiered, with China dominating mining and processing stages, and with many supply-chain nodes between ore and delivered defense components. Related policy/legal commentary also quantifies China's position as ~60% of rare earth mining and ~90% of rare earth processing (2023), which is a key driver of Western "re-shoring" and recycling push strategies.

A small set of defense-relevant suppliers recur across the sources for "defense-grade" magnets and assemblies. Examples include Electron Energy Corporation (EEC), which states it purchases pure raw materials (Sm, Co, Fe, Cu, Zr) for in-house production of SmCo alloys and is ITAR/DDTC registered with AS9100D/ISO 9001 certification, and Arnold, which cites ITAR-registered supply chain for its SmCo RECOMA products and lists AS9100/ISO 9001 certifications, signaling a defense-ready compliance posture. VACUUMSCHMELZE is positioned in the source set as a leading global manufacturer of advanced magnetic materials and as the source of VACOMAX SmCo material and downstream processing capabilities including wire-EDM, which matters because machining and tight tolerances are a central cost/yield driver for defense magnets.

2. The Waste Problem The machining stage is consistently described as the dominant driver of material loss and waste generation for sintered rare-earth magnets, because sintered magnets cannot be produced directly to final, precise shapes and are therefore made as bulk blocks that must undergo extensive machining and grinding to meet final geometry and tolerances. One public U.S. report states machining recovery rates of 60–90% depending on size/complexity, implying 10–40% of material becomes machining waste in many cases, and explicitly identifies machining as the step with the greatest material loss.

Across the broader magnet manufacturing literature, reported waste fractions can be even higher for certain shapes: multiple sources cite 6–73% of raw material becoming swarf depending on final magnet geometry, while "typical" NdFeB cutting and grinding scrap is frequently cited at ~20–30 wt% of starting alloy in manufacturing practice. Waste is not a minor side stream: one analysis estimates the swarf recycling stream in 2020 was ~9% of global NdFeB magnet production, indicating swarf-handling and recycling logistics are already a non-trivial industrial system.

Waste form also matters. A review focused on magnet waste states that more than 70% of manufacturing waste consists of sludge that is heavily oxidized and contaminated with carbon from lubrication, which directly complicates recovery and creates a wastewater and sludge-handling problem rather than a clean "metal scrap" problem. Additionally, magnet cuttings/swarf can contain machine cutting fluids and magnetic powder, and cutting is performed before magnetization, which has implications for handling (magnetic vs non-magnetic), contamination control, and downstream separations.

SmCo machining has an additional yield penalty because Sm-Co magnets are brittle. Sources attribute production loss up to 20–30% in volume to brittleness, and separately note Sm-Co machining failure rates up to ~20%. This brittleness-driven fracture risk is observable at the part level; one study reported 1 of 7 reference SmCo magnets (~14.3%) fractured during wire-EDM preparation, illustrating why high-reliability, tight-tolerance programs can see compounding scrap and rework costs in finishing operations.

The "value leaking" through this waste stream is amplified in high-temperature/high-coercivity NdFeB because Dy/Tb are expensive, and their use scales with temperature requirements. Sources note Dy/Tb additions are used to prevent demagnetization in the 150–200℃ regime, and non-defense examples report NdFeB magnets can contain substantial HREE additions (e.g., "about 10 wt% Dy/Tb" in certain high-temperature motor contexts) and "up to 11 wt% dysprosium" cited in another high-temperature application context, indicating the magnitude of the value-at-risk can rise substantially above baseline compositions in the highest-temperature NdFeB grades.

3. **The Regulatory Tailwind** The procurement environment is hardening around magnet provenance, with a single date acting as a forcing function for defense primes. GAO reports DoD completed DFARS rulemaking for the revised prohibition and that the expanded prohibition will go into effect January 1, 2027, and also flags that DoD will have to require contractors to disclose magnet sources in DoD systems. The DFARS final rule published May 30, 2024 is effective and sets the staged restriction framework in the DFARS clause; sources explicitly cite the rule's May 30, 2024 effective date and the January 1, 2027 expansion that prohibits covered materials mined/refined/separated/melted/produced in covered countries.

A key nuance for strategy is that DFARS covers both SmCo and NdFeB as "covered materials," and that the 2027 expansion explicitly extends to the full upstream supply chain for both magnet types, not only final magnet manufacturing steps. DFARS also contemplates exceptions via nonavailability determinations, and the Federal Register text states DoD can issue individual and class nonavailability determinations if compliant materials are unavailable (DFARS 225.7018-4).

Recycling becomes strategically important because DFARS provides a specific compliance pathway for NdFeB magnets manufactured from recycled material if (and only if) the milling of recycled material and the sintering of the final magnet occur in the United States. This "recycled NdFeB" pathway is a direct demand driver for U.S.-based recycling and refining capacity and increases the strategic value of domestically captured and traceably processed manufacturing scrap and end-of-life magnet feedstock.

The DoD response to upstream concentration risk is not only regulatory. GAO reports that since 2020, DoD has awarded approximately $439 million to vendors to reestablish domestic rare earth supply chains, and DoD messaging emphasizes the importance of keeping critical supply-chain nodes in the United States to secure magnet supply.

In Europe, the policy direction is similar in objectives but different in mechanism. The EU's Critical Raw Materials Act (CRMA) sets 2030 benchmarks to reduce dependence on third countries, including targets of at least 10% extraction, 40% processing, and 25% recycling capacity (of EU annual consumption) for strategic raw materials, and a diversification target to avoid >65% reliance on any single third country at any processing stage. For permanent magnets, CRMA includes labeling and recycled-content disclosure mechanisms for certain products, including requirements to indicate magnet type (NdFeB, SmCo, etc.) and disclosure of the share of key elements recovered from post-consumer waste by May 24, 2027, alongside a pathway for minimum recycled content rules by end-2031; however, the CRMA also explicitly exempts products primarily designed for defense or space applications from these Article requirements, so defense procurement effects may be more indirect (industrial capacity and recycling ecosystem) than direct contract flowdown in many cases.

EU dual-use export controls can further shape availability and cross-border logistics for high-performance magnets. Regulation (EU) 2021/821 establishes an export-control regime for dual-use items, requires export authorization for items listed in Annex I, and requires exporters to keep detailed records for at least five years, which increases compliance friction and can tighten the practical availability of certain high-performance magnet products for international programs.

4. The Wastewater Opportunity The "machining waste" problem is also a wastewater and coolant-management problem, because grinding/cutting slurries are mixtures of metallic swarf, grinding-media abrasion, and a relevant amount of cooling agent, and residual coolant can remain from <1% to >20% by weight even after on-site solid-liquid separation and coolant recirculation. In addition, swarf streams may contain cutting fluids and magnetic powder, which creates a combined solids-and-organics treatment challenge rather than a simple "settle out metal" problem.

From a treatment-chemistry standpoint, the literature indicates that pH-driven precipitation is a workhorse method for metal removal in industrial wastewaters. Sources describe hydroxide precipitation via pH adjustment as widely used for metal-ion removal, and emphasize that pH adjustment is commonly applied to precipitate metal hydroxides and facilitate removal of suspended solids. For REEs specifically, multiple sources converge that REE hydroxide precipitation typically occurs in the alkaline range (roughly pH ~8.6–10, matrix dependent).

Selectivity is typically achieved with staged pH control, especially when iron is abundant (as in NdFeB-related waste streams). Review-level guidance and EPA-oriented sources describe iron removal prior to REE precipitation by adjusting pH to about 3–4 (often cited ~3.5) to precipitate Fe(III) hydroxides/oxides while keeping REEs in solution, followed by raising pH for REE precipitation to avoid co-precipitation losses. This staged approach is directionally consistent with the chemistry reality that magnet-waste-derived solutions and slurries often contain large iron fractions (e.g., NdFeB magnets with ~69% iron by weight in one supply-chain report framing), which otherwise drives high sludge volumes and poor selectivity if treated in a single pH step.

A key operational constraint for machining coolants is organic load (oils, surfactants). A coolant-treatment source notes that water-mix cutting fluids are diluted to create oil-in-water emulsions (2–10% by volume oil in water), and a coolant-focused REE recovery review notes membrane-based pre-concentration is promising but fouling from oils/organics complicates direct membrane use and typically requires pretreatment to remove organics and emulsified oils before membrane/adsorption-based REE recovery is viable.

Within this chemistry and operations envelope, Algamol's stated positioning is that its technology supports recovery of rare earth metals from wastewater, which aligns with the market need for treatment-plus-recovery rather than disposal-only solutions in a DFARS- and CRMA-driven environment. At the same time, the research base cautions that algal sorbent research for REE recovery from real wastes is still in its infancy, and biosorption is commonly preferred under mildly acidic conditions (pH 3–6, with a commonly reported optimum around pH ~5), which matters

for how Algamol is integrated with staged precipitation and oil-removal steps (i.e., it is unlikely to be a "single-box" solution).

Economically, the strongest initial ROI lever is disposal avoidance. Offsite hazardous waste disposal is commonly cited at $0.50–$3.00/gal (about $130–$800/m³), with additional minimum pickup charges and substantial variability by site and classification. On the cost side of the solution, the compiled Algamol deployment inputs cite a total installed CAPEX estimate of $110k–$310k and annual OPEX components (chemicals, membrane replacement, energy, labor, sludge handling) that sum to about $28k–$73k per year.

The table below shows a simple disposal-avoidance payback framing using only cited disposal ranges and cited CAPEX/OPEX ranges (excluding potential recovered-metal revenues and excluding residual offsite disposal for concentrated residues).

Scenario	Wastewater volume	Disposal cost range	Annual disposal baseline	Annual treatment OPEX	CAPEX range	Simple payback estimate
Small facility	5 m³/day	$130–$800/m³	~$237k–$1.46M/yr (calc.)	~$28k–$73k/yr	$110k–$310k	~0.1–1.9 years (calc.)
Medium facility	20 m³/day	$130–$800/m³	~$949k–$5.84M/yr (calc.)	~$28k–$73k/yr	$110k–$310k	<1 year in most cases (calc.)

This framing is consistent with broader industry narratives that wastewater projects can achieve rapid ROI when hauling/disposal is the baseline, including claims of sub-year payback for certain evaporation systems versus hauling and typical ROIs in the ~1.5–4 year range for broader industrial wastewater investments (noting that these are general benchmarks and site-specific).

5. Go to Market The near-term commercial opportunity is to target magnet suppliers and machining houses where (a) high-value rare earth and cobalt material is being converted into contaminated sludge, and (b) DFARS-driven provenance and domestic-processing incentives increase the strategic value of keeping these byproducts controlled, characterized, and routed to compliant downstream recovery.

A practical first wave of targets named in the compiled business-case inputs includes EEC/Permag (Landisville, PA), Arnold (multiple sites), and VAC (Sumter, SC). These targets align with the supplier landscape in which EEC is ITAR/DDTC registered and sources/purchases raw materials for in-house SmCo alloy production, and Arnold states ITAR-registered supply chain locations for its SmCo RECOMA products and defense-focused quality system certifications.

The recommended commercialization path is a pilot-to-scale program that uses on-site sampling and controlled demonstration to convert uncertain wastewater composition into bankable operating parameters. This is supported by both (i) the general industrial wastewater guidance that field tests and sometimes pilot tests are important and can be coordinated with a reputable water-treatment firm, and (ii) the biosorption literature's explicit caution that lab-scale results need to be advanced and tested with actual wastewater samples from manufacturing sources.

From a system-architecture perspective, the best-supported "minimal viable product" is an integrated train that begins with pretreatment for oils/solids (to protect downstream concentration/capture steps) and then applies staged

chemistry and/or biosorption where it has a plausible operating window. This approach is consistent with (a) coolant pretreatment requirements before membrane/adsorption-based REE recovery and (b) the practical staged precipitation concept (Fe at ~pH 3–4; REE hydroxides at ~pH 8.6–10) for selectivity and sludge reduction.

6. Key Risks and Unknowns The largest technical and commercial unknown is that published data are still thin on site-specific coolant wastewater chemistry for defense magnet machining, particularly for dissolved Nd/Pr/Dy/Sm/Co concentration ranges in real coolant loops and the partitioning between aqueous phase, fines, and oily sludge under different coolant chemistries and separation practices. This matters because the revenue upside (metals recovery from wastewater) depends on actual loading and partitioning, while the cost downside (treatability, sludge disposal) depends on emulsion stability, organics, and contaminants, which are known to complicate REE recovery from real machining slurries.

A second risk is technology maturity for algae-based capture in real industrial matrices. The literature states research into algal sorbents for REE recovery from real wastes is still in its infancy, and this must be reflected in commercial claims and risk allocation (e.g., performance guarantees should initially be based on disposal reduction and compliance metrics rather than recovered-metal credits).

A third risk is regulatory and procurement interpretation drift as DFARS source-disclosure requirements are implemented. GAO indicates DoD is moving toward requiring contractors to disclose magnet sources, and DFARS-related implementation for provenance disclosure is expected between 2025 and 2027, which means supplier questionnaires and traceability expectations may tighten during the same period that primes are requalifying suppliers for the 2027 expanded prohibition.

Finally, the supply-chain transition remains schedule-critical. GAO notes supply chains are multi-tier and China-dominated, and DoD has assessed it can take a decade or longer to establish new domestic sources for rare earths, which implies a non-trivial risk that capacity and qualification lag the January 1, 2027 prohibition expansion for certain niche high-temperature grades unless recycling and allied sourcing pathways scale quickly.

Near-term diligence priorities therefore center on actions that directly de-risk both technical performance and investor-underwritable economics:

- A defensible sampling plan for coolant, sludge, and swarf streams (including oil content, pH, solids, and metals) to bound treatability and confirm what fraction of rare earth value is in the solid phase versus aqueous phase.
- A pilot design that explicitly includes oil/solids pretreatment before any membrane/adsorption/biosorption steps, consistent with coolant fouling constraints noted in the literature.
- A commercialization narrative tied to DFARS 2027 and the "recycled NdFeB milled and sintered in the U.S." exception, because those rules create a premium for domestic, traceable processing of magnet manufacturing byproducts into compliant feedstock pathways.

Chapter 9 EU defense procurement rules

EU Defense Procurement Rules for Rare Earth Permanent Magnets REACH Dual Use Controls and Critical Raw Materials Act

Executive Summary

The European Union's regulatory framework for rare earth permanent magnets in the defense sector is a multi-layered system designed to enhance strategic autonomy and de-risk supply chains heavily dependent on China. Unlike the direct procurement prohibitions found in the US Defense Federal Acquisition Regulation Supplement (DFARS), the EU's approach is centered on industrial policy, capacity building, and market-shaping incentives. Key

regulations include the European Critical Raw Materials Act (CRMA), which sets ambitious 2030 benchmarks for domestic extraction, processing, and recycling; the REACH regulation, which governs the use of chemical substances like cobalt; and the Dual-Use Export Control Regulation (2021/821), which manages the export of sensitive technologies including high-performance magnets.

This regulatory suite is complemented by the European Defence Industrial Strategy (EDIS) and funding instruments like the European Defence Fund (EDF) and the European Defence Industry Programme (EDIP), which aim to bolster the European Defence Technological and Industrial Base (EDTIB) and encourage intra-EU sourcing. While the EU's legislative framework is less defense-specific than that of the US, its overarching goal is to mitigate geopolitical supply risks and foster a resilient domestic value chain for materials critical to defense, aerospace, and the green transition. For technology providers in the waste treatment and material recovery sector, these regulations—particularly the CRMA's aggressive recycling targets—create a significant market opportunity by establishing both a regulatory mandate and a strategic premium for recycled rare earth materials sourced within the Union.

European Critical Raw Materials Act

Adopted by the Council in March 2024 and entering into force in May 2024, the European Critical Raw Materials Act (CRMA), designated as Regulation (EU) 2024/1252, is the cornerstone of the EU's strategy to secure a sustainable supply of materials essential for its strategic sectors, including defense and aerospace. The Act explicitly identifies "rare earth elements for permanent magnets (Nd, Pr, Tb, Dy, Gd, Sm, and Ce)" as "strategic raw materials," placing them at the center of EU industrial policy.

The CRMA establishes clear quantitative benchmarks to be achieved by 2030 to reduce the Union's external dependencies:

- **EU Extraction**: At least 10% of the EU's annual consumption from domestic extraction.
- **EU Processing**: At least 40% of the EU's annual consumption from domestic processing.
- **EU Recycling**: At least 25% of the EU's annual consumption from domestic recycling.
- **Diversification**: No more than 65% of the Union's annual consumption of each strategic raw material at any processing stage should come from a single third country.

To support these goals, the CRMA introduces "Strategic Projects" that can benefit from streamlined and predictable permitting timelines (e.g., 15 months for processing and recycling projects) and support in accessing finance.

For permanent magnets specifically, the CRMA introduces circularity measures. It mandates that certain products (e.g., MRI devices, motor vehicles, industrial robots) bear a label indicating the presence and type of permanent magnets (NdFeB, SmCo, etc.). Furthermore, it requires a "data carrier" (akin to a product passport) with detailed information on the magnet's composition to aid recyclers. By May 2027, companies must publicly disclose the share of recycled content (Nd, Dy, Pr, Tb, B, Sm, Ni, Co) in their magnets, and the Commission is empowered to set minimum recycled content shares by 31 December 2031. Crucially, products primarily designed for defense or space applications are explicitly exempt from these labeling and recycled content disclosure requirements.

REACH Regulation and Rare Earth Substances

The EU's regulation on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), Regulation (EC) No 1907/2006, governs the use of chemical substances and their potential impacts on human health and the environment. This framework intersects with the high-temperature magnet supply chain, particularly concerning cobalt, a key constituent of SmCo magnets.

Cobalt and its compounds have been under scrutiny within REACH, with some being placed on the Candidate List of substances of very high concern (SVHC). This can trigger obligations for suppliers and may lead to future restrictions. The CRMA acknowledges this intersection, stating that the Commission will consider the "specific operational realities of the extractive, recycling and processing sectors" in any revision of REACH to ensure policy coherence. The regulatory landscape also includes restrictions on other processing agents like boric acid and rules on workplace exposure, which affect magnet manufacturing processes. For defense applications, compliance with REACH is mandatory, and any future restrictions on cobalt or rare earth compounds could significantly impact the availability and cost of qualified magnetic materials.

Dual Use Export Controls

The export of high-performance permanent magnets from the EU is governed by Regulation (EU) 2021/821, which establishes a Union-wide regime for the control of "dual-use items"—goods, software, and technology that can be used for both civil and military purposes. An export authorisation is required for any items listed in Annex I of the regulation.

This regulation is directly relevant to high-performance samarium-cobalt (SmCo) and high-coercivity neodymium-iron-boron (NdFeB) magnets, which have critical applications in military systems like guided munitions, drones, and aerospace actuators. The control lists can be based on technical parameters, such as a magnet's maximum energy product (BHmax), which determines its strength. The Commission is empowered to update these control lists via delegated acts to reflect changes in international export control regimes like the Wassenaar Arrangement. Exporters of controlled magnets must navigate a complex licensing process, which may require individual, per-order authorizations and detailed end-use statements, and maintain detailed records for at least five years.

EU Defence Procurement Directive

Directive 2009/81/EC sets out specific EU-wide rules for the procurement of arms, munitions, and sensitive security equipment. A key objective of the directive is to balance the principles of the internal market with the security interests of Member States, including the security of supply. The directive allows contracting authorities to impose requirements related to security of supply in tender documents, which can be used to favor more resilient and local supply chains.

While the directive aims to create a more open and competitive European defense market, Member States frequently invoke Article 346 of the Treaty on the Functioning of the European Union (TFEU) to exempt certain defense procurements from EU rules, citing the protection of essential national security interests. The European Defence Industrial Strategy (EDIS) and related instruments like EDIRPA aim to encourage more coordinated procurement and strengthen the application of EU rules to bolster the European Defence Technological and Industrial Base (EDTIB). As part of the RESourceEU action plan, the Commission is exploring revisions to the Defence Procurement Directive to include incentives for diversifying critical raw material supplies.

EU Defence Industrial Strategy

The EU's industrial strategy for defense is centered on achieving "strategic autonomy" and de-risking from geopolitical dependencies, particularly on China for critical raw materials and components. This strategy is being implemented through a combination of policy initiatives and funding instruments.

The **European Defence Industrial Strategy (EDIS)** and the **European Defence Industry Programme (EDIP)** aim to enhance the readiness and resilience of the EDTIB. A key proposal invites Member States to procure at least 50% of their defense investments within the EU by 2030, rising to 60% by 2035. Instruments like the **European Defence**

Industry Reinforcement through common Procurement Act (EDIRPA), with a budget of €300 million until 2025, introduce strict sourcing requirements. For a project to receive support, contractors must be established in the EU and not controlled by a third country, and the cost of components originating in the Union must be no lower than 65% of the end product's value.

The **European Defence Fund (EDF)**, with a budget of over €1 billion per year (2021-2027), funds collaborative research and development to foster innovation within the EDTIB. These programs are designed to work in synergy with the CRMA, leveraging defense spending to create a stable demand for domestically produced critical materials like rare earth magnets.

European Rare Earth Magnet Production Capacity

The EU's domestic production capacity for high-performance rare earth magnets is limited, representing a significant vulnerability. Current European production is estimated at around 1,000 tonnes per year, compared to approximately 16,000 tonnes of magnets imported from China annually. This means the EU is over 90% dependent on China for these critical components.

The key established producer in Europe is **Vacuumschmelze (VAC)** in Germany, which has been making SmCo magnets since 1973 and NdFeB magnets since 1985. Other notable players include **Less Common Metals (LCM)** in the UK and a **Shin-Etsu** facility in Germany. However, the landscape is evolving. The European Raw Materials Alliance (ERMA) has identified 14 "mine-to-magnet" projects with a potential investment volume of €1.7 billion, aiming to recapture a significant share of global production. New entrants are also emerging, such as **Neo Performance Materials'** commercial magnet plant in Estonia and recycling initiatives by companies like **Solvay** and **Carester** in France. These recyclers could contribute an estimated 3,800 tonnes of magnet production in the coming years if sufficient feedstock is available.

Comparison EU vs US Regulatory Approaches

The EU and the US share the strategic goal of securing rare earth magnet supply chains but employ different primary regulatory mechanisms. The US approach is more direct and defense-focused, while the EU's is broader, more integrated into industrial policy, and relies more on market incentives and benchmarks.

The table below outlines the key differences:

Feature	EU Approach	US Approach
Primary Instrument	**Critical Raw Materials Act (CRMA)**: Sets EU-wide benchmarks for domestic extraction (10%), processing (40%), and recycling (25%) by 2030.	**DFARS 252.225-7052**: Direct procurement prohibition on SmCo and NdFeB magnets and alloys from "covered countries" (China, Russia, etc.).
Core Mechanism	**Industrial Policy and Market Shaping**: Uses targets, streamlined permitting for "Strategic Projects," and future recycled-content mandates to build domestic capacity.	**Procurement Prohibition**: Directly bans the acquisition of non-compliant materials for Department of Defense contracts, creating a protected domestic market.

Feature	EU Approach	US Approach
Timeline	**2030 Benchmarks**: Long-term capacity-building goals. Specific mandates like minimum recycled content are phased in by 2031.	**January 1, 2027 Deadline**: Hard deadline for the expanded prohibition to cover the entire supply chain from mine to magnet.
Recycling	**Central Pillar**: 25% recycling capacity target by 2030; upcoming restrictions on scrap exports to retain feedstock within the EU.	**Specific Exemption**: Allows the use of NdFeB magnets made from recycled material if the milling and sintering steps occur in the US.
Funding Approach	**Coordination of Funds**: Relies on leveraging existing EU funds (EDF, InvestEU) and national programs via a CRM financing hub. No new central fund was created by CRMA.	**Direct Investment**: The DoD has made direct investments (e.g., over $439 million) and equity stakes (e.g., MP Materials) via the Defense Production Act (DPA).
Defense Specific Rules	**Exemptions**: Defense and space products are *exempt* from CRMA's magnet labeling and recycled content disclosure rules. Sourcing rules are in separate defense programs (e.g., EDIRPA).	**Core Focus**: The DFARS rule is *exclusively* for defense procurement and imposes the strictest sourcing requirements on this sector.

Implications for Algamol and Waste Treatment

The EU's comprehensive regulatory framework, particularly the CRMA, creates a fertile market for advanced material recovery and wastewater treatment technologies like Algamol. The key opportunities are driven by several interconnected factors:

1. **Mandated Recycling Capacity**: The CRMA's benchmark for the EU to have the capacity to recycle 25% of its annual strategic raw material consumption by 2030 is a powerful market signal. With current REE recycling rates below 1%, achieving this target requires massive investment in new technologies and facilities, creating demand for efficient and environmentally sound recovery processes.

2. **Restrictions on Scrap Exports**: The Commission's plan to propose restrictions on the export of permanent magnet scrap and waste by Q2 2026 will trap valuable feedstock within the EU. This ensures that domestic recyclers and technology providers will have access to the raw materials needed to build a circular economy, turning a waste stream into a strategic asset.

3. **Future Minimum Recycled Content**: The CRMA empowers the Commission to set minimum recycled content shares for elements like neodymium, dysprosium, and samarium in new permanent magnets by 2031. This will create direct, legally binding demand for high-purity recycled materials that technologies like Algamol could recover from machining sludge and other waste streams.

4. **Strategic Value Premium**: Beyond commodity value, domestically recycled and recovered rare earths will carry a "strategic premium." For defense contractors and other critical industries, using materials recovered within the EU contributes directly to the strategic autonomy and supply chain resilience goals of the CRMA and EDIS. This can be a significant advantage in public procurement, where resilience and security of supply are increasingly important criteria.

5. **REACH Compliance**: As regulations like REACH place increasing scrutiny on substances like cobalt, recovering these materials from waste streams for reuse becomes more economically and regulatorily attractive than disposal or sourcing primary materials from regions with less stringent environmental oversight.

Together, these drivers create a compelling business case for deploying wastewater and material recovery solutions at magnet manufacturing and finishing facilities across the EU. The CRMA's targets transform waste treatment from a cost center into a strategic enabler for the entire European defense and high-tech industrial base.

Chapter 10 EEC Permag facility deployment model

Algamol Deployment Model EEC Permag SmCo Magnet Facility Landisville PA

Summary

Electron Energy Corporation (EEC) is a rare-earth permanent magnet manufacturer headquartered in Landisville, Pennsylvania, with a second listed Lancaster, Pennsylvania address at 300 Running Pump Road, and it has operated since 1970. EEC's product focus includes samarium-cobalt (SmCo) magnets and assemblies that serve high-reliability aerospace and defense applications, including publicly referenced programs such as JSF, F-18, Patriot, Predator, and Navy nuclear programs. EEC describes itself as vertically integrated and documents an in-house SmCo manufacturing chain that includes induction melting of elemental inputs, powder milling to ~3 µm, pressing (axial/transverse/isostatic), sinter/solution/aging heat treatments, and brittle-magnet machining steps such as grinding, slicing, and wire EDM.

From a wastewater and pretreatment perspective, the relevant local context in Lancaster County includes industrial-user permitting and "local limits" under a POTW pretreatment program (e.g., Lancaster Area Sewer Authority's program), which is designed to prevent pollutants that could interfere with the POTW or contaminate biosolids. However, the provided EEC-facing sources do not disclose site-specific wastewater volumes, coolant chemistries, or quantitative scrap/waste generation metrics, which constrains a fully parameterized ROI model and necessitates a measurement-first pilot plan for any on-site treatment and metals recovery concept.

Facility profile

EEC was founded in 1970 and is headquartered in Landisville, PA, with a primary listed address of 924 Links Avenue, Landisville, PA 17538, and an additional Lancaster, PA address at 300 Running Pump Road, Lancaster, PA 17603. The company describes itself as a leading producer of rare earth magnets specializing in SmCo magnet manufacturing and also lists NdFeB and magnet assemblies among its specialties. EEC's defense relevance is supported both by its industry positioning (defense and aerospace markets) and by explicit examples of programs and platforms served in public-facing descriptions (e.g., JSF, F-18, Patriot, Predator, Navy nuclear programs).

EEC facility footprint can be triangulated from multiple company materials that describe a 40,000 sq. ft. facility and the addition of a new ~45,000 sq. ft. facility to house magnet finishing and assembly operations, yielding a combined manufacturing footprint on the order of ~85,000 sq. ft. EEC's current workforce size is presented in public sources in the ~130–150 employee range, indicating a mid-sized specialty manufacturer in a highly regulated supply chain.

Ownership and corporate relationships EEC joined Magnetic Holdings, LLC in 2024, and Magnetic Holdings was subsequently rebranded as Permag, which also includes Dexter Magnetic Technologies and Magnetic Component Engineering (MCE). Separately, an investor profile notes that Tinicum acquired a controlling interest in Permag's parent company (Robert Family Holdings, LLC) in January 2021 and that the parent was renamed TSL Engineered Products, LLC.

SmCo manufacturing operations

EEC explicitly documents an in-house SmCo alloy and magnet production pathway beginning with induction melting, in which it purchases pure raw materials (Sm, Co, Fe, Cu, and Zr) for in-house production of SmCo 1:5 and 2:17 alloys and alloys these metals in an inert gas using an induction melting furnace. EEC also documents that it processes crushed alloy powder via ball or jet milling to reduce average particle size to approximately 3 microns, which is a key input specification for subsequent pressing and sintering steps. Green compacts are formed using axial, transverse, or isostatic presses, aligning with standard powder-metallurgy approaches used for high-performance rare-earth permanent magnets.

EEC describes densification and magnetic-property development via sinter, solution, and aging processes conducted under vacuum or inert gas, which corresponds to the canonical Sm2Co17 ("2:17") processing route requiring solution/aging to develop high coercivity and thermal stability. Downstream, EEC notes that sintered rare earth magnets are brittle and are machined to final dimensions using grinding, slicing, or wire EDM, and it also lists facility machining capabilities including WEDM, Ram EDM, CNC machining, grinders, and slicers. EEC further states that magnetization, stabilization, and testing are performed to satisfy customer requirements, implying that magnetization and verification are integrated into its finishing and quality flow for defense-grade deliverables.

Finished parts and assemblies EEC positions itself as a supplier not only of magnets, but also of assemblies (e.g., rotor assemblies and traveling wave tube assemblies), which is consistent with delivering finished-to-print components and subassemblies into defense/aerospace system BOMs rather than only shipping sintered blocks.

Products and grades

EEC's public materials emphasize multiple SmCo offerings, including a "34 Grade" SmCo and an "Ultra-High Temperature SmCo 2:17" family with operating temperatures up to 550℃. EEC also describes ultra-high temperature SmCo stability in the 400℃ to 550℃ range on its SmCo product materials, reinforcing the high-temperature defense niche for Sm2Co17-based magnet solutions.

The table below compiles the numeric magnetic-property anchors available in EEC's 34 Grade sell sheet.

Parameter	EEC 2:17-34 (Sintered Sm2Co17)	Source
(BH)max	34 MGOe	EEC 34 Grade sell sheet
Br	11.9 kG	EEC 34 Grade sell sheet
Hc	11.1 kOe	EEC 34 Grade sell sheet
iHc (min)	18 kOe	EEC 34 Grade sell sheet
Max operating temperature	300℃	EEC 34 Grade sell sheet
DFARS status	"DFARS compliant"	EEC 34 Grade sell sheet

EEC's own stated application set for SmCo includes high-speed motors, generators and actuators, microwave signal amplification, and aircraft/aerospace assemblies, which map directly onto defense-relevant subsystems such as precision actuation, high-temperature motors/generators, and RF/microwave assemblies used in radar/EW architectures.

Machining operations and waste streams

EEC identifies multiple machining methods used for rare-earth magnets and associated hardware, including grinding, slicing, wire EDM (WEDM), and other precision machining equipment, and it explicitly notes that sintered rare earth

magnets are brittle and therefore machined via these techniques to reach final dimensions. EEC also advertises in-house capabilities for wire EDM and precision machining when describing magnet assembly offerings, supporting the view that EEC performs significant downstream finishing internally rather than outsourcing all final machining.

The available sources do not provide quantitative waste fractions (kerf loss, grinding sludge mass, scrap rates) or coolant chemistry details for EEC's operations, so a deployment model must treat waste characterization as an early-stage data-gathering activity tied to the site's documented machining and finishing process steps.

Waste stream mapping for pilot scoping A practical way to scope wastewater-relevant streams without assuming unreported values is to map them to EEC's stated finishing operations (machining, magnetizing, testing, and shipping) and machining modalities (grinding, slicing, WEDM). For example, a pilot can explicitly segment sample points associated with (1) grinding operations, (2) EDM operations, and (3) post-machining magnetization/testing areas, because these are explicitly present in EEC's documented capabilities and finishing operations.

Wastewater characterization

EEC's sources establish that the company performs brittle-magnet machining (grinding, slicing, and wire EDM) and that it conducts magnetization, stabilization, and testing as part of its manufacturing flow, which are operations that typically drive industrial wastewater and residual streams that fall under local pretreatment programs when discharged to a POTW. For Lancaster County, LASA describes an approved Pretreatment Program, states that it issues Industrial User Permits for significant industrial users, and explains that "Local Limits" apply to industrial and other non-domestic facilities discharging to the treatment facility.

At a permitting-interface level, publicly available Pennsylvania wastewater permitting resources emphasize the role of NPDES permits and associated documentation for identifying treatment processes and discharge limits for local facilities, and they direct users to the linked permit documents for specific POTW discharge limits. As an example of the kind of constraints that appear in local wastewater permits in Lancaster County, an NPDES permit for a Lancaster County WWTP shows a pH effluent range of 6.0 to 9.0 and provides design-capacity context, illustrating the type of effluent constraints that industrial pretreatment programs are designed to protect.

Local regulatory context

Because EEC's manufacturing footprint is located in Lancaster County (Landisville and Lancaster addresses), an on-site wastewater treatment and recovery deployment needs to be designed to align with a POTW pretreatment framework and any applicable industrial-user permitting requirements. LASA's pretreatment program description emphasizes preventing the introduction of pollutants that could interfere with the POTW or contaminate resulting biosolids, which is directly relevant to any process that could generate metal-bearing residuals or sludges that would otherwise be discharged or hauled offsite.

Pennsylvania DEP materials emphasize that permitted-facility reports and individual NPDES permit documents are the mechanism for obtaining permit details, and DEP guidance highlights that permit conditions and discharge limits for industrial wastewater discharges in Landisville are influenced by state procedures and by the local POTW.

System concept

Any algae-based or bio-based treatment concept for EEC should be framed around EEC's documented process reality, namely that the facility performs (a) alloying and sintered magnet production steps and (b) downstream machining, magnetization, stabilization, and testing in-house, all within a regulated defense supply chain with ITAR and quality-management certifications. EEC states it is ITAR compliant and DDTC registered and certified to AS9100D and ISO

9001:2015, which implies that any deployed wastewater system would need to be compatible with controlled-access production areas and auditable quality and compliance systems.

Within those constraints, a deployment concept can be structured as a modular add-on to the existing finishing operations rather than a modification to EEC's alloy/powder/sinter process flow, because EEC's publicly described finishing operations already include machining and magnetization/testing, which are the most plausible touchpoints for wastewater-relevant streams at a magnet manufacturer.

Financial model

A rigorous facility-specific financial model cannot be completed from the provided sources because EEC's public materials do not disclose machining-waste volumes, coolant consumption, wastewater flow rates, or current disposal and treatment costs, even though they do document the presence of machining operations and finishing activities that would create those costs. A defensible approach is therefore to define a parametric model that uses measurable site inputs—such as hours of grinding/WEDM operation, mass of collected solids/residuals, and any applicable industrial-user permit fees or surcharges—rather than attempting to infer costs from unreported values.

To remain aligned with local regulatory reality, the model should also include a compliance cost/benefit component based on industrial-user permitting and local limits under the applicable pretreatment program, because LASA explicitly regulates significant industrial users and sets local limits for industrial dischargers to the POTW.

Strategic value

EEC's strategic value to defense supply chains is rooted in its position as a U.S. SmCo producer with ITAR and DFARS compliance claims and in its vertically integrated capability for SmCo alloy and magnet production, including in-house production of SmCo 1:5 and 2:17 alloys from elemental feedstocks. Public materials also emphasize Permag/EEC's investment in expanding alloying, pressing, and fabrication equipment and "more than doubling" production capacity, which indicates a growth trajectory that can increase the volume and operational importance of downstream finishing activities over time.

For an on-site wastewater treatment and recovery system provider, the strategic engagement thesis is that a defense-critical, vertically integrated SmCo producer with in-house machining and magnetization/testing presents an opportunity to (1) reduce compliance risk under industrial pretreatment/local limits and (2) create a traceable pathway for managing metal-bearing residuals within a controlled, auditable environment consistent with ITAR and AS9100-quality expectations.

Pilot proposal

A pilot program at EEC should be scoped first around (a) EEC's documented finishing operations (machining, magnetization/stabilization/testing) and (b) the local pretreatment program and industrial-user permitting framework, because these are the best-supported operational and regulatory anchors in the provided sources. The pilot should also explicitly include a sampling and analysis plan that links any measured wastewaters or residuals to the specific machining modalities EEC lists (e.g., grinders and WEDM), because these modalities are directly stated in EEC's manufacturing capability descriptions.

To ensure the pilot is implementable, the engagement should incorporate early coordination with the relevant local pretreatment authority, since LASA's program issues Industrial User Permits for significant industrial users and enforces local limits on industrial discharges to its POTW.

Chapter 11 Competitive technology analysis

Conventional Wastewater Treatment Technologies for Rare Earth and Cobalt Recovery in Defense Magnet Machining

Overview

Defense magnet machining and finishing operations can generate wastewater and metal-bearing process liquids that contain rare earth elements (REEs) and, in some cases, cobalt (Co), creating both a compliance burden and a potential recovery opportunity for critical materials. Conventional treatment and recovery approaches for these aqueous streams commonly include chemical precipitation, membrane filtration (typically as pre-treatment and/or concentration), ion exchange, solvent extraction (usually after concentration), and electrochemical approaches for selective removal or direct recovery in certain cases.

Across the conventional methods, the same engineering trade-off appears repeatedly: the most operationally straightforward approaches (notably precipitation) can be cost-effective at large volumes but tend to be non-selective and produce significant sludge requiring downstream handling, whereas the more selective approaches (ion exchange, some membranes, solvent extraction) can be more complex and/or cost-intensive and are often more compatible with lower-flow, cleaner, or more concentrated streams rather than oily emulsions or mixed-metal slurries.

Chemical precipitation

Chemical precipitation is widely described as simple, fast, and low cost, and is frequently cited for high removal performance when conditions are well controlled, making it an attractive first-line treatment for metal-bearing waters and leachates.

Mechanisms and pH windows Precipitation of REEs occurs when sparingly soluble compounds (e.g., hydroxides, fluorides, phosphates, oxalates, carbonates) form and exceed solubility, causing solids that can be separated by settling/filtration.

A common staged-precipitation concept is to drive precipitation in distinct pH windows where different metals remove preferentially, with one summary describing Fe removal at pH ~3–4, REE precipitation at pH ~8–9, and Co precipitation at pH ~9–11.

For oxalate precipitation specifically, multiple sources emphasize that oxalate is among the most effective precipitants for REEs and can work at low pH, with one thermodynamic study explicitly describing oxalate as the "most effective precipitant" with "outstanding precipitation behavior" even at low pH.

Detailed oxalate operating windows and dosing requirements are reported in experimental work: an "optimal pH of 1.25 to 2" is described as tending to maximize RE recovery and precipitate purity, and a pH of 1.5–2 with reaction time >2 h and oxalic acid dosage of 30–40 g/L is described as optimizing REE recovery at ~95% to nearly 100% for individual REEs in one study.

The same oxalate-focused work provides quantitative dose–response behavior at a moderate REE feed concentration (440 mg/L), noting that 10 g/L oxalic acid is a minimum capable of recovery while higher doses (≥20 g/L) significantly increase recovery, and that at pH 1.5 and 120 min, recovery of 95% was achieved at 20 g/L and approached 100% at 30 g/L.

In addition to oxalate, precipitation can be carried out via carbonate and double-sulfate routes depending on upstream chemistry; in one comparative precipitation study (for Nd-Fe-B leachate), oxalic-acid precipitation achieved total

REE precipitation efficiency of 99.94% with total recovery efficiency of 99.44%, while sodium carbonate precipitation achieved 99.93% precipitation efficiency and 99.12% overall recovery after transformation and washing.

For double-sulfate precipitation, the same study reports optimum conditions including a sodium sulfate amount ratio of 1.3:1.0, a reaction temperature of 80 ℃, and reaction time of 1 h, with individual precipitation efficiencies (Ce, Pr, Nd, Gd) around 98.7–98.8% and total REE precipitation efficiency of 98.75%.

Recovery performance and impurity co-precipitation High recovery via oxalate precipitation is repeatedly reported, including a hydrometallurgical study where precipitation of Nd, Pr, Dy, Gd and Ho reached ~99% using only 1.8× the stoichiometric amount of oxalic acid (following HCl leaching).

However, precipitation-based recovery is also frequently described as vulnerable to impurity effects and non-selective co-precipitation, with one study reporting that while >99% REE precipitation could be achieved with oxalate, product purity was "significantly affected by impurities like Fe and Co," and providing an example where Fe and Co content increased to 1.28% and 0.24% in the product, reducing final purity.

The same work notes a mechanistic cause: oxalate ions can complex strongly with dissolved iron, leading to inefficient precipitant utilization and low-purity REE products.

Some precipitation approaches are explicitly positioned to mitigate impurity impacts by changing the chemistry: for REE-enriched sulfate solutions, double-sulfate precipitation is described as achieving effective separation with recovery less affected by iron and cobalt impurities, although high Na2SO4 levels (8× stoichiometric) may be required to achieve >98% REE precipitation, particularly for heavy REEs.

Sequential pH adjustment has been used to exploit "well defined separation of removal pH ranges," with one study adjusting sequentially between pH 3.0 and 8.0 (including steps at pH 4.0, 5.0, and 8.0) and reporting efficient recovery due to these separations; the same study notes that flocculant can increase overall recovery but decrease the REE weight percent in the precipitate (i.e., a grade–recovery trade-off).

Sludge and downstream processing Multiple reviews describe precipitation as potentially producing impure precipitates and additional waste, with "large amounts of slurry" or "large sludge volumes" highlighted as recurring disadvantages that require dewatering, disposal, or further processing of mixed precipitates.

A further limitation is that precipitation can create secondary sludge even when used as part of otherwise high-efficiency hydrometallurgical recovery flowsheets, emphasizing that precipitation can shift the problem from dissolved metals to solid waste management if solids are not valorized into recoverable products.

Membrane filtration

Membrane filtration is often framed as a selective separation tool that can operate continuously and avoid chemical addition, but is also widely described as subject to fouling, maintenance demands, and potentially higher costs than precipitation for comparable bulk removal—especially when streams contain organic/inorganic foulants.

Roles in a treatment train Ultrafiltration (UF) is commonly positioned as pre-treatment for emulsion breaking and solids removal, which is particularly relevant to industrial streams that contain fine particulates or emulsified organics.

Nanofiltration (NF) is commonly presented as a route for partial rejection and concentration of REEs (or other multivalent metals), producing a retentate that may require additional recovery steps.

Example performance metrics Quantitative membrane performance is highly configuration- and chemistry-dependent, but some sources provide numeric examples. One review reports an MEUF/UF configuration (UP020 membrane) achieving 97% REE rejection under conditions including pH 3.5, 60 min contact time at 25 ℃, 3 bar pressure, and SDS concentration of 8.0 mM.

The same review reports NF270 achieving >98% rejection at pH 1 in a tabulated summary, illustrating the high rejection possible in strongly acidic conditions for certain membranes and conditions.

Limitations Membrane filtration is repeatedly described as prone to fouling and scaling that necessitate pre-treatment and periodic cleaning, with high operating costs attributed to fouling and maintenance in multiple sources.

In contexts involving radiation (relevant to some defense/nuclear-adjacent waste streams), polymeric membrane damage is explicitly cited as another potential operating risk and cost driver.

Ion exchange

Ion exchange is frequently positioned as well suited to solutions containing traces of metal ions and as a route to higher selectivity when appropriate resin chemistry is chosen, including the potential for selective separation of individual REEs with selective REE resins.

Selectivity and operating considerations Selectivity in ion exchange is described as dependent on resin choice and operating conditions, with one source noting that resins generally have better selectivity for ions with higher valence/charge and that selective separation of rare earths requires careful selection of resin characteristics (exchange groups, cross-linking, exchange capacity) and operating conditions.

While specific pH windows depend on resin chemistry and feed, one source notes cation exchangers "work at low pH (3–5)," indicating potential compatibility with moderately acidic streams when resin stability permits.

Constraints Despite its selectivity advantages, ion exchange is repeatedly described as limited by resin costs and the need for regeneration, and as susceptible to fouling by organics, especially for complex industrial wastewaters.

Solvent extraction

Solvent extraction is widely used in hydrometallurgy and is often described as a highly selective route for producing high-purity REE products, but is also commonly described as ill-suited for dilute wastewater because it relies on organic solvents and typically becomes more practical after upstream concentration or clarification.

Extractants and example operating configurations Organophosphorus extractants such as D2EHPA, Cyanex grades, and TBP are frequently cited as extractants used for rare earth separations from chloride and nitrate media.

One study explicitly describes a solvent-extraction system used upstream of precipitation, using Elixore 205 diluent with D2EHPA and TBP as extractants at an organic-to-aqueous ratio of 1:1, and stripping the loaded organic with 6 mol/L nitric acid, illustrating the acid-intensive nature of stripping for some systems and the production of concentrated acidic strip solutions that require downstream recovery steps.

A separate review describes solvent extraction as already used for other acidic REE-bearing leachates and as applicable in a wide concentration range compared to lower-concentration methods such as membrane separation or ion

exchange, consistent with its role as a workhorse for concentrated hydrometallurgical streams rather than dilute wastewater polishing.

Performance examples and dependencies Solvent extraction performance can depend strongly on aqueous chemistry (e.g., nitrate concentration), with one review reporting that under an "optimum condition" including 0.1 mol/L D2EHPA and 4 mol/L HNO3 (and feed pH 4.6), Nd3+ extraction was near 92.9%, and that increasing NO3− ions from 2 mol/L to 3 mol/L enabled 99% Nd3+ transport in a membrane-transport context using extractant chemistry (highlighting the importance of nitrate-driven speciation and transport).

Constraints and waste streams Solvent extraction is explicitly associated with organic solvent handling risks and potential VOC emissions and fire hazards, and is described as not suitable for dilute wastewater in one comparative summary, reinforcing its typical placement after pre-concentration rather than as a front-end wastewater treatment method.

Electrochemical recovery

Electrochemical approaches (including electrowinning/electrodeposition and related electro-driven separations) are often described as capable of direct metal recovery without chemical addition, but also as challenged by energy demands and operational limitations for dilute streams and mixed matrices.

Examples and constraints for dilute feeds A review of REE recovery methods reports that capacitive deionization achieved 36.5% Co removal efficiency at an initial cobalt concentration of 5 mg/L, illustrating both applicability to dilute metals and the potential limitation in single-pass removal depending on configuration and kinetics.

For electrochemical methods more broadly, electrode passivation and relatively high energy consumption are noted drawbacks for electrocoagulation, while scale-up challenges and energy consumption barriers are noted for industrial commercialization of some electrical-based methods.

Sludge considerations Some electrochemical modes can reduce or avoid sludge formation relative to chemical precipitation in certain configurations, with electrodeposition described as depositing targeted species on a cathode surface and noting that sludge is not formed in that treatment mode, although other electrical treatments (e.g., electrocoagulation) can increase sludge production.

Comparison

The table below summarizes the principal, source-supported "shape" of each conventional technology family as it is typically applied to REE/Co-bearing aqueous streams, emphasizing what each method produces (sludge, concentrate, eluate, or metal deposit) and the main operational constraints reported in the literature.

Technology	Primary output	Strengths	Common constraints
Chemical precipitation	Metal-bearing sludge/slurry requiring solid–liquid separation	Simple and low CAPEX with high removal often cited (>95%)	Non-selective, mixed precipitates, large sludge volumes requiring further processing/disposal

Technology	Primary output	Strengths	Common constraints
UF and NF membranes	Clarified permeate and metal-rich retentate/concentrate	UF supports emulsion/solids removal; NF can concentrate REEs without chemical addition	Fouling and maintenance costs; may yield concentrate rather than final product; high CAPEX described in some summaries
Ion exchange	Loaded resin and a regenerant eluate (concentrate)	Can be selective and regenerable; suitable for trace ions and can separate individual REEs with selective resins	Expensive resins and regeneration; fouling by organics; limited to low concentrations in some summaries
Solvent extraction	Separated aqueous/organic phases and concentrated strip solutions	Highly selective and established for concentrated hydrometallurgy with high purity	Requires organic solvents (fire/VOC risks) and is described as not suitable for dilute wastewater
Electrochemical recovery	Metal deposits and/or treated water, depending on mode	Direct metal recovery without chemical additions in some approaches	High energy for dilute streams and fouling/operational limits noted; some modes face passivation/energy barriers

Practical implications

From the available evidence, precipitation remains the most "default" conventional approach for bulk removal and recovery of REEs because it is repeatedly described as simple and cost-effective and can achieve high recovery when optimized (including near-quantitative precipitation in some oxalate/carbonate/double-sulfate systems).

The major engineering drawback of precipitation for a recovery-oriented process is the consistent emphasis on non-selectivity and sludge generation, which can create a second-stage separation problem (purifying a mixed precipitate) and a disposal problem if the solids are not further processed into saleable products or refined concentrates.

Membrane filtration, particularly UF, is best supported as an enabling pre-treatment for solids and emulsion removal, while NF can serve as a concentration step with high reported rejection in some conditions (e.g., >98% in acidic conditions in one summary).

Ion exchange is best supported as a selective approach for lower-concentration solutions and for separations requiring higher purity, but resin cost and organic fouling are widely cited constraints that may be critical for complex industrial streams unless upstream clarification and organic removal are effective.

Solvent extraction is strongly supported as a high-selectivity approach for concentrated hydrometallurgical streams using organophosphorus extractants, but multiple sources explicitly flag poor fit for dilute wastewater due to solvent requirements and associated hazards, indicating that it is most appropriately evaluated as a downstream polishing/purification step after concentration (e.g., NF, evaporation, or resin eluate concentration).

Electrochemical recovery shows promise for direct metal recovery in some configurations, but the cited examples and summaries emphasize performance and energy challenges for dilute feeds and operational issues such as passivation, suggesting it is most plausibly deployed either on concentrated streams or for targeted ions where electrochemical selectivity is strong and deposits are readily recovered.

Competitive Technology Analysis for Rare Earth and Cobalt Recovery from Magnet Machining Streams Part 2

Biosorption and bioaccumulation

Biosorption and bioaccumulation rely on the migration of dissolved REEs from water to algal biomass surfaces and/or into cells, followed by a second step that transfers the concentrated REEs back into a recoverable solution (e.g., via desorption or extraction). In practice, this means performance is governed by surface functional groups (cell-wall polysaccharides/proteins, extracellular polymeric substances (EPS), and engineered/modified sites), aqueous speciation, and competing ions that occupy binding sites or alter charge balance.

Reported biosorption performance varies widely by organism, metal, and test conditions, but the literature provides quantitative anchors that are relevant to defense-magnet wastewater scenarios spanning trace (µg/L) through moderate (mg/L to hundreds mg/L) metal concentrations. For neodymium uptake by the microalga *Chlorella vulgaris* at pH 5, batch tests with an initial Nd concentration of 250 mg/L reported maximum Nd uptakes of 126.13 mg/g at 21℃ and 157.40 mg/g at 35℃, with equilibrium reached within 30 minutes, and a Langmuir maximum capacity (qmax) of 188.68 mg/g. In another biological sorbent example, *Physcomitrella patens* demonstrated Nd sorption rates/capacities reported as ~0.74–0.75 mmol/g and showed strong pH dependence with maximum Nd biosorption around pH ~3 and near-constant performance up to pH 6. For cyanobacterial/EPS-derived sorbents, maximum adsorption capacities in the range of 89.12–138.2 mg/g were reported across strains, and EPS-extracted biomass retained substantial fractions of the untreated adsorption capacity (e.g., ~59% to ~84% depending on strain).

Biosorption frequently exhibits an "optimal window" in mildly acidic conditions for many REE systems, consistent with both adsorption site protonation behavior and REE speciation. A review focused on microorganisms and dilute REE feeds notes biosorption is commonly preferred at pH 3–6, with a frequently reported optimum around pH 5. Experimental work with the acidophilic red microalga *Galdieria sulphuraria* showed removal efficiency from solution increasing from below 9% at pH 2.5 to as high as 86% at pH 5.0 in mixed-metal (ppm-level) tests, and also cites prior work reporting >90% REE removal from a 5 ppm solution under highly acidic conditions (pH 1.5–2.5), indicating that both organism choice and nutrient/medium conditions can shift the usable pH window. In seaweed-based biosorption at trace concentrations, living *Ulva* sp. achieved removals up to 88% in 24 hours under optimized conditions (10 µg/L, salinity 15, dosage 5.0 g/L), and ≥90% removal at 96 hours with reduced biomass dosage (4.2 g/L).

Selectivity is a key uncertainty for algae-based processes when applied to mixed-metal machining coolants. For example, *Galdieria sulphuraria* biosorption data showed no particular selectivity towards REEs over Cu in mixed-metal tests, which is directly relevant to real-world coolants and rinse waters that often contain transition metals alongside REEs. Conversely, some literature argues for the potential of EPS and related biological materials as effective adsorbents for separation and recovery from wastewater, but without consistently reporting separation factors in realistic multicomponent streams. Where biosorption is used primarily as a concentration step, downstream recovery is commonly framed as desorption and/or conversion of the biomass to a denser, metal-rich intermediate. One seaweed study verified that REEs removed by *Ulva* sp. were mostly surface-bound and were easily desorbed with a 15-minute EDTA wash; it also notes that diluted acids and chelating agents can be efficient for regeneration and REE recovery from seaweed biomass. More generally, adsorption–desorption cycling and reversibility are described as key factors for environmentally friendly and cost-effective implementation, and EPS/biomass can be dried into a metal-enriched powder for further processing.

From an energy-and-waste perspective, biosorption is repeatedly positioned as a "green" alternative to conventional routes because it can avoid large volumes of chemical precipitation sludge and can operate at dilute concentrations where solvent extraction is inefficient. Biosorption has been described as efficient and cost-effective and as avoiding generation of chemical sludge in comparison with conventional methods. Multiple reviews contrast biological approaches with conventional hydrometallurgical/pyro/electrometallurgical recovery methods that can be energy-intensive and can generate secondary pollutants such as spent extractants/electrolytes.

Comparison matrix

The table below provides a side-by-side comparison of six technology families as they relate to dilute-to-moderate REE and cobalt streams typical of machining coolants, rinses, and associated wastewater. Where the reviewed sources did not provide quantitative CAPEX/OPEX ranges, the table records that limitation and cites qualitative cost statements.

Technology family	Removal efficiency evidence	Selectivity evidence	CAPEX and OPEX signal	TRL signal	Feed concentration fit	Key advantage	Key disadvantage
Chemical precipitation	Reported as high removal (e.g., >95%) in general descriptions, but commonly produces mixed precipitates.	Generally non-selective; overlapping precipitation windows can drive co-precipitation losses (e.g., REE with Al).	"Relatively low costs" and low energy use stated qualitatively; sludge can cause secondary pollution.	Commercially common for metals wastewater treatment and bulk removal.	Best for moderate-to-high metals where sludge handling is acceptable; not ideal for trace recovery value capture.	Simple, robust, high removal.	Large sludge volumes, low selectivity, downstream refining required.

Technology family	Removal efficiency evidence	Selectivity evidence	CAPEX and OPEX signal	TRL signal	Feed concentration fit	Key advantage	Key disadvantage
Membranes (UF/NF/RO and membrane distillation)	UF/NF described as pre-treatment and concentration; DCMD example reports flux 7.0±0.5 $L/m^2 \cdot h$ and modeled yield of 193.2 g of 99% $EuCl_3$ per 1000 m^3 AMD when integrated with adsorption/concentration.	NF can reject multivalent ions and has promise for REE separation; performance sensitive to fouling in practice.	High installation/maintenance costs and frequent replacement noted qualitatively; no universal $ ranges provided.	UF/NF used commercially for pre-treatment; advanced integrated schemes often bench/pilot in the cited studies.	Useful as front-end solids/emulsions removal and to concentrate dilute streams.	Produces a concentrate stream without immediate chemical sludge generation.	Fouling and maintenance burden; often still requires a downstream recovery step to make product.

Technology family	Removal efficiency evidence	Selectivity evidence	CAPEX and OPEX signal	TRL signal	Feed concentration fit	Key advantage	Key disadvantage
Ion exchange and chelating resins	High retention can be achieved under optimized pH; examples include 97.78% retention at pH 2 on a cation resin (AG 50W-X4) and >99% retention at pH 7 on an anion resin (reported in batch tests).	Selectivity depends on resin functional groups; cation resins can show higher affinity for heavy REEs; chelation and EDTA elution can yield high HREE purity (e.g., 98.4% HREE in eluate under specific column conditions).	No direct CAPEX/OPEX; positioned as suitable for dilute feeds and "high selectivity" with manageable maintenance in qualitative summaries.	Resins are commercial products; specific REE fractionation flowsheets may be lab-to-pilot in literature demonstrations.	Strong fit for dilute-to-moderate feeds where long contact time and regeneration are feasible.	High selectivity potential and regenerability; can produce concentrated eluates for downstream precipitation/refining.	Slow processing at low concentrations can lead to prolonged batch operations (reported up to a month in one discussion).

Technology family	Removal efficiency evidence	Selectivity evidence	CAPEX and OPEX signal	TRL signal	Feed concentration fit	Key advantage	Key disadvantage
Biosorption and bioaccumulation	Quantitative examples include: *C. vulgaris* Nd uptake up to 157.40 mg/g at pH 5 with equilibrium within 30 min; *Ulva* sp. removals up to 88% in 24 h and ≥90% at 96 h in trace solutions; EPS adsorption capacities 89.12–138.2 mg/g in single-metal tests.	Selectivity is variable; some systems show limited selectivity against competing ions (e.g., no particular selectivity of REEs over Cu for *G. sulphuraria*), while EPS is positioned as promising for separation without quantified separation factors in the cited excerpt.	Often framed as low-energy/low-cost and sustainable; avoids chemical sludge, but no system-level CAPEX/OPEX numbers provided in the reviewed excerpts.	Repeatedly described as needing further development toward practical deployment; positioned as lab-to-pilot with commercial potential.	Strong conceptual fit for dilute streams where solvent extraction is inefficient (<100 ppm).	Potentially lower energy and lower secondary chemical waste; yields metal-enriched biomass for downstream recovery.	Early TRL; sensitive to pH/speciation and competing ions; longer contact times in some systems and biomass handling required.

Technology family	Removal efficiency evidence	Selectivity evidence	CAPEX and OPEX signal	TRL signal	Feed concentration fit	Key advantage	Key disadvantage
Solvent extraction	High extraction is achievable at high concentrations (e.g., ~99% extraction of Nd/Tb/Dy using 0.8 M D2EHPA from a 1500 mg/L-each synthetic feed; D2EHPA equilibrium reached after ~10 min in one study).	Single-stage separation factors can be low for adjacent REEs, requiring many stages (e.g., 40–50 stages for separation factor ~2 noted in one discussion).	Characterized as having high capital and operating costs and operational challenges (emulsification, flooding, loading limits); quantitative $ not provided in cited text.	Commercially used and often the preferred industrial process for REE production and separations.	Best fit for concentrated leachates; limited applicability when REEs are dilute (<100 ppm).	Mature, scalable, continuous, high throughput at concentration.	Poor economics for dilute streams; solvent loss and secondary waste concerns; complex stage count for fine separations.

Technology family	Removal efficiency evidence	Selectivity evidence	CAPEX and OPEX signal	TRL signal	Feed concentration fit	Key advantage	Key disadvantage
Electrochemical and electromembrane	Electrodialysis examples include 99.52% Sc3+ removal from 20 ppm at 3 V and extraction/recovery values reported for Nd/Dy/Tb (e.g., Tb 90% extracted, 55% recovered) in a cited dataset; chelation-assisted electrodialysis achieved Dy yield 82% and purity 86% with energy sensitivity to voltage (SEC 48 kWh/mol at 14 V vs 14 kWh/mol at 4 V).	Selectivity can be tuned with chelants (EDTA) and membrane configuration; a separation factor of 56 was reported for Dy vs Nd/Pr under stated conditions.	Energy use and scale-up constraints are repeatedly emphasized; quantitative system cost not provided in the cited excerpts.	Many approaches are lab-to-pilot, with limited large-scale application experience in wastewater contexts.	More viable when a concentrate is available or when architectures overcome mass-transfer limits in dilute feeds.	Potential for selective recovery without chemical sludge; continuous operation possible.	Energy intensity and mass-transfer limits in dilute streams; REE electrodeposition is not feasible in aqueous electrolytes due to very negative standard potentials.

Positioning

Algamol's "algae-first" positioning is best supported when the wastewater is (a) too dilute for solvent extraction to be efficient, and/or (b) burdened by complex organic matrices where precipitation would create large mixed sludges and lose value in low-grade solids. A key technical rationale is that conventional organic-solvent extraction has limited applicability when REE concentrations are below ~100 ppm, which is consistent with many rinse waters and diluted coolant effluents (even if grinding sludge itself is solids-rich). In those dilute regimes, biosorption is repeatedly presented as a greener and cheaper tool to remove REEs from aqueous solutions and as having advantages such as lower sludge volumes and the potential for element recovery.

A practical differentiation thesis for Algamol is to focus less on "replacing" precipitation (which is described as simple, low-cost, and high-removal but non-selective) and more on building a hybrid, compliance-ready recovery train where algae-based capture produces a security-relevant, metal-enriched intermediate while minimizing secondary chemical sludge. In this framing, biosorption acts as a selective or semi-selective capture/concentration step, followed by desorption into a smaller-volume eluate suitable for conventional refining (e.g., oxalate precipitation or other downstream hydromet steps).

A recommended "default" hybrid architecture consistent with the reviewed evidence is:

- UF as an upfront step for emulsion breaking and solids removal in oily machining streams, reducing downstream fouling and biomass poisoning risk.
- Biosorption/bioaccumulation operated in a pH regime compatible with strong REE uptake (many studies use or cite pH ~5 as effective), while explicitly controlling pH because REE complexation changes across pH 4–8 and above pH 8 due to organic and carbonate complexes.
- A downstream desorption/regeneration step (e.g., EDTA wash or diluted acid) to transfer REEs from biomass into a smaller-volume solution, enabling a conventional polishing step if needed.

This positioning also helps manage a known weakness: biosorbents may not be selective for REEs over key transition metals (e.g., Cu) in real mixed wastewaters, which implies either upstream impurity control or downstream fractionation is required. Complementary "precision" capture technologies—such as lanmodulin-based supports that can separate binary Nd/Y or Nd/Dy pairs—exist as a longer-term pathway to true REE fractionation, but scale-up may be constrained by biomolecule production and purification considerations discussed in the literature.

Recommendations

Technology development priorities should be anchored in the demonstrated sensitivities of biosorption systems and in defensible test protocols that map to real machining wastewater complexity. The first recommendation is to generate performance curves across realistic pH and competing-ion conditions, because pH changes both microbial surface charge and metal speciation, and low pH can electrostatically repel cations while higher pH exposes deprotonated binding sites that enhance adsorption.

These pH studies should explicitly track REE complexation behavior (organic complexes at pH 4–8 and carbonate complexes above pH 8), since speciation shifts can change both uptake and desorption behavior even at constant total metal concentration.

The second recommendation is to prioritize experiments with both NdFeB-relevant REE and SmCo-relevant transition metals (notably cobalt) in the same matrix, and to quantify selectivity tradeoffs rather than relying on single-metal capacity tests. This is supported by evidence that some biosorbents show no selectivity for REEs over Cu in mixed feeds, and by separate demonstrations that algae can remove cobalt efficiently under specific conditions (e.g., 87.8% removal from a 50 mg/L Co feed in 60 minutes at pH 5).

The third recommendation is to design and validate a closed-loop regeneration approach early, because adsorption–desorption cycling and reversibility are flagged as central to cost-effective implementation. EDTA desorption (15 minutes in one *Ulva* study) and regeneration with diluted acids are concrete starting points for bench-scale recycle-loop demonstrations, and EPS/biomass drying into a metal-enriched powder provides an alternative pathway to a solid intermediate for downstream processing.

Finally, any pilot should be built as an integrated train rather than a "single-box" biology demo, because upstream solids/emulsion management and downstream recovery chemistry are critical determinants of operability. The reviewed literature explicitly supports UF as a pre-treatment for emulsion breaking and solids removal, and it repeatedly emphasizes that further research and optimization are necessary to determine feasibility and efficiency of EPS/biomass adsorption-based recovery in practical deployment contexts.

Chapter 12 Pitch deck framework

Algamol Pitch Deck Framework Rare Earth Recovery from Defense Magnet Machining Wastewater

Deck overview

This framework is designed as a slide-by-slide structure for a 16-slide investor and customer deck, plus appendix slides, focused on (1) defense-driven demand for high-performance rare-earth magnets, (2) value leakage from machining waste and coolant slurries, and (3) Algamol's positioning as a wastewater treatment and metals-recovery enabler in a tightening provenance/compliance environment.

Slide 1

Slide title: Algamol

Key message (1 sentence): Algamol enables recovery of rare earth metals from wastewater, turning a defense-critical waste stream into a domestic supply and compliance advantage.

Supporting data points to include:

- "supports the recovery of rare earth metals from wastewater."
- DoD emphasis on keeping "critical nodes" inside the U.S. for secure magnet supply.
- DFARS expanded prohibition effective January 1, 2027 for covered magnets/materials from covered countries.

Visual suggestion: A simple graphic showing "wastewater ⟶ recovered metals concentrate ⟶ compliant domestic value chain."

Slide 2

Slide title: The problem

Key message (1 sentence): Defense magnet production loses a material fraction as machining swarf/sludge, and the resulting coolant-contaminated waste is expensive to dispose of and hard to recycle cleanly.

Supporting data points to include:

- "the new solid scrap and cutting sludge 'swarf'… accounts for 20–30 % of the starting alloy."
- "About 20–30 wt.% scrap is generated during the NdFeB magnets cutting and grinding processes."
- "More than 70% of the waste generated in manufacturing consists of sludge, which is heavily oxidized and contaminated with carbon from lubrication."

- Offsite hazardous waste disposal "typically $0.50–$3.00/gallon ($130–$800/m3)."

Visual suggestion: Photo/illustration of grinding sludge and a callout showing 20–30% loss and $130–$800/m³ disposal range.

Slide 3

Slide title: The market

Key message (1 sentence): Rare earth magnets are required across major U.S. defense platforms at large scale, making magnet waste and provenance a strategic issue—not just an operational one.

Supporting data points to include:

- DoD statement that rare earth permanent magnets are essential across defense capabilities (e.g., F-35, submarines, UAVs).
- Platform REE requirements: F-35 "more than 900 pounds," DDG-51 "5,200 pounds," Virginia-class submarine "9,200 pounds."

Visual suggestion: Three platform images (F-35, DDG-51, submarine) with large numeric REE labels.

Slide 4

Slide title: The regulatory tailwind

Key message (1 sentence): DFARS expands to upstream "mined/refined/separated/melted/produced" restrictions on January 1, 2027, making traceable domestic recycling pathways a procurement lever for NdFeB in particular.

Supporting data points to include:

- DFARS final rule effective date anchor: "Effective May 30, 2024."
- The 2027 prohibition: "Effective January 1, 2027, do not acquire any covered material mined, refined, separated, melted, or produced in any covered country…"
- Covered magnets include "(1) Samarium-cobalt magnets; (2) Neodymium-iron-boron magnets."
- Recycled NdFeB exception: "manufactured from recycled material if the milling… and sintering… takes place in the United States."

Visual suggestion: A timeline from 2024 ⟶ 2027 showing the "scope expansion cliff," plus a branch highlighting "Recycled NdFeB (US milled + sintered) exception."

Slide 5

Slide title: The solution

Key message (1 sentence): Algamol's modular on-site treatment concept pairs pre-treatment for oily machining coolants with an algae-based capture step positioned to support rare earth recovery from wastewater.

Supporting data points to include:

- Membrane/adsorption recovery from coolants typically needs pretreatment to remove oils/organics.
- Water-mix cutting fluids form oil-in-water emulsions (2–10% oil by volume).
- Algamol positioning: "supports the recovery of rare earth metals from wastewater."

Visual suggestion: Process block diagram: (1) equalization + oil/solids removal → (2) algae-based capture → (3) polishing and concentrate/sludge handling.

Slide 6

Slide title: How it works

Key message (1 sentence): A staged treatment train can manage oily metalworking fluids and selectively separate iron vs rare earths using known pH windows, while concentrating REE-bearing residues for downstream recovery.

Supporting data points to include:

- Operational selectivity: Fe can be precipitated first at ~pH 3–4, then REE precipitation occurs at higher pH to avoid co-precipitation.
- REE hydroxide precipitation range: commonly achieved by increasing pH to ~8.6 and typically occurs ~8.6–10.
- Membrane-based pre-concentration is promising, but oil/organic fouling complicates direct use and implies pretreatment requirements.

Visual suggestion: A "pH cascade" diagram with three bands: Fe removal (~3–4), biosorption window (mildly acidic), REE precipitation (~8.6–10).

Slide 7

Slide title: Competitive advantage

Key message (1 sentence): Algamol is positioned as a "treatment + recovery" module that can complement standard precipitation and protect downstream membrane/recovery steps in oily coolant matrices where fouling is a known constraint.

Supporting data points to include:

- Chemical precipitation via pH adjustment is widely used for metal-ion removal.
- Membrane recovery is promising but fouling from oils/organics complicates direct membrane use; pretreatment is usually required.
- The algae/biomass pathway is still emerging for real wastes (risk and differentiation must be communicated honestly).

Visual suggestion: A 2×2 comparison matrix (CAPEX vs selectivity; or TRL vs waste compatibility) comparing precipitation, membranes, ion exchange (placeholder), and Algamol biosorption.

Slide 8

Slide title: Unit economics

Key message (1 sentence): Even before recovered-metal upside, avoiding hazardous hauling at \$130–\$800/m³ can support rapid payback against a \$110k–\$310k modular system with ~\$28k–\$73k/year OPEX drivers.

Supporting data points to include:

- Disposal benchmark: \$0.50–\$3.00/gal (\$130–\$800/m³).
- CAPEX estimate: \$110k–\$310k.
- OPEX components: chemicals \$5k–\$15k/yr; membrane replacement \$5k–\$10k/yr; energy \$3k–\$8k/yr; labor \$10k–\$25k/yr; sludge disposal/processing \$5k–\$15k/yr.

Visual suggestion: Waterfall chart showing "baseline hauling spend → on-site treatment OPEX → net savings," with CAPEX shown as an upfront bar.

Slide 9

Slide title: SmCo opportunity

Key message (1 sentence): High-temperature defense magnets often use SmCo, which is preferred for 200–550℃ applications, and its finishing waste is amplified by brittleness-driven loss and fracture risk.

Supporting data points to include:

- SmCo is described as preferred/ultimate for high-temperature applications "200–550℃."
- Brittle magnet loss "up to 20–30% in volume."
- Sm-Co machining failure rate "up to 20%."

Visual suggestion: Side-by-side graphic: "SmCo at high temperature" (200–550℃) vs "brittleness waste," with a callout to brittleness-driven loss range.

Slide 10

Slide title: Beachhead customer

Key message (1 sentence): EEC in Landisville, PA is a defense-relevant SmCo producer with a documented in-house process chain (melting → ~3 μm powder → pressing → sinter/solution/aging → grinding/slicing/WEDM) and certifications aligned with defense programs.

Supporting data points to include:

- Location: "Electron Energy Corporation, 924 Links Avenue, Landisville, PA 17538."
- Alloy inputs: purchases pure raw materials (Sm, Co, Fe, Cu, Zr) for in-house SmCo 1:5 and 2:17 alloys.
- Powder size: average particle size reduced to "≈ 3 microns."
- Pressing: axial, transverse, or isostatic presses.
- Heat treatment: sinter, solution, and aging under vacuum or inert gas.
- Machining: grinding, slicing, wire EDM for brittle sintered magnets.
- Compliance: "ITAR COMPLIANT AND DDTC REGISTERED" and "certified to AS9100D and ISO 9001:2015."

Visual suggestion: Map of Landisville/Lancaster County with an "EEC" pin, plus a vertical process-flow graphic ending in machining and shipping.

Slide 11

Slide title: Go to market

Key message (1 sentence): A pilot-first approach is supported by wastewater best practices and biosorption literature calling for validation with actual industrial samples, with priority pilots aimed at defense-relevant magnet suppliers and machinists.

Supporting data points to include:

- Priority targets named in compiled inputs: "EEC/Permag (Landisville PA), Arnold (multiple sites), VAC (Sumter SC)."
- Pilot importance: "Field tests and sometimes pilot tests are also important items…"

- Scale-up requirement: lab experiments "need to be advanced to industrial levels and tested with actual wastewater samples from manufacturing sources."

Visual suggestion: Three-phase roadmap: pilot at EEC → multi-site deployments → scale with DFARS timing.

Slide 12

Slide title: Business model

Key message (1 sentence): Because hauling/disposal costs can dominate, a service or lease model tied to verified disposal reduction can be underwritten even while metals-recovery performance is validated on real waste matrices.

Supporting data points to include:

- Disposal cost anchor for savings-based contracting: $0.50–$3.00/gal ($130–$800/m³).
- Technology maturity caveat: research into algal sorbents for REE recovery from real wastes is "still in its infancy."
- Pilot testing is a recognized pathway to derisk performance in industrial water treatment.

Visual suggestion: Pricing schematic with three options: (1) lease + monthly fee, (2) performance-based (share of hauling savings), (3) sale + support; include "pilot validation gate."

Slide 13

Slide title: Strategic value

Key message (1 sentence): Waste capture and recovery supports defense supply security goals, improves provenance readiness as disclosure requirements phase in, and can align with domestic "mine-to-magnet" investments underway since 2020.

Supporting data points to include:

- Supply-security rationale: "It's important that all those critical nodes happen inside the U.S."
- Provenance disclosure direction: DoD "will have to require that contractors disclose the source of certain rare earth permanent magnets…"
- DoD funding scale: "Since 2020, DOD has awarded approximately $439 million…" to reestablish domestic rare earth supply chains.

Visual suggestion: Four-pillar slide (Security, Compliance, Cost, Circularity) with each pillar anchored to one cited statement/data point.

Slide 14

Slide title: Market sizing

Key message (1 sentence): Magnet swarf and sludge is already a sizable stream relative to global production, and machining scrap fractions in the 20–30% range imply meaningful capture volumes where high-value magnets are finished to tight tolerances.

Supporting data points to include:

- Swarf stream estimate: "The volume of the swarf recycling stream in 2020 is estimated as 9% of the NdFeB magnets produced globally."

- Typical machining scrap: 20–30% of starting alloy as swarf in manufacturing and 20–30 wt% scrap during NdFeB cutting/grinding.
- Machining is called out as the step with the greatest material loss; machining recovery rate 60–90% depending on complexity (implying 10–40% loss).

Visual suggestion: TAM/SAM/SOM concentric circles with (a) global NdFeB production baseline, (b) swarf stream fraction (~9%), and (c) defense-relevant subset (placeholder).

Slide 15

Slide title: Risks

Key message (1 sentence): The core technical risk is variability and contamination in real machining slurries and coolants, and the core commercialization risk is proving performance on real streams early enough to match DFARS timing pressure.

Supporting data points to include:

- Real waste streams complicate recovery due to cutting fluids/organics; pretreatment is often required.
- Membrane-based REE recovery from coolants is complicated by fouling from oils/organics and typically needs pretreatment.
- DFARS 2027 expansion date is a fixed milestone that drives procurement urgency.

Visual suggestion: Risk matrix with "technical" (matrix effects, fouling, pH control) and "schedule" (2027 deadline) axes, plus a mitigation column tied to pilot testing.

Slide 16

Slide title: Team and ask

Key message (1 sentence): Raise $[X]M to run a pilot at a defense-relevant SmCo producer and mature the process to industrial validation using real wastewater samples, consistent with published scale-up needs for biosorption approaches.

Supporting data points to include:

- Validation requirement: lab-scale experiments must be tested with actual wastewater samples from manufacturing sources.
- Maturity caveat and why funding is needed: algal sorbents for real wastes are still "in its infancy."
- Named pilot targets include EEC/Permag, Arnold, VAC Sumter (beachhead list).

Visual suggestion: "Use of funds" pie chart (pilot build, sampling/analytics, field ops, BD), plus a 6–12 month pilot timeline ending with a validated performance report.

Appendix A1

Slide title: Detailed financial model

Key message (1 sentence): A transparent model should separate (1) disposal avoidance (bankable) from (2) metals recovery (upside), using cited disposal, CAPEX, and OPEX ranges as the base case.

Supporting data points to include:

- Disposal: $130–$800/m³ benchmark.
- CAPEX: $110k–$310k; OPEX line items as cited ranges.
- If needed, include third-party ROI benchmarks (e.g., ROIs as fast as 1½ years; typical 3–4 years).

Visual suggestion: Spreadsheet screenshot mockup with scenario toggles (flow rate, disposal cost, treatment OPEX) and outputs (payback, NPV).

Appendix A2

Slide title: SmCo grade anchor

Key message (1 sentence): EEC's 2:17-34 grade provides a defense-relevant numeric anchor for high-energy SmCo magnets and illustrates why finishing operations (and their waste streams) concentrate significant embedded value.

Supporting data points to include:

- EEC 2:17-34: (BH)max 34 MGOe; Br 11.9 kG; iHc min 18 kOe; max operating temp 300℃; "DFARS compliant."
- Machining to final dimensions via grinding/slicing/WEDM for brittle sintered magnets (connects performance to finishing waste).

Visual suggestion: Property radar chart (BHmax/Br/Hc/iHc/Tmax) plus a process note "sintered ⟶ machined to print."

Appendix A3

Slide title: DFARS details

Key message (1 sentence): DFARS covers both SmCo and NdFeB magnets as "covered materials," and the January 1, 2027 expansion reaches upstream mining/refining/separation, with a specific recycled-material exception only for NdFeB if U.S. milled and sintered.

Supporting data points to include:

- Covered material includes SmCo and NdFeB magnets.
- 2027 expansion language ("mined, refined, separated, melted, or produced").
- NdFeB recycled exception language (U.S. milling + sintering).
- SmCo rule scope includes "including recycled material… through production of finished magnets" (useful to frame SmCo compliance complexity).

Visual suggestion: One-page "what changes in 2027" schematic with two branches (NdFeB with recycling exception vs SmCo without such an exception in the cited text).

Appendix A4

Slide title: Treatment chemistry

Key message (1 sentence): The most defensible near-term architecture combines pretreatment for oily coolants with staged pH control for selective metals removal, consistent with documented precipitation and coolant-fouling constraints.

Supporting data points to include:

- Coolant emulsions: 2–10% oil-in-water (water-mix cutting fluids).

- Staged pH: Fe removal first (~3–4), then REE hydroxide precipitation at ~8.6–10.
- Membrane-based pre-concentration is promising but fouling from oils/organics complicates direct use; pretreatment is usually required.

Visual suggestion: A "treatment train" slide with unit operations and pH setpoints labeled, plus an "organics pretreatment" callout.

Appendix A5

Slide title: Waste form reality

Key message (1 sentence): Magnet-manufacturing waste is frequently sludge-like and coolant-contaminated rather than clean scrap, which affects both downstream recovery and wastewater treatment design constraints.

Supporting data points to include:

- "More than 70% of the waste… consists of sludge" that is oxidized and lubrication-contaminated.
- Grinding slurry description: mixture of metallic swarf, grinding-media abrasion, and cooling agent, with remaining coolant varying from <1% to >20%.

Visual suggestion: Cross-section illustration of slurry (solids + abrasive + coolant) with the <1% to >20% residual coolant band highlighted.

Appendix A6

Slide title: Pilot site interface

Key message (1 sentence): If the pilot discharges to a local POTW, the relevant interface is a pretreatment program with industrial user permits and local limits designed to prevent pollutant interference and biosolids contamination.

Supporting data points to include:

- LASA: "approved Pretreatment Program" and it "issues Industrial User Permits" for significant industrial users.
- Pretreatment goal: prevent pollutants that interfere with the POTW or contaminate biosolids.

Visual suggestion: A compliance flow diagram: sampling → pretreatment permit → local limits monitoring → discharge approval loop.

Chapter 13 Arnold facility model

Algamol Deployment Model Arnold Magnetic Technologies Multi Site Defense Magnet Operations

Arnold Magnetic Technologies is a multi-division, multi-site producer of permanent magnets and assemblies that explicitly targets aerospace and defense markets and emphasizes ITAR/DFARS compliance. Arnold's footprint spans U.S. manufacturing (including Rochester, NY and Marengo, IL), European operations (including the UK and Switzerland), and Asia (including Shenzhen, China and a newer Thailand site), creating both scale advantages and compliance complexity for defense magnet supply chains.

Company profile

Arnold is headquartered at 770 Linden Avenue, Rochester, New York 14625. Company materials describe Arnold as having an operating history of over ~130 years and long-standing supply of "defense sensitive materials" to U.S.

and allied markets. Arnold states it is a subsidiary of Compass Diversified (NYSE: CODI), and Compass Diversified reports acquiring Arnold in March 2012 for $129 million. Separately, an Arnold UK "History" page states that "as of the latest updates" Arnold operates under the ownership of TMS International.

Locations and capabilities

Arnold publicly lists a global manufacturing footprint that includes permanent magnet production and/or assemblies in Rochester, NY (U.S.), Sheffield (UK), Lupfig (Switzerland), Amata City (Thailand), and Shenzhen (China). Arnold also lists Woodstock, Illinois as a site for Alnico and Precision Thin Metals, and provides a Woodstock address at 1005 Courtaulds Dr, Woodstock, IL 60098.

Arnold has stated that its Rochester headquarters fabricates and tests SmCo and builds custom assemblies. Arnold also states that it manufactures RECOMA Samarium Cobalt magnets in Rochester, NY and Marengo, IL.

The table below consolidates the most directly evidenced site-level roles from the provided sources.

Site	Evidence of role relevant to defense magnet operations	Notes for Algamol targeting
Rochester, NY (770 Linden Ave)	Headquarters and permanent magnets; Rochester address listed. Fabrication/testing of SmCo and building custom assemblies stated.	Likely hub for RECOMA SmCo work and defense assembly work based on stated SmCo fabrication/testing and "custom assemblies."
Marengo, IL (300 West Street)	Listed as a company location (Marengo, IL 60152). RECOMA SmCo manufacturing stated in Rochester and Marengo.	Candidate site for SmCo production steps and/or machining to print associated with RECOMA magnets.
Woodstock, IL (1005 Courtaulds Dr)	Woodstock, IL address listed; quality certifications page ties Woodstock to Alnico and PTM. A 130,000 sq ft Woodstock facility is described as opening in January 2025 for Alnico and Precision Thin Metals.	Near-term commissioning/relocation window can enable "design-in" of waste/coolant handling and metal recovery processes for magnet-adjacent operations.
Lupfig, Switzerland	Address listed (Hubelacherstrasse 15, CH-5242 Lupfig). Listed as "Permanent Magnets and Assemblies."	EU/Swiss operations may be relevant for EU defense programs and non-U.S. supply chain work.
Sheffield, UK	Listed on the corporate site as "Permanent Magnets & Assemblies."	UK assembly site can be a second-stage integrator where magnet finishing/assembly waste streams may occur.

Site	Evidence of role relevant to defense magnet operations	Notes for Algamol targeting
Shenzhen, China	Shenzhen address listed (Guangming, Shenzhen). Quality certifications page lists "SmCo Group" and "Bonded Magnet Group" in Shenzhen. Shenzhen facility achieved AS9100D certification and is described as the sixth Arnold location to hold that certification.	China location creates DFARS/covered-country complexity for U.S. defense programs, even if it is AS9100D-certified.
Chonburi, Thailand	Arnold states it opened a 26,250 sq ft manufacturing site in 2024 producing magnet assemblies, rotors, and motors.	Newer site suggests greenfield opportunities for embedding water reuse and waste capture into assembly/rotor lines.
Ganzhou, China (JV mine)	Arnold materials list a "Joint Venture - Mine Ganzhou, CN."	Upstream JV highlights exposure to China-based raw material nodes that may be incompatible with DFARS pathways for certain contracts.

Arnold also reports maintaining "twelve manufacturing facilities, plus their joint venture," and lists a broader set of division sites including Ogallala, NE; Madison, WI; Marietta, OH; Norfolk, NE; and Greenville, OH.

Product portfolio

Arnold describes itself as a manufacturer of high-performance permanent magnets and explicitly lists RECOMA samarium cobalt, neodymium iron boron (NdFeB), Alnico, and precision magnetic assemblies among its offerings. Arnold markets RECOMA Samarium Cobalt in "standard grades 18–35E" with "custom grades for high temperature," and it states that RECOMA 35E is "the most power dense samarium cobalt magnet on the market." Arnold also states that it can adjust material composition "for performance at extreme temperatures up to 550℃."

Arnold's RECOMA documentation distinguishes two SmCo families, stating that Sm2Co17 magnets show the highest magnetic performance at elevated temperatures while SmCo5 offers easier magnetizing and strong corrosion resistance. For NdFeB, Arnold publishes a catalog of "Neodymium-Iron-Boron Magnet Grades" and notes these are also referred to as "Neo" or NdFeB magnets. Arnold's catalog also lists "Grain Boundary Dysprosium Diffused Grades" and "Grain Boundary Diffused Grades," signaling availability of diffusion-enhanced, high-coercivity NdFeB options relevant to high-temperature and/or demagnetization-resistance design constraints.

Certifications and programs

Arnold states that all of its facilities are AS9100 certified. Arnold's aerospace and defense market page lists ISO 9001:2015 and AS 9100 Rev D as certifications. Arnold's quality certifications materials list NADCAP as a certification, with scope indicated for Precision Thin Metals.

For export-control and defense procurement positioning, Arnold states that "all locations" in its RECOMA supply chain are registered with ITAR and can adhere to DFARS clauses. Arnold further positions itself as "registered with ITAR" and able to adhere to DFARS clauses for defense work.

On the customer/program side, Arnold's aerospace and defense materials state that its products are used in "commercial and defense aviation, satellites," and list defense-relevant application areas including rotor assemblies, electric motors, auxiliary power units, flap actuators, magnetic bearings, electromagnetic brakes, waveguides for microwave transmission (TWT), EMI/RFI shielding, fuel systems, and "black chaff." An Arnold acquisitions/mergers infographic also indicates key customers include defense primes such as Raytheon, Lockheed Martin, and Northrop Grumman.

Machining and waste

Arnold presents itself as a supplier of "precision magnetic systems to print" and emphasizes vertically integrated processes and contract manufacturing capabilities that span magnets and assemblies. On its precision magnetic assemblies page, Arnold lists machining and finishing capabilities including EDM, ID slicing and grinding, in-house machining of hubs/rotors/cylinders/sleeves, magnetic and thermal stabilization, rotor dynamic balancing, serializing, and magnetic characterization.

Arnold also states it provides "advanced machining, grinding, and finishing capabilities" to achieve tight tolerances suitable for defense-grade applications. The provided sources do not quantify the fraction of magnet material lost as kerf, grinding sludge, or rejected parts during Arnold's machining operations.

Wastewater characterization

The provided sources establish that Arnold performs EDM and grinding/slicing operations and manufactures magnetic assemblies, which are the kinds of operations that commonly drive the "waste stream" interface for any on-site capture or treatment initiative. However, none of the provided Arnold sources describe coolant chemistries, wastewater volumes, metal concentrations, or sludge handling methods at any Arnold site.

As a result, an Arnold-specific wastewater characterization for Algamol must be treated as a data-collection task rather than a literature-derived specification, and it must be anchored to Arnold's explicitly stated machining processes and sites.

DFARS compliance

Arnold states it can supply NdFeB magnets "fully compliant" with DFARS 225.7018. Arnold also states that RECOMA supply chain locations are ITAR-registered and can adhere to DFARS clauses, and it positions RECOMA samarium cobalt as "DFARS compliant."

Arnold's communications also describe active DFARS preparation for 2027, stating it is preparing for "DFARS Compliance updates starting Jan 1st 2027" and plans to establish a fully compliant supply chain for NdFeB and SmCo magnets by mid-2026. In parallel, Arnold states it has entered into strategic agreements to secure samarium raw material supply outside China and to reduce dependency on Chinese sources amid export-control uncertainties.

Algamol system design

Because Arnold's evidenced manufacturing and assembly processes include EDM, slicing, and grinding, any Algamol deployment concept for Arnold should be designed around capture points associated with these operations and with assembly-related washing/cleaning where applicable. Arnold explicitly highlights magnet production/fabrication, magnetizing, stabilization/calibration, and testing as part of its permanent magnet capabilities, which can be used to bound where in the process route a waste/water interface is most likely to exist.

At a program level, Arnold's multi-country footprint implies that any Algamol deployment strategy should be segmented into (a) U.S. defense-program-aligned sites and (b) non-U.S. sites that may serve different customer bases or export-control contexts, with separate data packages for ITAR/DFARS handling.

Financial model

The sources provided for Arnold do not include site-level wastewater disposal costs, coolant spend, sludge disposal costs, or magnet machining scrap rates, so a bottom-up ROI cannot be computed from the evidence set alone. The evidence does support that Arnold is investing "over $50 million" in facility improvements, manufacturing technology upgrades, and capacity expansion, indicating ongoing capital programs that could potentially be leveraged to integrate waste capture or treatment systems as part of broader manufacturing upgrades. The evidence also supports that Arnold is expanding capacity via a new Thailand permanent magnets manufacturing facility and by adding/modernizing facilities such as Woodstock, which can change both production scale and waste/water load profiles over time.

Strategic value and pilot proposal

Arnold's stated position as a DFARS- and ITAR-capable supplier to aerospace and defense markets makes traceable, domestic-aligned handling of rare-earth-bearing process streams strategically relevant, especially as the company explicitly prepares for DFARS updates effective January 1, 2027. Arnold's publicly listed application set in defense aviation, satellites, motors, and RF/microwave subsystems suggests that customer programs may demand both performance and supply-chain assurance, which can extend to how critical-material-bearing waste is handled inside controlled facilities.

A pragmatic pilot pathway, constrained to what the sources actually substantiate, is to start at a U.S. site with explicitly stated SmCo fabrication/testing and assembly work and documented machining capability (e.g., Rochester, NY and/or the Rochester-linked assemblies capability set), and build a measurement-first program focused on sampling and quantifying process-stream composition before selecting treatment architecture. A second pilot option is to engage during transition windows such as the January 2025 Woodstock facility opening (Alnico and Precision Thin Metals), where new processes and infrastructure could be instrumented from the outset even if the most REE-intensive streams remain concentrated in SmCo/NdFeB operations.

The largest gaps that must be closed to turn this facility model into an executable Algamol project are (1) which specific Arnold sites run the highest-volume rare-earth magnet machining steps (EDM/slicing/grinding) for SmCo and NdFeB, (2) what coolant and cleaning chemistries are used, and (3) what the current waste handling pathway is at each site. These gaps are not addressed in the provided sources and require direct site engagement and sampling under the relevant quality and export-control frameworks that Arnold states it can support.

Chapter 14 VAC Sumter facility model

Algamol Deployment Model Vacuumschmelze VAC Sumter SC New US Magnet Production Plant

Summary

Vacuumschmelze (VAC Group) is a Germany-based rare-earth permanent magnet producer headquartered in Hanau (Grüner Weg 37, D-63450 Hanau) and described as a global producer of advanced magnetic solutions and rare-earth permanent magnets. The company reports significant global scale (nearly 4,000 employees) and multiple production facilities, while also highlighting vertical-integration capability "from base element to end product."

VAC's U.S. expansion is executed through eVAC Magnetics LLC, a VAC Group subsidiary that built a new rare-earth permanent magnet manufacturing facility in Sumter County, South Carolina (Pocotaligo Industrial Park). Public reporting places operations coming online in late autumn 2025 / end of 2025 and describes the facility as having begun shipments after opening in November 2025. Public sources also describe a production capacity target on the order of ~1,600–2,000 metric tons per year (with various documents citing "up to 1,600 MT" NdFeB magnet capacity and "just under 2,000 metric tons a year," and local reporting stating 2,000 metric tons/year by spring 2026).

The Sumter facility is positioned as a dual-use asset supporting both commercial and military demand, with multiple sources explicitly linking the plant to magnets used in defense applications (including F-35 and UAVs) and to domestic supply chain resilience objectives. The plant's business model and capital stack are strongly shaped by public incentives and national-security-linked funding, including a reported $94.1 million U.S. Department of Defense agreement for manufacturing equipment and production line engineering and a $111.9 million federal tax credit (Section 48c).

Company

VAC Group is described as a leading global producer of advanced magnetic solutions, rare earth permanent magnets, and inductive components, with corporate operations based in Hanau, Germany. VAC reports a broad manufacturing footprint (facilities in Finland, Slovakia, China, and Malaysia) and maintains a model in which different plants are optimized for different product groups and scales (e.g., small-series precision work vs. flexible, partially automated production).

VAC's permanent-magnet portfolio is branded around VACOMAX (SmCo) and VACODYM (NdFeB), and the company states it has produced VACOMAX SmCo magnets (powder-metallurgical) since 1973 and VACODYM NdFeB magnets at industrial scale since 1986. VAC also links its aerospace/defense market access to quality management standards, including AS9100/EN 9100 (with EN 9100 referenced as a certification basis for permanent magnets).

Ownership has shifted recently, with multiple sources stating VAC was acquired by Ara Partners in October 2023.

Facility

The eVAC Magnetics Sumter facility is located in the Pocotaligo Industrial Park in Sumter County, South Carolina, and multiple announcements describe the project as a "more than half a billion-dollar investment," with reported capital investment of $506 million and planned creation of about 300 jobs. One local report describes the plant footprint as 410,000 square feet on 83 acres, while other public announcements describe the project site as 85 acres.

Public timelines consistently place the facility's operational start in late 2025 (e.g., "online in late autumn 2025," "complete in late fall 2025," and "come online in the second half of 2025"). A later local report states the facility opened in November 2025 and shipped its first NdFeB permanent magnets to customers.

Capacity is reported in several ways. Local reporting states the facility will reach a 2,000-metric-ton annual production capacity by spring 2026 and may scale output substantially thereafter. A separate policy-focused document states the magnet-making facility will be able to produce up to 1,600 metric tons of NdFeB permanent magnets, and notes a potential expansion to 4,000 metric tons within 12–18 months after start-up (conditional on agreements and lead times).

Products

Public reporting and VAC communications identify the Sumter facility as producing high-performance rare-earth permanent magnets, including NdFeB and (in some reporting) SmCo types, with end-markets spanning EV traction

motors and defense applications. Multiple sources also describe the facility as producing NdFeB magnets specifically and link those magnets to automotive and defense use cases.

At the corporate product-family level, VACODYM magnets are NdFeB-based (neodymium-iron-boron alloy basis) and are described as "highest energy density" grades, while VACOMAX magnets are rare-earth–cobalt (SmCo-class) and described as high-coercivity, temperature-stable magnets with corrosion stability benefits.

Grades

VAC's NdFeB portfolio includes VACODYM 902 TP, which VAC describes as heavy-rare-earth-free and manufactured entirely in the Western Hemisphere. A VAC press release on the VACODYM 902 TP family reports room-temperature remanence of at least 1.40 T and coercivity of at least 1190 kA/m for VACODYM 902 TP.

VAC's VACODYM product family page reports (across grades) Br minimum ranges of 1.03–1.42 T and typical (BH)max of 225–415 kJ/m³, and notes that an established grain boundary diffusion process can increase coercivity up to a maximum of 600 kA/m (for applicable magnets).

For SmCo, a trade source describing VAC's permanent magnet capabilities notes that Vacomax sintered SmCo alloys are used in high-performance rotors and cites Vacomax 262 HRP as enabling application temperatures up to 250 ℃ in a rotor system context.

The table below summarizes publicly stated grade-level performance anchors that are relevant for high-reliability motor, actuator, and defense/aerospace applications.

Product family	Magnet type	Example grade	Public performance anchors	Notes for defense procurement relevance
VACODYM	NdFeB	VACODYM 902 TP	Br ≥ 1.40 T; Hc ≥ 1190 kA/m (reported for 902 TP)	Marketed as heavy-rare-earth-free and made in the Western Hemisphere
VACODYM	NdFeB	Family ranges	Br min 1.03–1.42 T; (BH)max typ 225–415 kJ/m³; grain boundary diffusion can raise coercivity up to 600 kA/m	Process options cited that are commonly associated with high-coercivity/high-temp NdFeB needs
VACOMAX / Vacomax	SmCo	Vacomax 262 HRP	Application temperatures up to 250 ℃ in cited rotor context	SmCo is frequently positioned for high-temperature, high-reliability duty cycles in aerospace/defense motors

Manufacturing

VAC's NdFeB manufacturing is described (at the product-family level) as powder-metallurgical and sintering-based, with production routes that depend on form factors and tolerances and can involve larger isostatically pressed blocks or die-pressed parts. The same source describes die-press alignment as using strong magnetic fields oriented parallel (axial) or perpendicular (transverse) to the pressing direction depending on geometry and grade family (AP vs TP).

At the enterprise level, VAC describes itself as vertically integrated, with statements that it can "design and manufacture from base element to end product" and that it is "extensively vertically integrated." A technical brochure further states VAC can perform processing steps in-house, ranging from vacuum melting of alloys through coating and magnetization, assembly, and final inspection (with those statements made specifically about VAC's established European operations).

For Sumter specifically, one policy-focused description frames the facility as "onshoring the NdFeB permanent magnet supply chain" and states the magnet plant will produce up to 1,600 MT NdFeB magnets and is supported by DoD Title III Defense Production Act authorities. The same source states that a next-phase, co-located metal- and alloy-making facility is planned to produce rare earth metals and strip cast alloys, with capacity up to 2,000 MT of magnet alloy annually starting in early 2027 (sized to support the 1,600 MT magnet plant).

Compliance

The Sumter facility is repeatedly positioned as a contribution to a resilient, U.S.-based rare-earth permanent magnet supply chain and a step toward reducing dependence on China for rare-earth magnet production. A policy-focused document states that eVAC's vertically integrated campus would "eliminate reliance on foreign sources — particularly China" by processing non-Chinese rare earth oxides into finished magnets and explicitly links this to "compliance with … DFARS 225.7018," with a 2027 deadline for excluding adversarial sourcing for NdFeB-containing defense components.

In parallel, DFARS rules (as published on acquisition.gov) state that effective January 1, 2027 contractors shall not deliver "covered material" mined/refined/separated/melted/produced in "any covered country," and also include an exception for NdFeB magnets manufactured from recycled material if milling and sintering occur in the United States. DFARS also specifies that for samarium-cobalt magnets, the restriction includes the entire supply chain from mining (including recycled material) through production of finished magnets, except as provided by additional exceptions.

VAC also publicly positions itself as compliant with U.S. and EU regulatory expectations, including DFARS and ITAR (as a general compliance statement) and lists quality certifications such as DIN EN ISO 9001:2015 and EN 9100:2018 in corporate materials.

Waste

Public documentation indicates the Sumter facility is expected to generate magnet production by-products, explicitly described as "swarf," and VAC has contracted for recycling of these by-products. A Cyclic Materials announcement states that VAC and Cyclic signed a 10-year exclusive agreement to recycle 100% of the magnet production by-products (swarf) produced at the Sumter facility, which was described as scheduled to begin operations at the end of 2025. The same announcement states VAC has an option to reintegrate recycled mixed rare earth oxides (rMREOs) from Cyclic into its own production processes, connecting manufacturing waste streams to a circular feedstock pathway.

Separate reporting similarly notes a long-duration recycling agreement to process byproducts from the Sumter facility, reinforcing that by-product handling is part of the operating model rather than an afterthought. In addition, local

reporting indicates the plant includes quality control areas and automated/robotic inspection operations, consistent with a facility designed for tightly controlled product outputs where scrap and rework streams can be material (even though public sources do not quantify scrap rates).

At the corporate capability level (not necessarily Sumter-specific), VAC describes advanced machining for magnetic materials—including wire EDM and grinding—to meet stringent specifications, and also references EDM-based production approaches in its manufacturing capability descriptions.

Algamol fit

The public record demonstrates that (1) Sumter is intended to be a high-volume rare-earth magnet manufacturing site, (2) its by-products (swarf) are explicitly anticipated, and (3) there is already a contractual pathway to collect and recycle those by-products in North America. These three points define the minimum integration boundary conditions for any onsite water, coolant, or effluent treatment system intended to capture value from magnet-making waste streams at Sumter.

State and local incentives explicitly included "water and wastewater improvements" as eligible project-support uses, indicating that water and wastewater infrastructure is a recognized part of the site build-out and may be adaptable for future add-on treatment or capture systems (without implying any particular technology choice).

Financial model

A facility-specific financial model for Sumter must start from the disclosed project economics and public co-investments. Multiple sources describe the project as a more-than-half-billion-dollar investment, with one local report stating $506 million in capital investment for the site. Public incentives and federal support include a $111.9 million Section 48c Qualifying Advanced Energy Project Tax Credit, and a DoD agreement reported as $94.1 million to acquire/install manufacturing equipment, operationalize technical infrastructure, and engineer production lines.

A Reuters-syndicated business report states VAC received about $200 million in U.S. government funding and tax credits to build a $500 million plant and states the Sumter plant's capacity is just under 2,000 metric tons per year, with up to 90% of output going to General Motors and the remainder supplying the U.S. Department of Defense.

The table below consolidates publicly stated facility-scale financial anchors and support mechanisms.

Category	Publicly stated figure	Notes
Capital investment	$506M	Local reporting of total capital investment for Sumter
Federal tax credit	$111.9M (Section 48c)	Described as a Qualifying Advanced Energy Project Tax Credit supporting construction
DoD agreement support	$94.1M	Described as funding for equipment, infrastructure, and engineering production lines
State/local infrastructure support	$13M Rural Infrastructure Fund grant; $2M LocateSC grant	Explicitly includes "water and wastewater improvements" among eligible uses
Capacity anchor	Up to 1,600 MT/year NdFeB magnets	Policy-focused description of magnet-plant capacity

Category	Publicly stated figure	Notes
Capacity anchor	Just under 2,000 metric tons/year	Reuters-syndicated estimate of plant capacity

Strategic value

The Sumter facility is consistently framed as strategically significant for the United States because it increases domestic rare-earth permanent magnet production and reduces reliance on foreign (particularly Chinese) supply chains, a theme repeated in local reporting and federal policy discussions. VAC and partners also explicitly link U.S.-produced rare-earth magnets to defense platforms and systems, including the F-35 and UAVs, and DoD communications list a wider set of magnet-dependent platforms including submarines, missiles, radar systems, UAVs, and precision-guided munitions.

From a defense procurement standpoint, the facility's strategic value is also tied to the January 1, 2027 DFARS sourcing restrictions and the need for non-covered-country production pathways for both NdFeB and SmCo magnets in defense end-items.

Integration

Sumter's key integration premise is that waste streams are expected and contractually managed, with a declared intention to recycle 100% of magnet-production by-products (swarf) through a long-term agreement, and an option to reintegrate recycled mixed rare earth oxides into VAC's own production processes. Any onsite capture or treatment system (including a wastewater-oriented system) would therefore need to map cleanly onto the existing swarf collection and recycling logistics, rather than creating a parallel, conflicting waste disposition pathway.

To translate this into an execution plan without over-claiming beyond public sources, the immediate next step is a due-diligence package of questions that align with the published facility design and compliance goals:

- What are the physical forms of "swarf" generated at the plant (grinding sludge, EDM fines, saw kerf, coating overspray), and what proportion is currently routed to the contracted recycler under the "100%" agreement language?
- Which stages of the NdFeB supply chain are executed on the Sumter campus at start-up vs. after the planned early-2027 alloy/metal facility comes online (rare-earth metals and strip-cast alloy production), and how does that change the waste stream composition and security requirements?
- What quality system and audit artifacts are required for Sumter's defense-market deliveries under AS9100/EN9100-aligned expectations, and how will waste handling be documented to preserve traceability and compliance narratives?

These questions are directly motivated by the public claims of a China-independent supply chain, DFARS-linked compliance positioning, and contractual commitment to recycle manufacturing by-products, which collectively define the operational constraints for any onsite waste or wastewater intervention at the facility.

PART IV: Defense Platform Analysis, Surge Demand, and Policy

Chapter 15

Israeli Defense Platform Permanent Magnet Requirements

Open sources provide strong evidence that Israeli missile-defense and precision-weapon ecosystems rely on subsystems where permanent magnets are commonly required (actuators, control surfaces, guidance-related motors, and radar/RF hardware), but they rarely disclose magnet grades, quantities, or suppliers at platform level. Public reporting also shows that the highest-volume driver is the Iron Dome/Tamir interceptor enterprise, which has been fielded since 2011 and is supported by accelerating serial production and a growing U.S.–Israeli co-production base.

At the strategic-materials level, multiple sources describe significant exposure to China's dominance in rare-earth mining, refining/processing, and magnet manufacturing, creating a supply-chain vulnerability that affects Israel as well as the U.S. and other allies. This exposure is particularly important for high-temperature and high-coercivity permanent magnets (SmCo and Dy/Tb-enhanced NdFeB), which are widely referenced as necessary in defense actuators, radar, and missile guidance contexts, even when platform-specific details remain classified or proprietary.

Iron Dome

Iron Dome is described as a Rafael-developed air defense system designed to intercept short-range threats (including rockets and artillery), with the Tamir interceptor as the missile effector and a radar and command-and-control architecture supporting engagement decisions and midcourse updates. Several sources specify that a typical Iron Dome battery comprises an EL/M-2084 multi-mission radar (Elta/IAI), a battle management/weapon control element (mPrest is frequently cited for the control function), and three to four launchers, with each launcher carrying up to 20 Tamir interceptors (i.e., ~60–80 ready missiles per battery).

Tamir's publicly described functional components include a solid rocket motor, datalink for midcourse updates, an onboard terminal seeker described as active radar in multiple open sources, and a proximity-fuzed warhead; several sources also describe electro-optical sensors and steering fins for maneuvering. These descriptions do not enumerate actuators, servo architectures, or magnet materials; however, multiple defense supply-chain and technical references explicitly tie permanent magnets to guided-missile actuators and flight-control surface manipulation (i.e., the class of hardware that would be required to drive steering fins).

Operational tempo and replenishment dynamics are visible in open reporting, including statements that Iron Dome has intercepted thousands of rockets since October 2023 and that the system has intercepted over 9,000 rockets since activation with an approximately 95% success rate (Rafael corporate reporting). This high usage rate underpins the strong linkage between Iron Dome interceptor procurement and magnet demand in the guided-munition supply chain (even if magnet quantities per interceptor are not publicly disclosed).

Tamir specifications that matter for magnets

Tamir's published physical envelope is consistently reported at approximately 3 m length, ~160 mm diameter, and ~90 kg weight, with an engagement range up to ~70 km, and performance references such as Mach 2.2 appearing in some open sources. These constraints suggest that any embedded electromechanical actuation for steering fins (or seeker gimbal mechanisms, if present) must be compact and power-dense, which is the regime where rare-earth permanent magnets are generally used in military actuation and guidance contexts.

Some sources go further and claim or imply "permanent magnet" use in missile guidance for Tamir, but without providing technical substantiation such as magnet composition, grade, coercivity, or operating temperature.

David's Sling

David's Sling is widely described as the middle layer of Israel's multi-layered air defense architecture, designed to intercept rockets, missiles (including ballistic and cruise missiles), aircraft, and drones, and jointly developed by Rafael and Raytheon. Open sources describe the Stunner interceptor as a two-stage solid-propellant missile with an operational range cited around 70–300 km and maximum speed around Mach 7.5, which implies high-g maneuver demands and tight mass/volume constraints for guidance and control hardware.

Guidance is described as integrating the EL/M-2084 AESA radar with a dual electro-optical seeker suite (imaging infrared plus CCD), plus datalink and counter-countermeasure capabilities in some summaries. One open source explicitly states that David's Sling actuator motors "utilize permanent magnet components," but it does not identify whether these are SmCo versus high-temperature NdFeB grades, nor provide quantities or temperature requirements.

Cost signals also indicate high per-interceptor value density: open sources cite Stunner unit costs around $700,000 and note U.S. financial support for the program (e.g., $1.39B in one recent report; $1.99B FY2006–2020 in another). These economics typically motivate conservative material selections, qualification rigor, and second-source strategies for components like rare-earth magnets, even when procurement specifics are not public.

Arrow 2 and Arrow 3

Open reporting describes Arrow 2 as an upper-atmosphere interceptor (around ~50 km altitude) and Arrow 3 as an exo-atmospheric interceptor designed to engage long-range ballistic missiles at the top of their trajectory, with Arrow 3 credited to Israel Aerospace Industries (state-owned) with U.S. support (e.g., Boeing). These operational regimes strongly imply extreme thermal cycling (cold-soak at altitude/space-like conditions and potential heating during atmospheric flight), which is the regime where defense systems often prefer SmCo or specially engineered high-temperature NdFeB, but open sources reviewed here do not disclose Arrow magnet types, quantities, or suppliers.

Trophy and other active protection systems

Trophy (Windbreaker) is described by Rafael as a combat-proven active protection system intended to enhance survivability of armored vehicles (tanks and APCs). System-level descriptions emphasize 360-degree detection and autonomous neutralization, but do not describe internal motor/actuator architectures or magnet materials for turreted sensors, countermeasure launcher drives, or stabilization motors.

Open sources on another Israeli APS (Elbit's Iron Fist) emphasize compactness (low volume, weight, and power) and multi-sensor cueing to turreted countermeasure launchers, which are application spaces where permanent-magnet motors are common, but the cited materials remain non-specific on magnet selection and sourcing.

Spike and other Rafael precision weapons

Rafael describes the Spike family (multiple variants) as electro-optically guided missiles and provides an aggregate volume indicator: as of 2023, more than 38,000 missiles have been delivered or are under order, with over 7,000 deployed in combat and training. While the guidance suite is described at a high level (electro-optical guidance), the reviewed sources do not disclose magnet types, actuator designs, or magnet counts per missile for Spike variants.

In adjacent Israeli precision munitions, Elbit's "Iron Sting" guidance descriptions (GPS/laser guidance, deployable fins, multi-mode fuze concepts) indicate the presence of guidance fins and seekers, but open sources do not provide magnet-level specifications for the fin deployment or any actuation mechanisms.

Radar and electronic warfare hardware

Multiple defense supply-chain sources explicitly cite traveling-wave tubes (TWTs) as components that use rare earth permanent magnets to amplify radio-frequency signals, and they also cite radar systems as major magnet-using assets (including samarium-cobalt magnet components in long-lived radar systems such as Aegis SPY-1). Separate technical descriptions of microwave tubes reinforce that magnetrons can incorporate "an extremely powerful permanent magnet" and that electron beams in some tube types are guided by magnets, which is relevant to high-power radar/EW transmitter architectures even when specific Israeli radar bill-of-materials details are not public.

For Israeli radar specifically, open sources identify the EL/M-2084 as an AESA radar used in Iron Dome and David's Sling, and provide performance descriptors (e.g., tracking ~1,200 rockets at once; surveillance tracking to 474 km in one description), but none of the reviewed sources breaks down whether the radar uses solid-state T/R modules only, or whether any vacuum electronics (and thus permanent-magnet focusing assemblies) are used in associated transmit/receive or ancillary subsystems.

Elbit publicly describes an EW suite deployed on "hundreds" of aircraft and enumerates functional blocks (EW controller, radar warning receiver, IR missile warning, countermeasures, jammer), but it does not discuss permanent magnets, nor does it disclose rare-earth magnet suppliers or grades used in any actuators or cooling/power subsystems within those products.

Magnet materials and performance requirements

The open-source record in this dataset supports a materials-level conclusion: SmCo magnets are preferred where high-temperature demagnetization resistance is required, while NdFeB magnets are used where higher energy density is needed but temperature limits must be managed, sometimes through heavy rare earth (Dy/Tb) additions for coercivity at elevated temperature. One DOE reference states NdFeB magnets are used up to about 180℃ and that operation at higher temperatures can require replacing some Nd/Pr with heavy rare earths (Dy or Tb). Separately, a U.S. government-linked supply chain assessment notes sintered NdFeB magnet alloys may include ~0.5% to 11% Dy or Tb by weight to improve high-temperature resistance to demagnetization (a range that covers common "high-temp" grades and heavy-REE-enhanced defense grades).

For SmCo, multiple sources emphasize high-temperature robustness and suitability above 300℃ for some weapon-system contexts, while also noting higher cost and processing complexity compared to NdFeB in general market descriptions. Industrial supplier datasheets reinforce the high-temperature envelope: VACOMAX SmCo is described as suitable up to 350℃ depending on alloy, with Br and (BH)max ranges given, and Arnold describes RECOMA 35E as engineered for high temperature applications around 150–250℃ in high-speed/high-corrosion environments.

Because missile-defense interceptors and seekers can encounter elevated internal temperatures from power electronics and proximity to propulsion sections (as well as external aerodynamic heating), these materials considerations strongly influence the "design space" for guidance and actuation magnets; however, platform-to-material mapping (e.g., "Tamir seeker uses SmCo") is not supported by the reviewed open sources and should be treated as an open question pending program-specific disclosures or supplier documentation.

Supply chain and co-production implications

Multiple sources document that Iron Dome/Tamir has shifted from Israel-only production to a joint U.S.–Israeli industrial base, including references to a U.S. Tamir production facility and a Raytheon–Rafael joint venture producing Tamir rounds and the U.S. SkyHunter variant. Several sources identify a $1.25B production contract for Tamir interceptors at a Camden/East Camden, Arkansas facility and describe investments reported at $33M (and in some

reporting increased to $63M) to establish and expand production capacity.

From a magnet supply standpoint, this co-production structure matters because one RTX/Raytheon source states that "the majority of Tamir missile components are procured through the Raytheon supply chain in the United States," while also describing SkyHunter as producible in the U.S. for the U.S. and its allies. At the same time, none of the cited sources identifies which magnet suppliers (or which countries of origin for magnet raw materials) are used for Tamir or SkyHunter, which is a critical data gap given upcoming provenance and disclosure requirements for rare earth permanent magnets in U.S. defense procurement.

For David's Sling, open sources state that roughly 50% of components are built in the United States and that some parts of the interceptor and launcher are manufactured in the U.S., reinforcing that magnet sourcing decisions may be influenced by both Israeli and U.S. compliance/traceability frameworks.

Rare earth dependency and risk to Israel

Several sources explicitly describe Israel's vulnerability stemming from dependency on Chinese rare earth supplies for defense technologies including magnets, with Chinese export restrictions (including restrictions applying to medium/heavy rare earths such as samarium, terbium, and dysprosium) cited as a significant disruption factor. More broadly, U.S. government sources and analyses document that most rare earth materials are mined and processed abroad, with China identified as a dominant actor across mining, processing, and magnet manufacturing, which implies that Israeli defense platforms embedded in global supply chains inherit this dependency unless mitigated by allied sourcing and domestic capacity initiatives.

On the mitigation side, U.S. initiatives to establish a "mine-to-magnet" industrial base explicitly include permanent magnets used in weapon systems "including the Iron Dome missile interceptor," and note substantial investment scale since 2020. The same U.S. effort cites domestic manufacturing nodes such as a rare earth magnet facility in San Marcos, Texas (Noveon Magnetics) funded via a $28.8M award, which may become relevant to U.S.-based co-production supply chains for interceptors and for allied procurement ecosystems over time.

Suppliers and procurement signals in open sources

Open sources in this dataset provide partial but meaningful signals about potential magnet supplier categories relevant to Israeli systems. Permag (a group comprising Dexter Magnetic Technologies, Electron Energy Corporation, and Magnetic Component Engineering) positions itself as a DFARS-capable supplier for defense magnetics, states that "through EEC" it is "the only U.S. producer of samarium cobalt (SmCo) magnets," and lists certifications including ITAR registration and AS9100D, which are prerequisites for many U.S. defense magnet supply chains. These statements do not establish actual supply into Iron Dome/David's Sling, but they highlight a limited U.S. SmCo industrial base and the scarcity of qualified sources for high-temperature magnets.

Within Europe, VAC positions VACOMAX (SmCo) and VACODYM (NdFeB) as vertically integrated product families, including in-house processing "from the melting of the alloy" through coating, magnetization, assembly, and final inspection, which aligns with the traceability and reliability requirements typical of defense procurement, even though Israeli program participation is not stated in the sources provided.

Within Israel, one open catalog confirms that Magma Magnetic Technologies is based in Israel and offers SmCo and NdFeB magnets with stated heat resistance ranges (SmCo 250–350℃; NdFeB 80–240℃), which suggests a local channel for some magnet forms, although defense certifications and actual program tie-ins are not provided in the cited material.

Production scale indicators and implied magnet demand

While most sources avoid publishing annual interceptor production rates, procurement and inventory numbers provide a basis to understand potential scale. A U.S. Marine Corps plan to acquire three Iron Dome batteries and 1,840 Tamir/SkyHunter missiles is explicitly cited in multiple open sources, and a U.S. Army acquisition for test and evaluation is cited as 480 missiles (with associated battery hardware). Independent reporting also emphasizes that the Arkansas facility and related contracts are designed to "accelerate serial output" and bolster production capacity for both Iron Dome and SkyHunter variants, implying increased throughput and sustained demand for components such as permanent magnets used in actuators and guidance-relevant motors.

At the same time, the publicly available sources do not provide magnets-per-interceptor counts or kilograms of magnet material per interceptor, which prevents a fully grounded bottom-up magnet-demand calculation in this report.

Implications for Algamol

The most direct implication for a wastewater treatment and recovery technology provider is that Israel's defense industrial activity spans high-volume interceptors (Tamir), high-value interceptors (Stunner), and very high-value sensor and EW ecosystems, all of which are associated with permanent-magnet-containing subsystems (actuators, control surfaces, and RF vacuum electronics in some architectures). The open-source record also shows a tightening regulatory environment on magnet provenance and disclosure in U.S. defense procurement (implementation time windows cited between 2025 and 2027; expanded prohibitions effective January 1, 2027), which could increase the value of domestic and traceable recycling/recovery pathways for magnet-bearing waste streams in U.S. co-production facilities supporting Israeli programs.

Because U.S. co-production nodes exist for Tamir/SkyHunter (Camden/East Camden, Arkansas) and because a U.S. rare earth magnet facility exists in San Marcos, Texas, a plausible commercialization pathway for wastewater and waste-stream treatment solutions is to target U.S.-based machining, finishing, and subassembly operations tied to these supply chains (where traceability constraints and disposal economics can be aligned).

Unknowns

This review finds that several critical, decision-relevant data elements are either absent from open sources or explicitly deferred to direct engagement with OEMs. Specifically, sources generally do not disclose which magnet families (SmCo versus Dy/Tb-enhanced NdFeB) are used in each Israeli platform's actuators, seekers, and radar subsystems, nor do they provide magnet counts per interceptor or magnet mass per unit. In addition, detailed procurement channels and supplier lists for magnets into Rafael, IAI/Elta, and Elbit are not provided in the cited materials, and at least one Elbit product page explicitly indicates that inquiries about rare earth magnets or sensors require direct follow-up through Elbit or other technical resources.

Finally, none of the reviewed platform sources quantify manufacturing or machining scrap rates, coolant volumes, or wastewater compositions associated with magnet production or finishing in Israel or in U.S. co-production facilities, which is a prerequisite for sizing and valuing an Algamol-type treatment deployment. As a result, primary research (supplier interviews, site visits, or analysis of publicly permitted industrial discharge data where accessible) is required to translate the clear strategic importance of magnets in these systems into a site-specific waste and wastewater opportunity model.

Chapter 16

US Iron Dome Co-Production and DFARS Magnet Sourcing

Executive Summary

The establishment of a U.S. production line for Israel's Iron Dome missile defense system and its American variant, SkyHunter, creates a significant new demand signal for domestically sourced, high-performance permanent magnets compliant with U.S. defense regulations. Through the Raytheon-Rafael Area Protection Systems (R2S) joint venture, a new "all-up-round" interceptor production facility in East Camden, Arkansas, is now operational, supported by over a billion dollars in contracts and U.S. aid designated for replenishing Israeli stockpiles and equipping U.S. forces. This domestic manufacturing effort directly implicates the Defense Federal Acquisition Regulation Supplement (DFARS), which mandates that critical materials, including rare earth magnets, be sourced from non-adversarial nations.

With an expanded DFARS prohibition taking full effect on January 1, 2027—covering the entire supply chain from mining to finished magnet—prime contractors like Raytheon (RTX) must secure compliant sources for the magnets essential to interceptor guidance and control systems. These systems rely on high-temperature samarium-cobalt (SmCo) magnets for seeker head gimbals and high-coercivity neodymium-iron-boron (NdFeB) magnets for fin actuators. While DFARS provides an exemption for NdFeB magnets made from recycled material milled and sintered in the U.S., no such exemption exists for SmCo magnets, creating a significant sourcing challenge. This report analyzes the emerging U.S. Iron Dome production ecosystem, maps the DFARS compliance requirements onto the magnet supply chain, and identifies the key domestic magnet suppliers positioned to support this critical defense industrial base initiative.

Iron Dome Co-Production Program Overview

The U.S.-Israeli collaboration on the Iron Dome system has evolved from financial support to direct industrial base integration. Over the last decade, the U.S. has provided billions in funding, including $1.6 billion between 2011 and 2021 and an additional $1 billion approved by Congress in 2022. This partnership deepened with the formation of the Raytheon-Rafael Area Protection Systems (R2S) joint venture in 2020, created explicitly to establish an Iron Dome production facility in the United States.

This initiative is driven by several factors:

- **Replenishment Needs**: Increased operational use of the system, particularly since October 2023, created an urgent need to replenish Israel's stockpile of Tamir interceptors, prompting a "multi-billion dollar contract" to accelerate production. A U.S. aid package approved in April 2024 allocated $5.2 billion (out of an $8.7 billion total) specifically for Israeli air and missile defense systems.
- **U.S. Military Adoption**: The U.S. military is procuring its own Iron Dome derivatives. The U.S. Army acquired two batteries for evaluation, and the U.S. Marine Corps is integrating the system as its Medium-Range Intercept Capability (MRIC) program, using the U.S. variant of the Tamir interceptor, known as SkyHunter. The Marine Corps' FY25 budget requested $111 million for 12 launchers and 242 missiles, and a separate planned order may include 1,840 interceptors.
- **Industrial Base Security**: The establishment of a U.S. production line is a strategic move to create a "robust American production base" and a shared U.S.-Israeli supply chain, enhancing resilience and ensuring interceptors can be delivered "efficiently and at scale".

The culmination of this effort is the new R2S facility in Arkansas, which received a $1.25 billion contract in late 2025 to begin supplying Tamir interceptors to Israel.

US Production Facilities and Capabilities

The domestic production of the Tamir and SkyHunter interceptors is centered around a growing network of Raytheon/RTX facilities, with the new R2S plant in Arkansas as the centerpiece.

- **R2S Facility (East Camden, Arkansas)**: Opened in November 2025, this is the first "all-up-round" production facility for Iron Dome interceptors in the United States. The joint venture's investment in the site grew from an initial $33 million to $63 million. The plant, which broke ground in February 2024, is slated to begin missile production in 2025 and will create up to 60 jobs. It is located at the Highland Industrial Park, building on RTX's existing presence in the area.
- **Raytheon Facility (Tucson, Arizona)**: Tucson is a major hub for Raytheon's missile defense activities. Sources from 2014 and later confirm that Raytheon was producing components for the Tamir interceptor at its Tucson facility, which entered full production by July 2015. In 2020, the joint venture also agreed to a separate plant investment in Tucson.
- **Raytheon Facility (Andover, Massachusetts)**: This facility is mentioned as being involved in the production of the seeker and guidance sections for missile systems, making it a likely contributor of critical components for the U.S.-built interceptors.
- **Huntsville, Alabama**: This location was noted as one of the sites that would benefit from the co-production arrangement, suggesting a role in engineering, prototyping, or component manufacturing.

This distributed manufacturing footprint indicates that while final assembly and integration of all-up-rounds will occur in Camden, critical subsystems and components are being produced across multiple specialized Raytheon sites.

US vs Israeli-Built Components

The co-production model involves a deep integration of U.S. and Israeli supply chains. While the new Camden facility is capable of all-up-round production, the existing supply chain already features significant U.S. content.

Raytheon states that the "majority of Tamir missile components are procured through the Raytheon supply chain in the United States". Another source notes that the Israeli production line historically receives approximately half of its components from Raytheon. The U.S. version of the interceptor, SkyHunter, shares 95-99% commonality with the Israeli Tamir, meaning the designs are nearly identical.

The establishment of a U.S. production line for systems like the seeker, guidance, and control sections is the critical factor for magnet sourcing. The production of these magnet-intensive subsystems in U.S. facilities (like Tucson for missile integration and Andover for seekers) means the components used within them are subject to U.S. procurement law, specifically DFARS.

Contract Values and Production Quantities

The scale of U.S. and Israeli demand is driving significant investment and large-scale production contracts.

- **R2S Production Contract**: In November 2025, the R2S joint venture was awarded a $1.25 billion contract to produce Tamir interceptors at the new Camden facility to supply Israel. This is the first production contract for the R2S partnership.
- **U.S. Aid Package**: The production surge is substantially funded by an $8.7 billion U.S. aid package approved in April 2024, which allocated $5.2 billion specifically for Israeli air and missile defense systems. One report

suggests this funding could cover the production of "about 100 Iron Dome batteries and at least 14,000 Tamir interceptor missiles".

- **U.S. Military Procurements**: The U.S. Army previously acquired two Iron Dome batteries with 480 missiles in a 2019 deal valued at $373 million. The U.S. Marine Corps has a planned order for three batteries, 44 launchers, and 1,840 interceptors for its MRIC program. The FY25 budget request for MRIC includes $111 million for 242 missiles.

Permanent Magnet Requirements for US-Built Interceptors

The Tamir and SkyHunter interceptors rely on high-performance permanent magnets for their agility and precision. These magnets are critical components within the guidance and control subsystems.

- **Seeker Head**: The interceptor uses an active radar and/or electro-optical seeker for terminal guidance. This seeker is mounted on a gimbal system that requires small, powerful motors to rapidly articulate the sensor. These gimbal motors use permanent magnets. Due to the high temperatures generated by aerodynamic heating on the missile's nose cone, these are likely high-temperature SmCo magnets.
- **Fin Actuators**: The missile is equipped with several steering fins for high-g maneuvering. Each fin is controlled by an actuator, typically a brushless DC motor that relies on a permanent magnet rotor. These actuators must perform under extreme thermal and vibratory stress, requiring high-temperature NdFeB or SmCo magnets.
- **Control Section Electronics**: Other sensors and electronic components within the guidance and control section may also contain permanent magnets.

Given the high-volume production plans, with thousands of interceptors required annually, the demand for these specialized, defense-grade magnets will be substantial.

DFARS Compliance for US-Built Interceptor Magnets

Any interceptor or component produced in the U.S. under a Department of Defense contract must comply with DFARS. The final DFARS rule published on May 30, 2024, implements and expands upon these restrictions.

- **The Regulation**: DFARS clause 252.225-7052 explicitly lists SmCo and NdFeB magnets as "covered materials".
- **The Deadline**: Effective January 1, 2027, the rule prohibits the acquisition of any covered material that was "mined, refined, separated, melted, or produced" in a covered country (China, Russia, North Korea, Iran). This covers the entire upstream supply chain.
- **Application**: Because the US-built Tamir/SkyHunter interceptors are part of a U.S. defense procurement (for the USMC and funded by U.S. aid for Israel), the magnets within their U.S.-made components must be DFARS-compliant. Prime contractors like Raytheon are required to flow this clause down to their subcontractors.

There is a critical distinction in compliance pathways for the two key magnet types:

- **NdFeB Magnets**: The rule provides an exception for NdFeB magnets manufactured from recycled material, but only if the milling of the recycled material and the sintering of the final magnet occur in the United States.
- **SmCo Magnets**: There is **no recycling exemption** for SmCo magnets. As of January 1, 2027, the entire supply chain—from the mining of samarium and cobalt ore—must be free of covered countries. This presents a much higher compliance hurdle.

Magnet Supply Chain for US Iron Dome

Given the stringent DFARS requirements, Raytheon and its subcontractors must source magnets for U.S. Iron Dome production from a very narrow list of qualified domestic suppliers. The likely Tier-1 suppliers for these magnet

assemblies include:

- **Electron Energy Corporation (EEC) / Permag**: EEC, part of the Permag group, is the only vertically integrated U.S. producer of SmCo magnets, controlling the process from alloy melt to finished magnet. This makes them a critical supplier for the SmCo magnets likely used in the interceptor's seeker head.
- **Arnold Magnetic Technologies**: Arnold is a key U.S. defense supplier with a long history of producing high-performance SmCo (RECOMA series) and other magnets. They have publicly stated their commitment to establishing a fully DFARS-compliant supply chain for both NdFeB and SmCo magnets ahead of the 2027 deadline.
- **Vacuumschmelze (VAC)**: While traditionally a German company, VAC has received a $94.1 million DoD award to establish a rare earth magnet manufacturing facility in Sumter, South Carolina, specifically to support defense needs and DFARS compliance. This plant will become a new domestic source once fully operational.

The extreme stress on the supply chain from surge demand for Iron Dome, combined with demand from other platforms like the F-35 (900 lbs of REEs) and DDG-51 destroyers (5,200 lbs of REEs), creates a significant risk of bottlenecks. Qualifying a second source for a critical missile component can take 12-24 months, making the existing capacity of these few suppliers a key strategic concern.

Production Surge and Supply Chain Stress

The surge in demand for Iron Dome interceptors following October 2023 places immense pressure on an already constrained global supply chain for high-performance magnets. Israel's need to replenish its stocks, combined with new U.S. production, will test the limits of domestic magnet producers.

The SmCo supply chain is particularly vulnerable. Samarium is almost exclusively sourced from China, and cobalt is also a critical material with a concentrated supply chain. The DFARS rule's lack of a recycling exemption for SmCo means that, post-2027, SmCo magnets for U.S. defense must originate from fully compliant ore-to-magnet chains, which are still nascent outside of China. This regulatory pressure, combined with high demand, creates a significant risk for programs like the US-built Iron Dome that rely on these materials for high-temperature applications.

Implications for Algamol

The ramp-up of U.S. Iron Dome co-production creates a concentrated and valuable opportunity for wastewater treatment and resource recovery technologies. The precision machining of SmCo and NdFeB magnets into final parts for seeker gimbals and fin actuators is a waste-intensive process. This waste stream, generated at Raytheon subcontractor facilities, is a key target.

- **Concentrated Waste Streams**: Instead of being diffuse, magnet machining waste for the Iron Dome program will be concentrated at a few Tier-2 supplier locations that specialize in grinding and finishing rare earth magnets.
- **High-Value Waste**: This machining sludge contains high concentrations of valuable and strategic materials, including samarium, cobalt, neodymium, and dysprosium.
- **DFARS Compliance Driver**: For NdFeB magnets, on-site recovery of machining waste provides a direct pathway to creating a DFARS-compliant feedstock. Under DFARS 252.225-7052, NdFeB magnets manufactured from recycled material are exempt, provided the milling and sintering occur in the U.S. An on-site recovery system would enable a Tier-2 supplier to create a secure, domestic source of recycled powder.
- **Strategic Value for SmCo**: While SmCo lacks a recycling exemption, the strategic importance of cobalt and samarium, coupled with the supply chain risks, makes on-site recovery and purification highly valuable for securing a resilient internal supply.

- **Ideal Customer Profile**: A large defense prime contractor like Raytheon has established Environmental, Health, and Safety (EHS) infrastructure and would be receptive to a pilot program that offers both cost savings and a tangible solution to a pressing supply chain and regulatory challenge. The priority target is the Tier-2 machining house supplying these finished magnet components to Raytheon's integration facilities in Tucson or Andover.

Key Unknowns and Intelligence Gaps

While the strategic direction is clear, several critical pieces of information remain proprietary or classified:

- The specific Tier-1 and Tier-2 magnet suppliers and machining subcontractors to Raytheon for the U.S.-built Tamir/SkyHunter program.
- The exact bill of materials for the interceptor, including the precise quantity, grade, and weight of SmCo and NdFeB magnets used in each seeker and actuator assembly.
- The division of labor for component manufacturing between U.S. and Israeli facilities under the new co-production agreements.
- Whether the technical specifications for magnets in the U.S.-built SkyHunter are identical to those in the Israeli-built Tamir, or if they have been "Americanized" with different materials.

US Iron Dome Co-Production and DFARS Magnet Sourcing: The Permanent Magnet Supply Chain for US-Built Interceptors

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Given the high-volume production plans, with thousands of interceptors required annually, the demand for these specialized, defense-grade magnets will be substantial.

DFARS Compliance for US-Built Interceptor Magnets

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There is a critical distinction in compliance pathways for the two key magnet types:

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Magnet Supply Chain for US Iron Dome

Given the stringent DFARS requirements, Raytheon and its subcontractors must source magnets for U.S. Iron Dome production from a very narrow list of qualified domestic suppliers. The likely Tier-1 suppliers for these magnet assemblies include:

- **Electron Energy Corporation (EEC) / Permag**: EEC, part of the Permag group, is the only vertically integrated U.S. producer of SmCo magnets, controlling the process from alloy melt to finished magnet. This makes them a critical supplier for the SmCo magnets likely used in the interceptor's seeker head.
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Production Surge and Supply Chain Stress

The surge in demand for Iron Dome interceptors following October 2023 places immense pressure on an already constrained global supply chain for high-performance magnets. Israel's need to replenish its stocks, combined with new U.S. production, will test the limits of domestic magnet producers.

The SmCo supply chain is particularly vulnerable. Samarium is almost exclusively sourced from China, and cobalt is also a critical material with a concentrated supply chain. The DFARS rule's lack of a recycling exemption for SmCo means that, post-2027, SmCo magnets for U.S. defense must originate from fully compliant ore-to-magnet chains, which are still nascent outside of China. This regulatory pressure, combined with high demand, creates a significant risk for programs like the US-built Iron Dome that rely on these materials for high-temperature applications.

Implications for Algamol

The ramp-up of U.S. Iron Dome co-production creates a concentrated and valuable opportunity for wastewater treatment and resource recovery technologies. The precision machining of SmCo and NdFeB magnets into final parts for seeker gimbals and fin actuators is a waste-intensive process. This waste stream, generated at Raytheon subcontractor facilities, is a key target.

- **Concentrated Waste Streams**: Instead of being diffuse, magnet machining waste for the Iron Dome program will be concentrated at a few Tier-2 supplier locations that specialize in grinding and finishing rare earth magnets.
- **High-Value Waste**: This machining sludge contains high concentrations of valuable and strategic materials, including samarium, cobalt, neodymium, and dysprosium.
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- **Strategic Value for SmCo**: While SmCo lacks a recycling exemption, the strategic importance of cobalt and samarium, coupled with the supply chain risks, makes on-site recovery and purification highly valuable for securing a resilient internal supply.
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- The exact bill of materials for the interceptor, including the precise quantity, grade, and weight of SmCo and NdFeB magnets used in each seeker and actuator assembly.
- The division of labor for component manufacturing between U.S. and Israeli facilities under the new co-production agreements.

- Whether the technical specifications for magnets in the U.S.-built SkyHunter are identical to those in the Israeli-built Tamir, or if they have been "Americanized" with different materials.

Chapter 17

Combined US-Israeli Interceptor Magnet Demand Under Surge

Executive Summary

Post–October 2023 operational demand and subsequent procurement actions have driven expansion of Iron Dome Tamir interceptor production in both Israel and the United States, with a new U.S. "all-up-round" Tamir/SkyHunter facility in East Camden/Camden, Arkansas established explicitly to accelerate serial output and serve both Israeli Iron Dome and the U.S. Marine Corps' MRIC program. Multiple sources tie this expansion to a U.S. aid package approved in April 2024 totaling $8.7B and including $5.2B dedicated to air and missile defense (Iron Dome, David's Sling, and laser defense). The practical implication for permanent magnets is that surge missile output increases demand for high-temperature and high-reliability permanent magnets used in guidance and control actuators, and that these magnets are subject to tightened sourcing rules effective January 1, 2027 under DFARS-related restrictions and provenance expectations.

The binding constraint is likely not magnet design knowledge (which is well understood in broad terms) but supply-chain capacity and compliance for high-temperature SmCo magnets, because U.S. industrial statements and government reporting emphasize dependence on foreign rare-earth processing and limited domestic specialty magnet production, with samarium supply and SmCo market share heavily concentrated in China and only a very small U.S. producer base for SmCo magnets. In contrast, NdFeB magnets are also strategically constrained but have more pathways for domestic capacity expansion (e.g., Mine-to-Magnet investments) and recycling-related compliance strategies than SmCo, as reflected by DoD investment programs and the attention to NdFeB processing-chain completeness in China.

Baseline Production

Public sources used in this dataset provide only limited, sometimes secondary estimates for pre-surge annual Tamir production, with one internal "input document" estimate citing approximately 1,000–2,000 Tamirs per year pre–October 2023. Several public sources instead anchor baseline demand through stockpile and procurement indicators (e.g., U.S. Army acquisition of 480 missiles in the 2019–2021 timeframe, and U.S. Marine Corps plans to acquire 1,840 Tamir/SkyHunter missiles for MRIC). These indicators establish that U.S. demand existed prior to the current surge and that Israel–U.S. co-production channels were already in place before the Arkansas plant opened, including U.S. manufacture of a majority share of components in some periods.

Baseline unit-cost anchors (useful for inferring procurement volumes when totals are known) put Tamir interceptor costs in the $40,000–$50,000 range in multiple references, though broader ranges also appear depending on accounting conventions and urgency premiums.

Surge Scenarios

The surge scenarios in open reporting are most defensible when framed as (i) a production-capacity step change (new U.S. all-up-round plant plus accelerated Israeli serial lines) and (ii) a range of annual output targets and procurement quantities inferred from budgets and public contract statements, because many official announcements do not disclose annual unit production numbers. One estimate in the dataset explicitly frames post–October 2023 surge demand as "8,000–12,000+ interceptors per year" for Israel, reflecting the intensity and duration of high-tempo

engagements and stockpile replenishment pressures. A separate internal estimate in the dataset states Tamir production could "surge to potentially 10,000+/year" post–October 2023, reinforcing the possibility of five-to-tenfold increases over a ~1,000–2,000/year baseline estimate.

For U.S. demand, MRIC procurement provides a bounded near-term (FY25) quantity signal, with a Marine Corps FY-25 budget request that includes $111M to support purchase of 12 launchers and 242 missiles, and a separate contract action covering 80 missiles in 16 all-up-round magazines with delivery scheduled for April 2027 (work in Tucson, AZ). Separately, Israeli–U.S. industrial reporting cites a U.S. Marine Corps order for 2,000 SkyHunter missiles as the intended load for the U.S. production line, indicating an upper-bound programmatic target even where annual profiles are not disclosed.

Given these constraints, this report defines three surge scenarios as planning constructs rather than confirmed production commitments:

- Scenario A (Moderate Surge): Israel replenishment plus early U.S. MRIC fielding, bounded by FY25 MRIC quantities and the ramp-up implied by "accelerating serial output."
- Scenario B (High Surge): Israel demand toward the 8,000–12,000+/year range plus continued U.S. MRIC procurement growth during 2024–2026 ramp period.
- Scenario C (Maximum Surge plus U.S. adoption): High Israeli demand plus U.S. MRIC full fielding across three battalions during 2026–2028, using published fielding windows and contract growth to approximate sustained demand.

Procurement Signals

Multiple U.S. and Israeli public statements connect accelerated interceptor production to large funding packages and contracts, even when unit quantities are not disclosed. U.S. Congress-approved April 2024 aid is repeatedly cited as $8.7B total, with $5.2B earmarked for air and missile defense programs including Iron Dome and David's Sling, establishing a multi-program funding pool that can drive magnet demand across multiple interceptor families beyond Tamir alone. On the industrial side, the Raytheon–Rafael joint venture (R2S) received a $1.25B contract to supply Tamir interceptors to Israel, and the contract disclosure coincided with inauguration/opening of the U.S. Tamir production facility. Several sources also record a $33M capital investment for the new U.S. facility, while one state economic development release describes the project as an approximately $63M investment, implying possible scope growth or additional related investments beyond the originally cited figure.

MRIC-specific procurement signals include contract values reaching $412.5M in total (including a potential $380M order), which indicates an expanding U.S. demand curve for Tamir-derived interceptors even if annual unit numbers beyond FY25 are not stated in those sources.

Magnet-Intensive Subsystems

Open sources in this dataset generally do not disclose per-interceptor permanent magnet mass, magnet grade, or exact magnet counts, and at least one technical source explicitly notes it does not address "the weight of permanent magnets or the quantity of actuator motors in guided munitions." Nonetheless, multiple sources support the qualitative mapping that guided interceptors use permanent magnets in actuator motors and fin actuation, and that missile subsystems can use SmCo magnets (often chosen for thermal stability and resistance to demagnetization) as well as NdFeB magnets (common in fin actuators).

Within the dataset, one internal synthesis asserts that Tamir includes permanent-magnet-containing components in the guidance section (seeker head motor and gimbal actuators) and control section (fin actuators), with a possibility

of additional motor-related magnets closer to propulsion-related structures. A separate internal table-level statement ties SmCo magnets to missile propulsion-system magnets and control-actuator magnets, supporting the inference that at least some interceptor subsystems in the relevant performance envelope may require SmCo rather than standard NdFeB grades.

Parametric Magnet Demand Model

Because the dataset lacks publicly supported values for magnet mass per interceptor or magnet mass per actuator, this report models magnet demand parametrically and ties the model to confirmed interceptor quantities and credible surge ranges. Let:

- N_T = annual Tamir-equivalent interceptor output (Israel + U.S. co-production).
- m_{SmCo} = average SmCo magnet mass per interceptor (kg/interceptor), covering seeker/gimbal and any high-temperature actuator use where SmCo is selected.
- m_{NdFeB} = average NdFeB magnet mass per interceptor (kg/interceptor), covering fin actuators and other permanent-magnet motor functions where NdFeB is used.

Then annual magnet demand in mass terms is:

- SmCo demand (kg/year) = $N_T \times m_{SmCo}$.
- NdFeB demand (kg/year) = $N_T \times m_{NdFeB}$.

The table below summarizes the interceptor quantity inputs available from sources and how they map into the three scenarios; it intentionally keeps magnet mass outputs symbolic due to absent public per-interceptor magnet mass data.

Scenario	Israel Tamir demand anchor	U.S. MRIC SkyHunter demand anchor	Combined annual output basis	Magnet mass output form
A	No official annual rate disclosed; constrained by stated intent to accelerate serial output and replenish stockpiles.	FY25 request: 242 missiles (plus 12 launchers) and early fielding timeline.	N_T treated as "Israel replenishment + 242 U.S. missiles" for minimum planning case.	SmCo = $N_T m_{SmCo}$; NdFeB = $N_T m_{NdFeB}$.
B	Estimate of 8,000–12,000+ interceptors/year under surge conditions is explicitly stated in one source set.	MRIC contract growth to $412.5M suggests expanding procurement beyond initial buys, but annual quantities not stated in those contract summaries.	N_T approximated by Israel surge range plus incremental U.S. production capacity ramp during 2024–2026.	SmCo = $N_T m_{SmCo}$; NdFeB = $N_T m_{NdFeB}$.

Scenario	Israel Tamir demand anchor	U.S. MRIC SkyHunter demand anchor	Combined annual output basis	Magnet mass output form
C	Israel surge range as in Scenario B (upper bound), recognizing that official quantities are often undisclosed.	Full fielding across three battalions scheduled 2026–2028 and delivery window signals sustained procurement, with "full-rate production begins this year" in one MRIC description.	(N_T) treated as Israel surge range plus sustained U.S. MRIC missile deliveries through 2026–2028.	SmCo = (N_T m_{SmCo}); NdFeB = (N_T m_{NdFeB}).

SmCo Supply Chain Capacity

Multiple sources describe SmCo supply as unusually concentrated and therefore vulnerable under surge conditions, including a claim that China accounts for "more than 90 percent of the global samarium–cobalt magnet market." A market estimate in the dataset puts global SmCo magnet production capacity at approximately 2,400 tonnes per year and indicates aerospace and defense account for over 30% of SmCo consumption, which implies that a defense surge can materially compete for a constrained global capacity pool even if defense is not the dominant end market in every segment. Separately, U.S. industrial statements emphasize that through EEC, Permag is "the only U.S. producer" of SmCo magnets, reinforcing the thinness of domestic SmCo capacity and the likely need for allied/foreign upstream sourcing even when final manufacturing is domestic.

From a surge-production perspective, two additional supply-side constraints are salient in the dataset: (1) samarium supply dependence (with one source stating "Samarium supply: 95%+ from China"), and (2) cobalt supply concentration and disruption risk (with one internal summary stating DRC accounted for 76% of mined output in 2024 and referencing a February 2025 DRC export ban).

NdFeB Supply Chain Considerations

NdFeB magnets are described in government reporting as strategically important and "irreplaceable" in key defense applications including missile guidance and fin actuators, and the U.S. is shown as having imported 75% of sintered NdFeB magnet supply from China in 2021 in a congressional report. A GAO report also notes that China is the only country with processing capabilities for each stage of the NdFeB permanent magnet supply chain, which creates systemic risk when defense missile production surges. However, the dataset also documents significant U.S. industrial-policy investments aimed at expanding domestic rare-earth processing and magnet manufacturing, including DoD awards totaling more than $439M since 2020 and specific funding to MP Materials (with projected full-scale milestones through 2025), Lynas USA (oxide capability by 2026), Noveon Magnetics (San Marcos, TX magnet facility), and E-VAC Magnetics (commercial-scale magnet capability by 2025).

From a manufacturing yield perspective (relevant to effective supply), one U.S. government report states that "final magnet machining has a recovery rate of 60 to 90 percent" depending on magnet size and complexity, which implies significant swarf/sludge and scrap generation that can tighten effective supply during surge periods.

Bottlenecks

Several sources directly connect defense missile production surges (2024–2025) to increased demand for rare-earth permanent magnets, and also emphasize that concentrated overseas supply chains for rare-earth magnets are a limiting factor for precision-guided munitions output under surge. Separately, one source identifies "specific bottlenecks like rocket motors" as constraints across missile lines, underscoring that magnets compete with other constrained subsystems for industrial attention and qualification bandwidth.

For magnets specifically, the dataset supports five bottleneck categories that align with the user's framing, though not all are quantified in the public record:

- Raw materials bottleneck: samarium supply dependence and cobalt supply concentration create risk for SmCo production under surge conditions.
- SmCo manufacturing bottleneck: a narrow supplier base and limited domestic production footprint for SmCo magnets raises the likelihood that SmCo becomes the binding constraint for high-temperature actuator and seeker assemblies.
- NdFeB processing bottleneck: China's unique completeness across NdFeB processing stages and high import dependence for sintered NdFeB creates vulnerability during periods of heightened demand and/or export restrictions.
- Qualification bottleneck: DFARS-related sourcing and provenance requirements are tightening into 2025–2027 with an expanded prohibition effective January 1, 2027, raising the time and cost to qualify alternative sources for magnets and their upstream materials.
- Compliance asymmetry for SmCo: one dataset source explicitly states "SmCo has NO recycling exemption under DFARS," which implies that recycling-driven compliance pathways may be more constrained for SmCo than for NdFeB under defense procurement rules as interpreted in this dataset.

Machining Waste Implications

Even though per-interceptor magnet mass is not publicly supported in the provided sources, machining yield data implies that a substantial fraction of magnet material can become machining swarf/sludge, which scales with interceptor production rates and precision requirements. One dataset source provides SmCo swarf composition (approximately 23% Sm and 50% Co by weight, with Fe, Cu, and Zr constituents), implying that SmCo machining waste is not only an environmental liability but also a concentrated strategic-metal stream whose value and compliance relevance increase during surge production.

This matters for surge conditions because a higher missile output implies higher magnet machining throughput at Tier-2/Tier-3 suppliers, which in turn implies higher volumes of magnet-bearing waste streams and greater incentives to treat and recover valuable metals domestically—especially when procurement rules and supply chain security concerns are tightening in parallel with surge demand.

Assumptions and Uncertainties

Most platform-specific magnet composition, grade selection, and per-interceptor magnet mass values are not available in the cited public sources, and at least one technical source explicitly notes it does not provide permanent magnet weights or actuator motor quantities in guided munitions, limiting the defensibility of any fixed "kg of magnets per interceptor" estimate from this dataset alone. Many production announcements (including large contracts and aid-funded procurements) do not disclose interceptor quantities, and multiple sources explicitly state that quantities were not provided for security or procurement reasons, which forces scenario modeling to rely on bounded procurement signals (e.g., FY25 MRIC missiles) and a small number of surge-demand estimates (e.g., 8,000–12,000+

interceptors/year). Finally, supply-side capacity for SmCo is described as highly concentrated and vulnerable to disruption, but the dataset provides limited, potentially inconsistent global capacity figures across market reports, so capacity analysis should be treated as indicative rather than definitive unless validated against primary industry production statistics.

Chapter 18

The SmCo DFARS Compliance Gap

Executive Summary

Samarium-cobalt (SmCo) permanent magnets are irreplaceable in high-temperature defense and aerospace applications, including missile guidance systems, radar traveling-wave tubes (TWTs), and satellite actuators, where performance above 300℃ is required. However, the U.S. defense industrial base faces a critical compliance gap concerning these magnets under the Defense Federal Acquisition Regulation Supplement (DFARS). The final rule, published May 30, 2024, expands sourcing prohibitions to the entire supply chain—from mine to magnet—effective January 1, 2027, for "covered materials" sourced from China, Russia, Iran, or North Korea. While SmCo magnets are explicitly defined as a covered material, they lack a critical compliance pathway available to their lower-temperature counterparts, neodymium-iron-boron (NdFeB) magnets.

The central issue is a significant asymmetry in the regulation: DFARS provides an exemption for NdFeB magnets manufactured from recycled feedstock that is milled and sintered in the United States. SmCo magnets have no such recycling exemption. This gap is compounded by a precarious supply chain; approximately 95% of the world's samarium is processed in China, and while non-China processing is emerging in Europe, domestic U.S. capacity is nascent. The Department of Defense (DoD) has invested over $439 million in its "Mine-to-Magnet" initiative, but these efforts have predominantly focused on the NdFeB supply chain.

Without a policy remedy, defense programs reliant on SmCo magnets will face an untenable choice between sourcing non-compliant materials or facing significant production disruptions after the 2027 deadline. This report analyzes the specifics of the SmCo compliance gap, evaluates the state of the global supply chain, and proposes actionable policy remedies, including extending the recycling exemption to SmCo, issuing a class nonavailability determination (NAD), and strategically investing in domestic SmCo recycling and stockpiling.

DFARS Treatment of Samarium-Cobalt Alloys

Samarium-cobalt (SmCo) magnets are explicitly identified as a "covered material" under DFARS 225.7018 and the corresponding contract clause 252.225-7052. These regulations prohibit the DoD from procuring certain sensitive materials that are mined, refined, separated, melted, or manufactured in "covered countries," defined as China, Russia, Iran, and North Korea.

The prohibition is being implemented in phases. According to the final rule published on May 30, 2024, the restriction through December 31, 2026, covers the later stages of production, such as "Melting samarium with cobalt to produce the samarium-cobalt alloy" and all subsequent manufacturing steps like powder formation, pressing, sintering, and magnetization.

Effective January 1, 2027, this restriction expands dramatically to cover the entire supply chain. From that date, it will be prohibited to acquire any SmCo magnet if any part of its production—from the mining of cobalt and samarium ore or feedstock, through all stages of processing and manufacturing—occurred in a covered country. This expanded prohibition also explicitly includes recycled material in its scope, a critical distinction from the treatment of NdFeB

magnets. This restriction is in addition to any applicable restrictions on the melting of specialty metals under DFARS 252.225-7009.

The Recycling Exemption Asymmetry

A central challenge for SmCo compliance is a critical asymmetry in the DFARS rules compared to NdFeB magnets. The statute (10 U.S.C. 4872) and the implementing DFARS clause 252.225-7052 provide a specific exemption for NdFeB magnets manufactured from recycled material. This exemption allows for the procurement of an end item containing an NdFeB magnet made from recycled stock, provided that the "milling of the recycled material and sintering of the final magnet takes place in the United States".

This pathway does not exist for SmCo magnets. The Federal Register explicitly notes that the legislative amendments creating the NdFeB recycling exemption "did not add exceptions for the remaining covered materials," which includes SmCo. This means that even if SmCo scrap is recovered and re-processed entirely within the United States, the resulting magnet would be non-compliant if the original samarium or cobalt ore was mined or initially processed in a covered country. The lack of a recycling exemption for SmCo is a significant policy gap, likely stemming from the fact that the SmCo recycling market is less commercially established than that for NdFeB. This asymmetry effectively closes off a vital potential compliance pathway for SmCo, forcing complete reliance on a compliant primary supply chain that is not yet mature.

Samarium Supply Chain Analysis

The viability of a DFARS-compliant SmCo supply chain is fundamentally constrained by the global geography of samarium processing. Samarium, a middle-heavy rare earth element, is almost entirely processed in China. Various sources estimate that China controls between 70% and 99% of global rare earth refining and processing, with a near-monopoly on heavy rare earths. This dominance has been identified as a significant national security risk, as demonstrated by China's use of export controls on rare earths, including samarium. A USGS analysis concluded that a trade disruption of samarium from China would have the largest probability-weighted negative impact on U.S. GDP of all rare earths, estimated at nearly $4.5 billion.

While the situation is critical, a non-Chinese supply chain is beginning to emerge. Key players include:

- **Solvay** at its facility in La Rochelle, France, which is now the largest separation facility outside of China and has begun producing samarium oxide.
- **Neo Performance Materials** (Silmet) in Estonia, which is the only other commercial rare earth separation plant in Europe.
- **Lynas Rare Earths**, which operates in Australia and Malaysia and is developing heavy rare earth separation capacity in Texas with DoD funding.

However, these efforts will take years to reach a scale sufficient to displace Chinese production, with DoD assessments suggesting it could take a decade or more to establish fully independent domestic sources.

Cobalt Supply Chain Analysis

The cobalt supply chain presents a different but related set of challenges. While cobalt is not sourced from DFARS-prohibited countries, its supply chain is highly concentrated. The Democratic Republic of the Congo (DRC) accounts for approximately 70-76% of the world's mined cobalt. This concentration creates significant supply risk, which is compounded by the fact that an estimated 80-90% of DRC-origin cobalt is further refined in China.

Furthermore, the DRC cobalt sector is plagued by ethical concerns, including documented cases of forced labor and human rights abuses, which present reputational and legal risks for defense contractors. The U.S. government has recognized this vulnerability and is taking steps to reshape cobalt supply chains, including supporting new refining capacity and considering additions to the National Defense Stockpile.

US SmCo Manufacturer Sourcing Practices

The United States has a very limited domestic industrial base for SmCo magnet production. **Electron Energy Corporation (EEC)**, now part of the unified **Permag** brand, is the only U.S.-based, vertically integrated domestic supplier of SmCo magnets, meaning it produces the alloy in-house in Landisville, PA. **Arnold Magnetic Technologies** also produces its RECOMA® SmCo grades in the U.S. (Rochester, NY) and describes itself as an integrated supplier, though its level of vertical integration for SmCo alloy is less clear.

Both companies are actively pursuing DFARS compliance. Arnold has announced a target of mid-2026 for its supply chain to meet the 2027 requirements and has entered strategic agreements with non-China sources like Solvay and Less Common Metals (LCM) to secure samarium oxide. Similarly, Permag (EEC) is working with compliant global suppliers and is part of a tri-national partnership with Solvay and LCM to establish a secure samarium supply chain.

However, a critical gap remains: while these companies can melt alloy in the U.S. to meet current DFARS requirements, the ultimate origin of the samarium metal they use is a significant vulnerability. Publicly available information does not confirm whether their samarium feedstock can be traced to a non-covered country mine and separation facility, which will be required after the January 1, 2027 deadline.

Nonavailability Determination for SmCo

DFARS provides a mechanism to address supply gaps through nonavailability determinations (NADs). Under DFARS 225.7018-4, DoD can issue both individual NADs for a single contract and class NADs that apply more broadly. An NAD can be granted if compliant materials ”of satisfactory quality and quantity, in the required form, cannot be procured as and when needed at a reasonable price”.

- **Individual NADs** can be approved by the head of a contracting activity.
- **Class NADs** require approval from the Under Secretary of Defense for Acquisition and Sustainment (USD(A&S)) and must undergo a public notice period of at least 30 days to solicit information from interested parties.

Historically, DoD has issued national security waivers for non-compliant, high-performance SmCo magnets, acknowledging the persistent lack of compliant sources. Given the looming 2027 deadline and the immature state of the domestic SmCo supply chain, many industry stakeholders have recommended that DoD proactively issue a class NAD for SmCo magnets to prevent widespread procurement disruptions. Without a class NAD, every contract requiring SmCo magnets would need an individual determination, creating an unsustainable administrative burden for high-volume defense programs.

DoD Mine-to-Magnet Investments

The Department of Defense has committed over $439 million since 2020 to re-establish a domestic ”Mine-to-Magnet” supply chain for rare earth materials under authorities like the Defense Production Act. This initiative is a direct policy response to the supply chain vulnerabilities highlighted by China's market dominance.

However, a review of the funded projects reveals a strong emphasis on the NdFeB magnet supply chain. Major awards have gone to:

- **MP Materials** for light and heavy rare earth separation and NdFeB magnet production.
- **Lynas USA** to establish light and heavy rare earth separation facilities in Texas.
- **Noveon Magnetics** and **E-VAC Magnetics (a subsidiary of VAC)** for domestic NdFeB magnet manufacturing and recycling.

While these investments are critical, there is a comparative lack of large-scale, publicly announced DoD funding specifically targeting the SmCo supply chain. Although President Trump issued a DPA Title III Determination that included "samarium-cobalt magnet manufacturing," subsequent large-scale investments have largely prioritized NdFeB. This funding gap overlooks the fact that for many high-temperature defense applications, SmCo is more critical and has fewer substitutes than NdFeB.

SmCo Recycling

Technical literature demonstrates that recycling SmCo magnets is feasible, with multiple processes capable of achieving high recovery rates for both samarium and cobalt. Promising methods include:

- **Hydrometallurgy**: Acid leaching followed by solvent extraction or precipitation can recover over 95% of Sm and Co in laboratory settings.
- **Pyrometallurgy**: Processes like selective chlorination have shown samarium recovery yields as high as 99.7% at the bench scale.
- **Ionic Liquids**: Oxidative dissolution using ionic liquids has been demonstrated to recover metals effectively, with the solvent being reusable over multiple cycles.

Despite this technical feasibility, commercial-scale SmCo recycling is not well-established. This is primarily due to economic factors. The market for SmCo magnets is significantly smaller than for NdFeB, meaning the volume of manufacturing scrap and end-of-life magnets is lower. This makes it difficult to achieve the economies of scale necessary for a dedicated SmCo recycling facility to be profitable, especially when competing with primary raw materials. While companies like Cyclic Materials are building capacity to process recycled magnets, their initial focus is on the more abundant NdFeB-containing feedstock.

Policy Remedies

The SmCo DFARS compliance gap requires proactive policy intervention to avoid significant disruption to critical defense programs. Based on the analysis, several remedies should be considered:

1. **Extend the DFARS Recycling Exemption to SmCo**: Congress and the DoD should amend 10 U.S.C. 4872 and the corresponding DFARS clause to create a recycling exemption for SmCo magnets, analogous to the one for NdFeB. This would enable U.S. companies like EEC/Permag to use recycled SmCo feedstock (from any origin) as long as the final milling and sintering occurs domestically, immediately creating a viable compliance pathway.
2. **Issue a Class Nonavailability Determination (NAD) for SmCo**: As an interim measure, the USD(A&S) should issue a class NAD for SmCo magnets and alloys. This would provide temporary relief for all defense contractors, acknowledging that a fully compliant primary supply chain does not currently exist and will take years to build.
3. **Fund Domestic SmCo-Specific Infrastructure**: DoD's Mine-to-Magnet and DPA Title III investments should specifically target gaps in the SmCo supply chain. This includes funding for domestic samarium separation from U.S. or allied rare earth concentrates and co-funding commercial-scale SmCo recycling facilities, potentially co-located with existing magnet producers.
4. **Expand the National Defense Stockpile (NDS)**: Congress should authorize and fund an expansion of the NDS to acquire and hold not just samarium oxide, but also DFARS-compliant SmCo alloy and finished magnet

blocks. This would create a strategic buffer to mitigate supply shocks and provide a ready source of material for defense surges.

5. **Qualify Allied Sources:** The U.S. should formalize agreements that recognize SmCo materials produced and processed in allied nations with secure supply chains (e.g., samarium oxide from Solvay in France, alloy from VAC in Germany or LCM in the UK) as compliant sources under DFARS, promoting "friend-shoring" of critical capabilities.

Compliance Timeline Analysis

The timeline for establishing a fully DFARS-compliant SmCo supply chain extends well beyond the January 1, 2027, deadline. While individual companies like Arnold and Permag are targeting compliance by mid-2026, this relies on a nascent and fragile non-Chinese primary supply chain.

Key timeline constraints include:

- **Domestic Separation:** DoD assessments indicate it can take a decade or longer to establish new domestic sources for rare earths. Even with investment, facilities for separating samarium from concentrate are unlikely to be operating at scale in the U.S. before the late 2020s or early 2030s.
- **Regulatory Action:** Extending the recycling exemption would require a federal rulemaking process, which typically takes 12-18 months at a minimum. Issuing a class NAD requires a 30-day public notice period but could be accomplished more quickly if prioritized by DoD leadership.
- **Facility Construction:** New processing plants and magnet factories, such as those being built by MP Materials and Lynas, have construction and commissioning timelines of 2-4 years.

Given these realities, it is highly probable that the defense industry will not have access to sufficient volumes of fully compliant SmCo magnets by 2027 without interim policy relief, such as a class nonavailability determination.

Implications for Algamol

The SmCo DFARS compliance gap creates a significant strategic opportunity for wastewater treatment and resource recovery technology providers like Algamol. The challenges outlined in this analysis directly enhance the value proposition for on-site recovery of samarium and cobalt from magnet manufacturing waste streams.

- **Direct Compliance Enabler (with Policy Change):** If DFARS is amended to include a recycling exemption for SmCo (Remedy 1), Algamol's technology becomes a direct compliance enabler. A magnet manufacturer like EEC/Permag could use an on-site Algamol system to recover samarium and cobalt from its own machining swarf, re-introduce it into its alloy production, and certify the final magnet as DFARS-compliant under the recycling provision.
- **Strategic Value Creation (Current Policy):** Even without a recycling exemption, on-site recovery of SmCo waste has immense strategic value. The recovered cobalt (~50 wt% of SmCo scrap) and samarium have significant economic value, estimated at $12,000-$17,000 per tonne of swarf, primarily from cobalt. By recovering these materials, a defense contractor can hedge against price volatility, reduce reliance on external suppliers, and improve supply chain resilience.
- **Policy-Relevant Capability:** Algamol can position itself as a provider of a key enabling technology that supports DoD's industrial base resilience goals. By demonstrating a commercially viable method for SmCo recycling at the point of waste generation, Algamol can provide data to support policy changes, such as the extension of the recycling exemption. Engaging with DoD's Industrial Base Policy office and advocating for such changes, with EEC as a potential demonstration partner, could be a powerful go-to-market strategy.

Key Unknowns and Intelligence Gaps

Despite extensive analysis of public records, several critical questions remain that are likely answered only in proprietary or classified domains:

- Is the Department of Defense actively considering or drafting a class nonavailability determination specifically for samarium-cobalt magnets ahead of the 2027 deadline?
- What are the precise primary sourcing chains for samarium and cobalt metal for the key U.S. SmCo producers (EEC/Permag and Arnold)? Can they currently trace their feedstock to a non-covered country mine?
- What is the true volume of SmCo manufacturing scrap and swarf generated annually within the U.S. defense industrial base, which would define the total addressable market for a recycling technology?
- Will the January 1, 2027, deadline be strictly enforced for SmCo-dependent programs, or is it likely to be extended or modified as the date approaches and supply gaps remain?
- What are the current holdings, if any, of samarium metal, samarium oxide, or SmCo alloy within the National Defense Stockpile?

Answering these questions would require direct engagement with DoD program offices, defense prime contractors, and the magnet manufacturers themselves.

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## Chapter 19

## Bench-Scale Validation Protocol

### Objective and scope

This protocol is designed to validate an algae-based biosorption approach for removing and recovering r

The experimental scope explicitly includes batch sorption testing (pH effects, dose-
response, kinetics, and equilibrium/isotherms) run in triplicate with controls, followed by column tes
up assessment.

### Test matrix design

The work should be executed as three phases to separate fundamental biosorption behavior from real-
matrix effects (oils/emulsions, suspended solids, and competing ions) and then to confirm continuous-
flow performance.

Phase 1 uses synthetic wastewaters to map pH, contact time, temperature, and biosorbent dosage over the
9, 5–360 min, 0.5–10 g/L, and 20–40 °C).

Phase 2 repeats the key Phase 1 conditions using real machining coolant (or coolant-
derived aqueous phase) as the feed and includes pre-treatment for oil/emulsion removal prior to biosorp

Phase 3 uses fixed-bed column tests to generate breakthrough curves and dynamic capacities, because col
bed) testing is the standard pathway for continuous flow and breakthrough studies after batch equilibri

### Recommended experimental matrix
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The table below provides a practical minimum matrix aligned to the required screening windows and commo
60 min with equilibrium typically within 2–6 h).

Sub-test	Purpose	Conditions to run	Primary response variables
pH edge (batch)	Identify pH window where uptake is maximized without confounding hydroxide precipit
40 °C	% removal, q (mg/g), evidence of precipitation (turbidity/solids)
Kinetics (batch)	Determine time to reach near-equilibrium and fit kinetic models	t = 5, 10, 15, 30
360 min window	PFO/PSO rate constants; time to 90% of equilibrium uptake
Isotherms (batch)	Determine qmax and affinity constants	Multiple C0 points per metal system; fit I
Dose-response (batch)	Determine minimum dose needed to meet removal targets	0.5–
10 g/L (screen) at fixed pH and time	% removal vs g/L, q (mg/g) vs g/L
Regeneration cycling	Determine reusability and metal recovery	≥5 adsorption/desorption cycles wi
1.0 M HCl/HNO3	% desorption, capacity retention, eluate quality

Synthetic wastewater preparation

Synthetic matrices should be prepared from analytical-grade parent salts and made fresh as working solu

Two synthetic wastewater families are required to represent the two defense-
relevant magnet waste streams targeted in this protocol: SmCo-dominant streams (cobalt and samarium) a
dominant streams (neodymium/praseodymium with dysprosium as the high-temperature performance driver).

SmCo-type synthetic matrix

Prepare an SmCo-type solution using SmCl3, CoCl2, FeCl3, and CuSO4, targeting Sm = 50–
500 mg/L and Co = 200–2000 mg/L as the primary concentration ranges for the bench challenge tests.

If the test plan includes multicomponent interference studies (recommended for real coolants), Fe and C
solutes because multicomponent systems can show reduced uptake due to competition for limited binding s

NdFeB-type synthetic matrix

Prepare an NdFeB-type solution using NdCl3, PrCl3, DyCl3, and FeCl3, targeting Nd = 100–
1000 mg/L and Dy = 10–100 mg/L for bench challenge tests.

Because many biosorption studies on REEs report optimal performance near pH ≈5 for dried biomass system
6 even though the protocol-wide screening window extends from pH 3–9.

Algae biomass preparation

Biomass should be produced and handled in ways that support both (i) bench screening with dried or other

If the Algamol concept uses acid-tolerant extremophile algae, one demonstrated approach is to grow Galo
metal solutions.

Because multiple sources report that inactive (dead) biomass can be effective and often has faster kine
dried) to quantify achievable capacities without growth-dependent confounders.

Pre-treatment module

Real machining coolants can contain oils and stable emulsions, so the protocol should include a pre-
treatment step to remove oil/emulsion before biosorption (especially before column tests) using an ult
end unit operation.

If pre-treatment is used, evaluate biosorption on both (i) raw coolant aqueous phase after coarse settl
treated permeate to quantify how emulsion removal changes apparent uptake and analytical stability.

Batch sorption tests

Batch testing should be conducted in triplicate and include blanks and controls, because triplicate bat

Batch setup

A representative REE batch format is to contact dried biosorbent dosages in the 0.5–
3 g/L range with fixed-volume solutions under agitation and temperature control; one detailed example u

A representative smaller-volume batch format (suitable for high-throughput screening) incubates 10–
20 mg dry biomass in 2 mL metal solutions for ~3 h under constant shaking at room temperature, with cent

pH effects

pH should be treated as a primary control variable because uptake is strongly pH-
dependent across biosorption systems, and higher uptake of cationic metals is often observed as pH incr
6 range (until precipitation becomes significant).

For Nd(III) on Chlorella vulgaris using magnet-derived leachate, one study found an optimum pH of 5 at 2
35°C, while at 50°C the optimum pH shifted to ~4 due to Nd precipitation risk at higher pH combined with

Independent work on REE biosorption also identified an optimum equilibrium pH of ~4.9 ± 0.2 for favorab

Kinetics and equilibrium

Kinetic tests should capture early-time uptake because many systems exhibit rapid initial adsorption (
60 minutes) followed by a slower approach to equilibrium, and equilibrium for cationic metals is common
6 h in biosorption studies.

For Nd(III) biosorption on C. vulgaris, equilibrium was reported to be reached within 30 minutes across
second-order model with R2 > 0.99.

For some microbial/algal systems, much faster equilibration is also plausible; for example, Ce(III) ad

Equilibrium and isotherm modeling

Equilibrium isotherms should be fit using standard models such as Langmuir and Freundlich, which are co

For Nd(III) on C. vulgaris, a Langmuir fit with R2 = 0.98 and qmax = 188.68 mg/g (KL = 0.029 L/mg) was re
performing" REE uptake can look like in a controlled system.

For REE mixtures (La, Ce, Eu, Yb), maximum biosorption uptakes from a Langmuir model were reported as 15

Real coolant testing

Real coolant tests are required because the literature emphasizes that many studies focus on idealized
realistic matrices can exhibit reduced sorption efficiency due to competing ions and matrix effects.

Real coolant samples should be pre-characterized for major water-quality parameters and matrix load be
treatment prior to ICP-MS measurement.

For real-matrix REE testing, one demonstrator approach used a complex multi-
element mixture intended to mimic industrial effluents, illustrating the value of including multiple c

Desorption and regeneration

Regeneration should be tested by stripping metals from loaded biomass using acid solutions and evaluati
desorption cycles, because regeneration is necessary for reuse and enables production of a metal-
bearing eluate for downstream recovery.

This protocol's required regeneration screen is acid stripping over 0.01–
1.0 M HCl or HNO3 with a minimum of five cycles, with regeneration efficiency targeted at >80%.

Where rapid desorption kinetics are desired, one reported approach achieved desorption equilibrium in
20 minutes using either 0.03 M HCl (pH 1.5) or 0.1 M Na2EDTA (pH 5), enabling relatively short elution s

Where an HNO3-based stripping approach is preferred, another reported desorption procedure treated 0.5

Analytical methods

Analytical method selection should be driven by matrix strength (TDS/suspended solids) and required de
MS generally provides much lower detection limits than ICP-OES but has lower tolerance for high-
TDS matrices, often requiring dilution or matrix separation.

Sample collection and preservation

For dissolved analyte measurements, aqueous samples should be filtered through a 0.45 µm membrane and a
metals methods.

For ICP-MS workflows, acid-preservation after collection is also standard practice; one application no

Metals quantification

ICP-OES is appropriate for mg/L-level quantification of metals in aqueous samples after suitable filtr

ICP-MS is appropriate when sub-μg/L detection is required; EPA Method 200.8 specifies filtration/acidi
59 entries).

For very low-level REE detection (e.g., to confirm residuals after high removal), triple-
quadrupole ICP-MS methods report sub-ng/L instrumental detection limits; one example reports an Nd ins
MS system.

Calibration and QC

QC should include spike-recovery tests and internal standards because matrix effects can bias trace REE

For ICP-MS sample preparation in an acid matrix, one EPA 200.8 application describes preparing calibrat

Data analysis

Removal efficiency should be calculated from initial and final concentrations, and uptake capacity sho
balance basis as q (mg metal per g dry biomass), consistent with standard biosorption reporting and com

Kinetic curves should be fit to pseudo-first-order and pseudo-second-order models because these are wi
second-order behavior for REE uptake.

Equilibrium data should be fit to Langmuir and Freundlich isotherms to report qmax and affinity constan

Success criteria

Success should be defined using explicit performance thresholds that include metal removal, loading ca

Where selectivity is relevant (e.g., separating REEs from non-REE metals), report selectivity factors
based separation report observed separation factors of at least 30 for Nd relative to nearly all non-
REE metals after a single adsorption/desorption cycle, showing that quantitative selectivity metrics a

Safety

Acid handling is integral to this protocol (sample preservation and regeneration), so laboratory safet
resistant PPE, secondary containment, and procedures for nitric acid use in preservation and in digesti

Because some analytical workflows and sample digestions use strong acids (including concentrated nitri

Timeline and budget

The achievable project cadence is driven primarily by the number of experimental conditions, replicate
MS workflows can reduce total analysis time to under 3 minutes per sample for multi-
element panels in some configurations.

For high-throughput screening, ICP-OES and ICP-MS platforms are reported to support very high daily sar
OES and up to ~1200 samples/day for ICP-MS in one vendor comparison), which can be used to plan sample b
phase program.

Budget should be built bottom-up from the number of samples (conditions × timepoints × replicates) and
MS often requires dilution and careful matrix management for high-TDS coolants, while ICP-
OES is more robust for high-TDS and suspended-solids matrices.

Chapter 20

TAM/SAM/SOM Market Sizing

Executive summary

The addressable market for wastewater treatment and metal recovery in defense-
grade rare-earth magnet machining is best treated as a *subset* of the broader industrial wastewater tr
site treatment markets, because public sources rarely quantify magnet-machining wastewater volumes dir
specific consumption and waste details.

On the “demand pull” side, rare-earth permanent magnets are described as the widest use of rare earth el
guided munitions). On the “waste supply” side, multiple sources emphasize that swarf is generated when
40% for brittle sintered blocks in some contexts.

Because defense programs represent a small share of total magnet demand and because DOD purchases are d
only wastewater treatment market is structurally niche, but concentrated into a small set of high-
value machining and finishing operations that are currently being expanded or onshored via DOD investme

TAM

A practical top-down TAM framing starts from the industrial/on-site wastewater treatment markets, whic
machining wastewater treatment modules. The industrial wastewater treatment equipment market is valued
site wastewater treatment systems market is cited at USD 23.4B in 2025, forecast to USD 59.4B by 2034 (1

A second, magnet-specific TAM lens anchors the scale of magnet production and machining waste that coul
bearing coolants, rinse waters, and sludge. Global rare-earth magnet market value is repeatedly cited a

However, global volume estimates vary by source and definition. For example, one recycling/economics p
150,000 metric tons/year, while other commentary reports China alone producing ~300,000 tonnes of NdFe
earth magnet demand and that downstream machining is a major loss point that produces swarf and related
site treatment and recovery.

In public DOE work, the swarf recycling stream in 2020 is estimated as ~9% of NdFeB magnets produced glo

40%, which provides a plausible “high-loss” bound for machining-driven waste generation where tight to

SAM

The SAM here is defined as magnet-machining wastewater treatment opportunities in U.S. and allied defer
reliability SmCo and NdFeB magnets for defense platforms. Public sources emphasize that defense demand
driven volume but does not eliminate the need for dedicated, compliant, traceable manufacturing and was

Despite the small global share, U.S. defense magnet requirements are described in the thousands of tons
of-magnitude anchor for potential machining throughput. One source states that “the US military requir
4,000 tons of highly specialized magnets annually with projected demand up to 10,000 tons by 2030.

In parallel, the U.S. and allied SAM is shaped by “mine-to-magnet” investments aimed at onshoring produ
China supply chains. DOD reports that since 2020 it has awarded more than USD 439M to establish domestic

SOM

The SOM is defined as the realistically serviceable market reachable by deploying at a small number of U
relevant magnet production and finishing sites (3–5 facilities) over a five-year period, consistent with the near-term concentration of domestic capacity expansion into a limited

Public sources identify a short list of U.S.-based facilities or facility-scale initiatives that plausibly concentrate swarf generation and coolant-handling needs:

- Noveon Magnetics is stated to have established a rare earth magnet manufacturing facility in San Marc
- MP Materials is described as commissioning a magnetics facility in Texas (“Independence”) that is poi
- E-VAC Magnetics is described as establishing a commercial-scale rare earth magnet manufacturing capa

Additional defense-relevant machining capacity may sit within established specialty magnet suppliers a

Per-platform consumption

Per-platform metrics are useful for explaining why defense magnets are strategically important even if
level rare-earth content examples: an F-35 requires more than ~900 pounds of rare earth elements, an Ar
class submarine requires ~9,200 pounds.

Some sources add a more magnet-specific anchor by stating that an F-35 contains about 50 pounds of samar
cobalt magnets, which is relevant because SmCo machining waste streams may have different chemistry and
bearing waste).

These per-platform numbers are not a direct proxy for magnet-machining wastewater volume, but they are
material-bearing wastes within secure, compliant supply chains.

US defense facility map

A facility map for the U.S. defense-relevant rare-earth magnet and upstream processing ecosystem can be
term capacity additions supported by DOD funding. As of a 2023 Federal Register summary, there was “cur

In DOD’s public “mine-to-magnet” disclosures, the department highlights U.S.-based participants and funded efforts including MP Materials, Lynas USA (oxide production capability b
scale magnet manufacturing by 2025).

Separately, upstream mining/separation constraints shape where compliant magnet-making and recycling can locate. GAO cites USGS estimates that China mined ~240,000 metric tons of rare

Growth drivers

The strongest macro drivers of rare-earth magnet demand (and therefore machining throughput and waste g
intensive technology transitions that support growth in magnet production, finishing, and recycling in

On the magnet-market side, DOE’s NdFeB supply-chain report projects rapid total global demand growth fo
zero scenarios (12.5% average annual growth through 2030 and 6.3% through 2050) and also projects U.S.

Defense is repeatedly treated as a smaller-but-sticky segment: the Federal Register notes defense dema

Finally, policy and supply-chain security are explicit growth drivers for domestic magnet production a
to-magnet supply chain capable of supporting U.S. defense requirements by 2027.

Revenue scenarios

Publicly available sources do not provide a direct, defensible dollar figure for “defense magnet machin
specific consumption or for machining waste quantification at the level needed to monetize wastewater s

Therefore, the most supportable way to present conservative/base/aggressive revenue scenarios is as *d
number forecasts. These scenarios are grounded in (i) a small number of identifiable near-term U.S. magnet production and finishing sites, (ii) evidence that machining generates swarf and can h
site treatment markets.

Conservative scenario: focus on one or two early-adopter sites in the U.S. mine-to-magnet ecosystem (e.g., a magnet manufacturer or finisher associated with DOD awards) where swarf gener

Base scenario: expand to multiple U.S. facilities (3–5) as domestic capacity ramps, including sites tha
scale magnet manufacturing capability (e.g., E-VAC Magnetics).

Aggressive scenario: scale alongside the projected expansion of U.S. rare-earth magnet manufacturing capacity (e.g., MP Materials’ projected ~10,000 metric tons capacity after
recovery and reuse solutions are advancing in the wastewater equipment market.

Adjacent markets

Even if defense demand is structurally small relative to global totals, adjacent commercial markets tha

growing. These include EV traction motors and wind turbines, both of which are repeatedly cited as majo

For example, DOE reports that each electric vehicle motor typically requires 1–2 kg of permanent magnet material, and wind turbine permanent magnet mass is cited as 2.7–3.2 tonnes per MW of wind turbine capacity, which implies substantial machining throughput in the broad

These adjacent markets matter for wastewater treatment TAM because they (a) grow the installed base of site treatment systems, a category that is described as growing and increasingly important in industria

Assumptions

This market sizing synthesis relies on several explicit assumptions and constraints imposed by the publ source evidence base. First, public sources frequently quantify platform-level rare earth usage but do not provide platform-level magnet masses for most systems, and at least on adjacent summary explicitly notes that further detailed data may be required for quantities of rare ear

Second, the literature and policy documents clearly identify swarf generation during shaping and empha facing supply-chain reports do not provide a single, definitive machining waste fraction for magnet man 40% in cutting/grinding, swarf streams estimated at 9% of global NdFeB production, and research report

Third, defense share of the overall permanent magnet market is generally described qualitatively as sma level procurement and wastewater-flow data.

Finally, while multiple wastewater market reports provide market sizes and growth rates for industrial site systems, they do not break out magnet machining as a distinct end-use. As a result, any attempt to translate these top-down market sizes into a magnet-machining-specific dollar TAM requires additional bottom-up facility-level data collection (e.g., coo